Supplementary Material: Additional context to all results presented

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1 Introduction

This document aims to detail all the analyses conducted for the article "Fertility rate among adolescents in Brazil in the period 2012-2021: effects of human development, primary care coverage and the COVID-19 pandemic", providing additional context to the results shown in the paper.

2 Importing the necessary packages

```
library(knitr)
library(kableExtra)
library(reactable)
library(dplyr)
library(tidyr)
library(ggplot2)
library(trend)
library(factoextra)
library(clusterSim)
library(clValid)
library(htmltools)
library(gridExtra)
library(grid)
library(rstatix)
library(sf)
library(geobr)
library(corrplot)
library(gamlss)
```

3 About the dataset

```r
# Importing the dataset
df_indicadores <- read.csv("databases/data_fertility_rates_article.csv") |>
  mutate(
    regiao = factor(
      regiao,
      levels = c("Norte", "Nordeste", "Centro-Oeste", "Sudeste", "Sul")
    )
  )
```

3.1 Variables description

The dataset used in this study included variables available on several different databases. To obtain the variables from SINASC (more specifically, the number of births of mothers in each age group), we used the `{microdatasus}` package in R. To obtain the population estimates for each age group, we used Tabnet DATASUS (using the updated estimates after the 2022 census). To obtain the number of women beneficiaries of private health plans (used for calculating the percentage of women who exclusively use the SUS), we used Tabnet ANS. To obtain the variables used to calculate the Primary Care coverage, we used the "e-Gestor Atenção Primária à Saúde" platform. Finally, to obtain the HDI-M, we used Atlas Brasil. All data was obtained for the period 2012 to 2021. A description of the variables included in the file `data_fertility_rates_article.csv` can be seen in **Table 1**.

Table 1: Description of the variables in the dataset.

Variable	Description	Source
codmunres	Municipality code	SINASC
municipio	Municipality name	SINASC
uf	Federative Unit	SINASC
regiao	Brazilian Region	SINASC
ano	Year	—
nvm_10_a_14	Live births of mothers aged 10 to 14 years	SINASC
nvm_15_a_19	Live births of mothers aged 15 to 19 years	SINASC
nvm_10_a_19	Live births of mothers aged 10 to 19 years	SINASC
pop_feminina_10_a_14	Estimated female population aged 10 to 14 years	Tabnet DATASUS
pop_feminina_15_a_19	Estimated female population aged 15 to 19 years	Tabnet DATASUS
pop_feminina_10_a_19	Estimated female population aged 10 to 19 years	Tabnet DATASUS
pop_feminina_10_a_49	Estimated female population aged 10 to 49 years	Tabnet DATASUS
idhm	HDI-M	Atlas Brasil
media_cobertura_ab	Population covered by Primary Care teams	e-Gestor Atenção Primária à Saúde
populacao_total	Total population	e-Gestor Atenção Primária à Saúde
pop_fem_10_49_com_plano_saude	Female population aged 10 to 49 years with private health insurance	Tabnet ANS

4 Phase 1: Time Series analysis

This stage of the article included a time series analysis of the evolution of fertility rates for girls aged 10 to 14 and 15 to 19 from 2012 to 2021. We used both the country as a whole and its 27 federative units as the unit of analysis. For the country as a whole, we visualized the evolution of both rates using a line graph, in addition to using the nonparametric Mann-Kendall test to detect trends in the time series. For the federative units, only the Mann-Kendall test was used. A negative value of the statistic test indicates a downward trend, a positive value indicates an upward trend and a statistic close to zero indicates no trend, with a p-value of less than 0.05 indicating statistical significance.

```
# Preparing the data
## Aggregating the data for the whole country
df_fecundidade_br <- df_indicadores |>
  group_by(ano) |>
  summarise(
    nvm_10_a_14 = sum(nvm_10_a_14),
    pop_feminina_10_a_14 = sum(pop_feminina_10_a_14),
    tx_fecundidade_10_a_14 = round(
      nvm_10_a_14 / pop_feminina_10_a_14 * 1000,
      1
    ),
    nvm_15_a_19 = sum(nvm_15_a_19),
    pop_feminina_15_a_19 = sum(pop_feminina_15_a_19),
    tx_fecundidade_15_a_19 = round(nvm_15_a_19 / pop_feminina_15_a_19 * 1000, 1)
  ) |>
  ungroup()

## Aggregating the data for each state
df_fecundidade_ufs <- df_indicadores |>
  group_by(ano, uf, regiao) |>
  summarise(
    nvm_10_a_14 = sum(nvm_10_a_14),
    pop_feminina_10_a_14 = sum(pop_feminina_10_a_14),
    tx_fecundidade_10_a_14 = round(
      nvm_10_a_14 / pop_feminina_10_a_14 * 1000,
      1
    ),
    nvm_15_a_19 = sum(nvm_15_a_19),
    pop_feminina_15_a_19 = sum(pop_feminina_15_a_19),
    tx_fecundidade_15_a_19 = round(nvm_15_a_19 / pop_feminina_15_a_19 * 1000, 1)
  ) |>
  ungroup() |>
  mutate(
    uf = factor(
      uf,
      levels = df_indicadores |>
        dplyr::select(uf, regiao) |>
        arrange(regiao, uf) |>
        pull(uf) |>
        unique()
    )
  ) |>
  arrange(uf)
```

Figure 1 presents the time series from 2012 to 2021 of fertility rates for the age groups 10 to 14 years and 15 to 19 years in Brazil. For the 10 to 14 year old age group, there is an average decrease of 7.69% in the fertility rate per year in the pre-pandemic period (2018 and 2019). During the pandemic, there was a decrease of 8.33% from 2019 to 2020, and the rate was stationary from 2020 to 2021. For the age group from 15 to 19 years old, the average decrease in the fertility rate per year before the pandemic was 6.55%. During the pandemic, there was a 7.92% decrease from 2019 to 2020 and a 3.09% decrease from 2020 to 2021.

```r
# Ploting Brazil's time series for both indicators
## Transforming the data.frame to the long format
df_fecundidade_br_unico <- df_fecundidade_br |>
  pivot_longer(
    cols = starts_with("tx"),
    names_to = "faixa_etaria",
    values_to = "tx_fecundidade"
  ) |>
  mutate(
    faixa_etaria = case_when(
      faixa_etaria == "tx_fecundidade_10_a_14" ~ "10 to 14 years",
      faixa_etaria == "tx_fecundidade_15_a_19" ~ "15 to 19 years"
    )
  )

## Ploting the time series
ggplot(
  data = df_fecundidade_br_unico,
  mapping = aes(x = ano, y = tx_fecundidade, color = faixa_etaria)
) +
  geom_line(linewidth = 1) +
  geom_point(aes(shape = faixa_etaria)) +
  scale_x_continuous(
    breaks = unique(df_fecundidade_br_unico$ano),
    guide = guide_axis(angle = 45)
  ) +
  theme_bw(base_size = 14) +
  labs(
    color = "Age groups",
    shape = "Age groups",
    x = "Year",
    y = "Fertility rate"
  ) +
  geom_text(
    label = df_fecundidade_br_unico$tx_fecundidade,
    nudge_y = 2,
    show.legend = FALSE
  ) +
  theme(legend.position = "bottom") +
  scale_color_manual(values = c("Salmon", "DodgerBlue"))
```

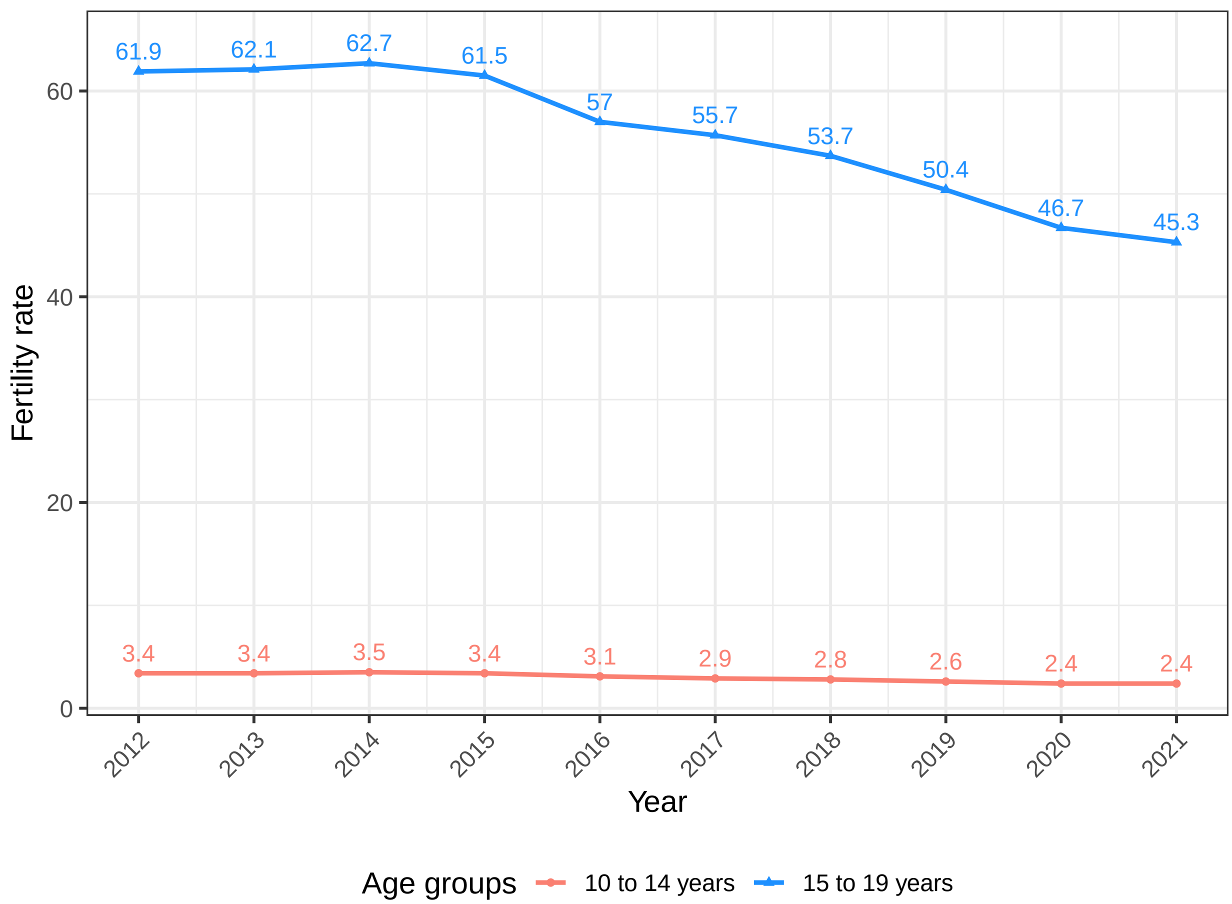

Figure 1: Evolution of fertility rates in the age groups of 10 to 14 years and 15 to 19 years in Brazil, 2012 – 2021.

Furthermore, in the period from 2012 to 2021, as it can be seen in **Table 2**, fertility rates for girls aged 10 to 14 and 15 to 19 years old showed statistically significant decreases in Brazil, going from 3.4 to 2.4 and from 61.9 to 45.3 per 1,000, respectively. The reduction in these rates was also observed in most Federative Units, except in the age group of 10 to 14 years in the Federative Units of Acre, Piauí and Roraima, where they showed non-significant decreasing trends, and in the age group of 15 to 19 years in the Federative Unit of Roraima, where it showed a non-significant decreasing trend (decrease from 94.7 to 92.6, p value = 0.720).

```r
# Mann-Kendall tests for trend analysis
## Joining the country and the states data
df_fecundidade_tendencia <- full_join(
  df_fecundidade_br |> mutate(local = "Brasil", .before = "ano"),
  df_fecundidade_ufs |> rename(local = uf)
)

## Selecting the desired variables
variaveis <- c("tx_fecundidade_10_a_14", "tx_fecundidade_15_a_19")

## Creating a data.frame with the Mann-Kendall tests results
results_table_completa <- data.frame()

for (i in 1:length(unique(df_fecundidade_tendencia$local))) {
  localidade <- unique(df_fecundidade_tendencia$local)[i]
```

```r
  mann_kendall_results <- lapply(
    df_fecundidade_tendencia |>
      filter(local == localidade) |>
      dplyr::select(all_of(variaveis)),
    function(x) {
      mann_kendall_test <- mk.test(x)
      return(c(
        local = localidade,
        p_value = mann_kendall_test$p.value,
        z_value = mann_kendall_test$statistic
      ))
    }
  )

  p_value <- round(
    as.numeric(unlist(lapply(mann_kendall_results, `[[`, "p_value"))),
    5
  )

  results_table <- data.frame(
    local = as.vector(unlist(lapply(mann_kendall_results, `[[`, "local"))),
    Variavel = names(mann_kendall_results),
    valor_2012 = lapply(
      variaveis,
      function(variavel) {
        df_fecundidade_tendencia |>
          filter(local == localidade, ano == 2012) |>
          pull(variavel)
      }
    ) |>
      as.numeric(),
    valor_2021 = lapply(
      variaveis,
      function(variavel) {
        df_fecundidade_tendencia |>
          filter(local == localidade, ano == 2021) |>
          pull(variavel)
      }
    ) |>
      as.numeric(),
    Mann_Kendall_z = round(
      as.numeric(unlist(lapply(mann_kendall_results, `[[`, "z_value.z"))),
      3
    ),
    Mann_Kendall_p = p_value
  )

  results_table_completa <- bind_rows(
    results_table_completa,
    results_table
  )
}

## Organizing the resulting table
results_table_completa_organizada <- results_table_completa |>
  filter(startsWith(Variavel, "tx")) |>
  pivot_wider(
    names_from = Variavel,
```

```r
    values_from = c(
      valor_2012,
      valor_2021,
      Mann_Kendall_z,
      Mann_Kendall_p
    )
  ) |>
  select_at(vars("local", ends_with("10_a_14"), ends_with("15_a_19")))

## Printing the table
results_table_completa_organizada |>
  kbl(
    format = "latex",
    col.names = c(
      "",
      "FR 2012",
      "FR 2021",
      "Mann-Kendall",
      "p-value",
      "FR 2012",
      "FR 2021",
      "Mann-Kendall",
      "p-value"
    ),
    align = "lcccccccc",
    caption = "Test statistics and p-value of the Mann-Kendall test for the
      fertility rate of the age groups 10 to 14 years and 15 to 19 years. Brazil
      and Federative Units, 2012 - 2021."
  ) |>
  add_header_above(c(
    "Location/age" = 1,
    "10-14 years old" = 4,
    "15-19 years old" = 4
  )) |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE)
```

Table 2: Test statistics and p-value of the Mann-Kendall test for the fertility rate of the age groups 10 to 14 years and 15 to 19 years. Brazil and Federative Units, 2012 - 2021.

Location/age	10–14 years old				15–19 years old			
	FR 2012	**FR 2021**	**Mann-Kendall**	**p-value**	**FR 2012**	**FR 2021**	**Mann-Kendall**	**p-value**
Brasil	3.4	2.4	-3.282	0.00103	61.9	45.3	-3.399	0.00068
Acre	7.1	6.4	-1.443	0.14911	101.3	79.4	-3.143	0.00167
Amapá	5.8	5.1	-2.525	0.01158	94.9	78.5	-3.041	0.00236
Amazonas	7.2	6.2	-3.066	0.00217	101.9	88.1	-3.041	0.00236
Pará	5.7	4.9	-2.735	0.00624	88.9	73.0	-3.323	0.00089
Rondônia	3.6	2.7	-2.326	0.02004	73.6	54.6	-3.757	0.00017
Roraima	8.3	7.1	-1.803	0.07133	94.7	92.6	-0.358	0.72051
Tocantins	5.3	4.3	-2.451	0.01424	79.8	65.0	-3.220	0.00128
Alagoas	6.1	4.3	-3.607	0.00031	81.5	68.5	-2.784	0.00537
Bahia	4.0	3.0	-3.387	0.00071	65.4	48.3	-3.935	0.00008
Ceará	3.9	3.0	-2.735	0.00624	59.7	45.2	-3.220	0.00128
Maranhão	5.3	4.7	-2.917	0.00353	84.0	70.2	-2.504	0.01227
Paraíba	3.7	2.6	-2.705	0.00683	64.8	53.4	-3.041	0.00236
Pernambuco	4.3	2.9	-3.323	0.00089	71.5	52.4	-3.578	0.00035
Piauí	3.7	3.2	-1.823	0.06827	67.8	55.2	-2.504	0.01227
Rio Grande do Norte	4.0	2.6	-3.143	0.00167	62.5	44.9	-3.578	0.00035
Sergipe	4.1	3.3	-2.525	0.01158	67.7	52.1	-2.862	0.00421
Distrito Federal	2.1	1.1	-3.099	0.00194	49.0	29.5	-3.757	0.00017
Goiás	3.1	1.9	-3.426	0.00061	59.9	43.3	-3.220	0.00128
Mato Grosso	4.1	3.7	-2.735	0.00624	73.8	61.6	-2.963	0.00304
Mato Grosso do Sul	5.0	3.2	-3.399	0.00068	78.1	55.9	-3.399	0.00068
Espírito Santo	2.8	2.3	-2.425	0.01532	57.7	43.2	-3.220	0.00128
Minas Gerais	2.0	1.4	-3.099	0.00194	48.2	34.9	-3.399	0.00068
Rio de Janeiro	2.9	1.9	-3.041	0.00236	56.6	40.3	-2.963	0.00304
São Paulo	2.2	1.1	-3.502	0.00046	51.7	30.8	-3.722	0.00020
Paraná	3.0	1.6	-3.426	0.00061	59.1	36.9	-3.578	0.00035
Rio Grande do Sul	2.2	1.3	-3.066	0.00217	50.2	31.2	-3.502	0.00046
Santa Catarina	2.2	1.1	-3.350	0.00081	48.2	35.0	-2.963	0.00304

5 Phase 2: Cluster analysis

This stage of the analysis aimed to find four different groupings for Brazilian municipalities: one that took into account the fertility rate of women aged 10 to 14 in the period from 2018 to 2019; one that took into account this same rate, but in the period from 2020 to 2021; one that took into account the fertility rate of women aged 15 to 19 in the period from 2018 to 2019; and one that considered this same rate, but in the period from 2020 to 2021. To this end, we adjusted non-hierarchical methods (K-means and K-medoids) and hierarchical methods (Ward, single linkage, complete linkage and average linkage). For the non-hierarchical methods, we chose candidates for the number of groups using the knee plot. For the hierarchical methods, we chose candidates for the number of groups using dendrograms. We compared the chosen candidates using the Davies-Bouldin, Dunn, Silhouette and Calinski-Harabasz indices, and chose as final clusters those that presented the best average performance in these metrics (as long as the number of observations in each cluster is adequate). Finally, after determining the four different groupings, we compared, for each grouping, the behavior of some variables of interest in each cluster formed.

```r
# Preparing the data
## Filtering and calculating the fertility rates for each period
df_indicadores_18_19 <- df_indicadores |>
  filter(ano %in% c(2018, 2019)) |>
  group_by(codmunres) |>
  summarise(
    tx_fecundidade_10_a_14 = round(
      sum(nvm_10_a_14) / sum(pop_feminina_10_a_14) * 1000,
      1
    ),
```

```
    tx_fecundidade_15_a_19 = round(
      sum(nvm_15_a_19) / sum(pop_feminina_15_a_19) * 1000,
      1
    ),
    cobertura_ab = round(
      sum(media_cobertura_ab) / sum(populacao_total) * 100,
      1
    ),
    porc_exclusivas_sus = round(
      sum(pop_fem_10_49_com_plano_saude) / sum(pop_feminina_10_a_49) * 100,
      1
    ),
    idhm = mean(idhm)
  ) |>
  ungroup()

df_indicadores_20_21 <- df_indicadores |>
  filter(ano %in% c(2020, 2021)) |>
  group_by(codmunres) |>
  summarise(
    tx_fecundidade_10_a_14 = round(
      sum(nvm_10_a_14) / sum(pop_feminina_10_a_14) * 1000,
      1
    ),
    tx_fecundidade_15_a_19 = round(
      sum(nvm_15_a_19) / sum(pop_feminina_15_a_19) * 1000,
      1
    ),
    cobertura_ab = round(
      sum(media_cobertura_ab) / sum(populacao_total) * 100,
      1
    ),
    porc_exclusivas_sus = round(
      sum(pop_fem_10_49_com_plano_saude) / sum(pop_feminina_10_a_49) * 100,
      1
    ),
    idhm = mean(idhm)
  ) |>
  ungroup()

## Standardizing the data for each of the four cluster analysis
dados1_18_19 <- df_indicadores_18_19 |>
  select(tx_fecundidade_10_a_14) |>
  scale()
dados2_18_19 <- df_indicadores_18_19 |>
  select(tx_fecundidade_15_a_19) |>
  scale()

dados1_20_21 <- df_indicadores_20_21 |>
  select(tx_fecundidade_10_a_14) |>
  scale()
dados2_20_21 <- df_indicadores_20_21 |>
  select(tx_fecundidade_15_a_19) |>
  scale()

## Calculating the distance matrices
dados1_18_19_dist <- dist(dados1_18_19, method = "euclidean")
dados2_18_19_dist <- dist(dados2_18_19, method = "euclidean")
```

```
dados1_20_21_dist <- dist(dados1_20_21, method = "euclidean")
dados2_20_21_dist <- dist(dados2_20_21, method = "euclidean")
```

5.1 For the 10 to 14 age group

5.1.1 K-Means

5.1.1.1 2018-2019

```
# Ploting the k-means knee-plot for the 2018-2019 period
fviz_nbclust(dados1_18_19, kmeans, method = "wss")
```

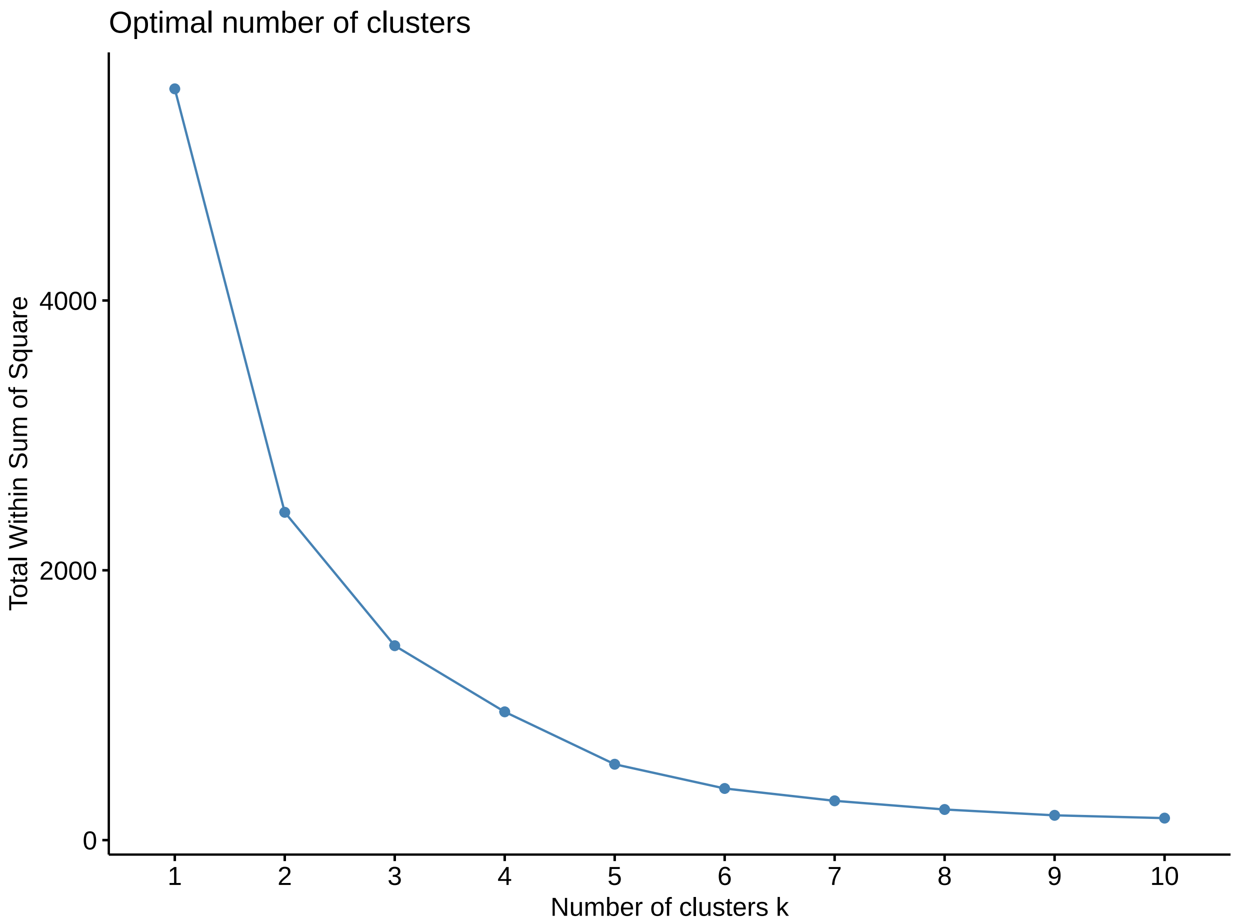

Figure 2: K-Means knee-plot for the 10 to 14 age group and 2018-2019 period

- **Candidates:** K = 3 and K = 4 (an argument that the total within-group variance does not decrease significantly after these values can be made for both numbers of groups).

```
# Adjusting the k-means method with the chosen numbers of clusters
set.seed(1504)
d1_kmeans3_18_19 <- kmeans(dados1_18_19, 3)
```

```r
set.seed(1504)
d1_kmeans4_18_19 <- kmeans(dados1_18_19, 4)
```

5.1.1.2 2020-2021

```r
# Ploting the k-means knee-plot for the 2020-2021 period
fviz_nbclust(dados1_20_21, kmeans, method = "wss")
```

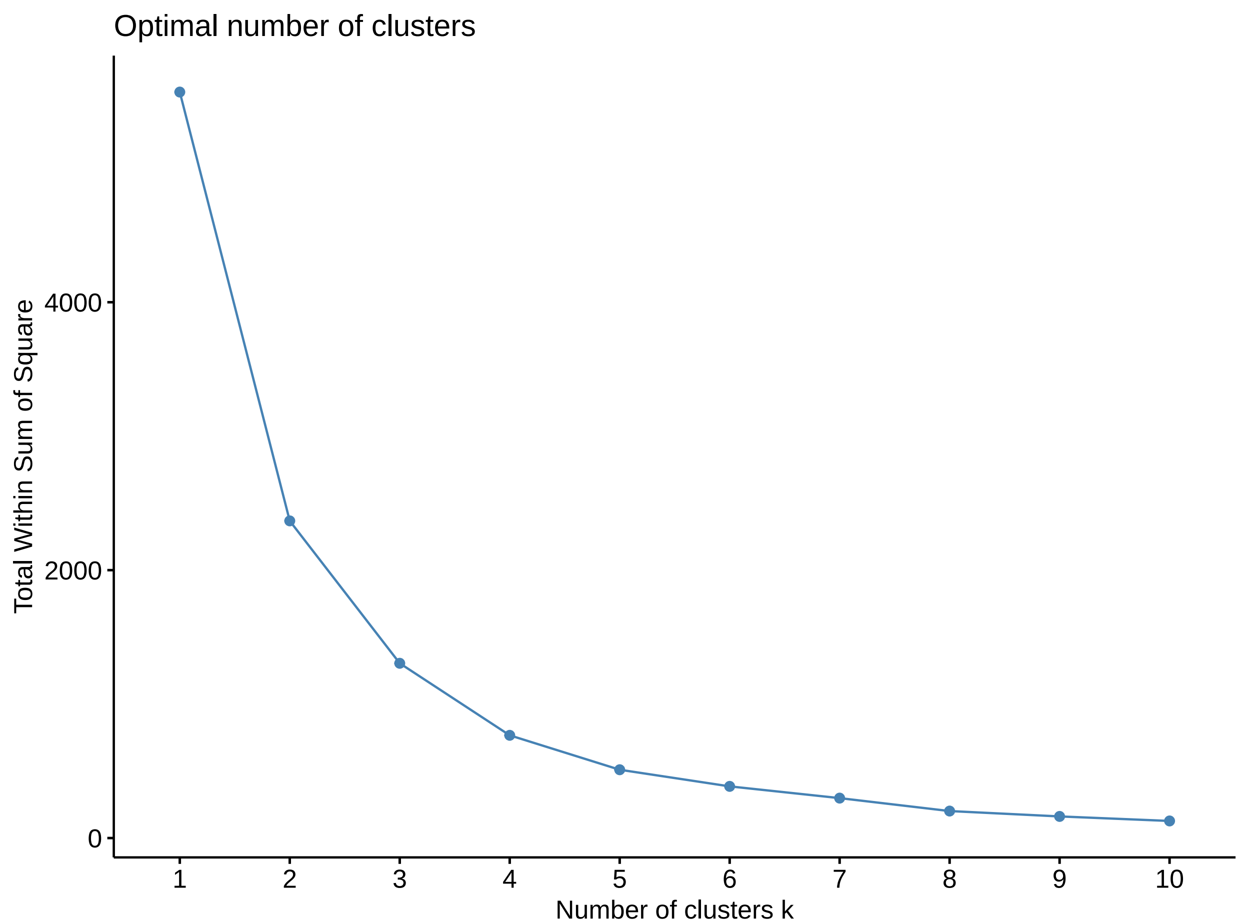

Figure 3: K-Means knee-plot for the 10 to 14 age group and 2020-2021 period

- **Candidates:** K = 3 and K = 4 (an argument that the total within-group variance does not decrease significantly after these values can be made for both numbers of groups).

```r
# Adjusting the k-means method with the chosen numbers of clusters
set.seed(1504)
d1_kmeans3_20_21 <- kmeans(dados1_20_21, 3)

set.seed(1504)
d1_kmeans4_20_21 <- kmeans(dados1_20_21, 4)
```

5.1.2 K-Medians

5.1.2.1 2018-2019

```r
# Ploting the k-medians knee-plots for the 2018-2019 period
fviz_nbclust(dados1_18_19, cluster::pam, method = "wss")
```

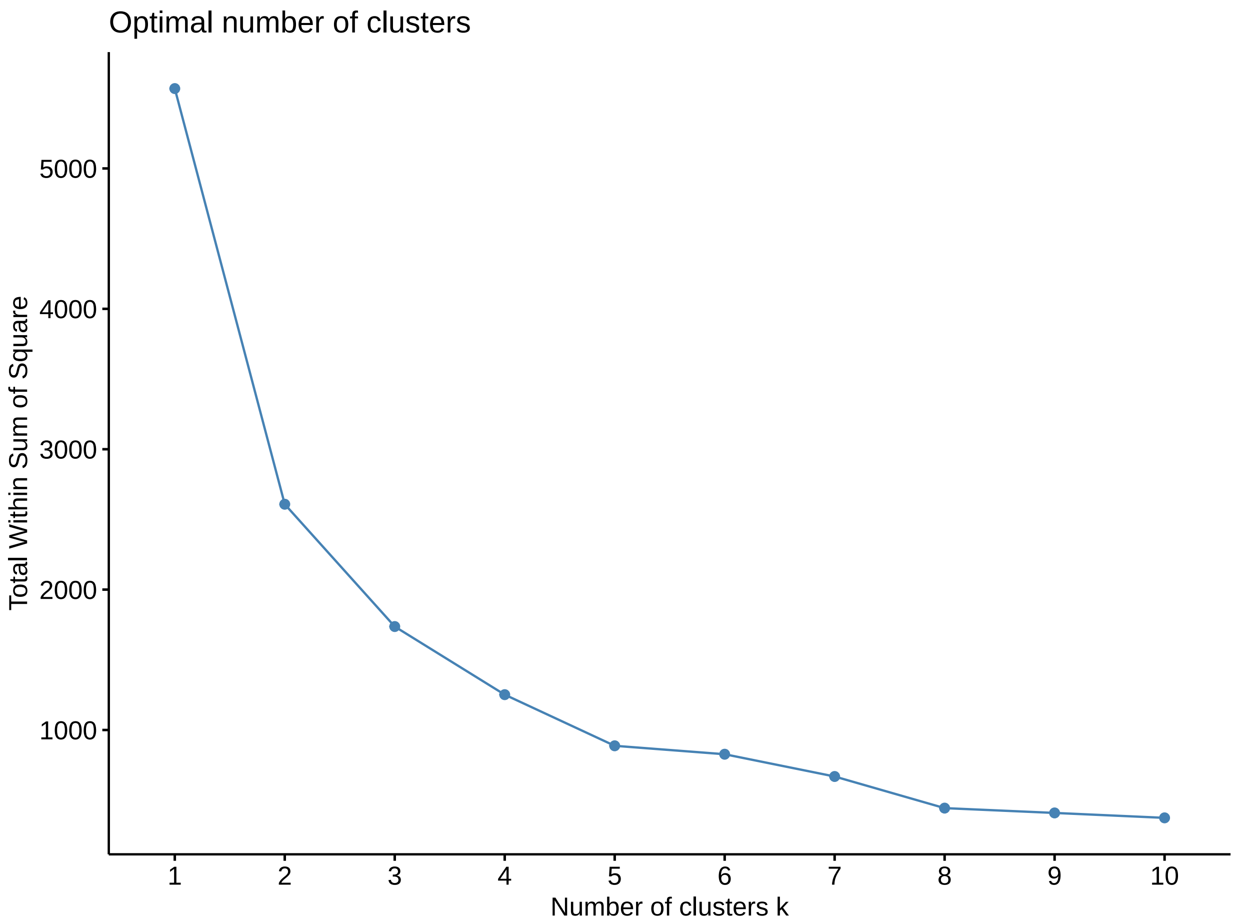

Figure 4: K-Medians knee-plot for the 10 to 14 age group and 2018-2019 period

- **Candidates:** K = 3 and K = 4 (an argument that the total within-group variance does not decrease significantly after these values can be made for both numbers of groups).

```r
# Adjusting the k-medians method with the chosen numbers of clusters
set.seed(1504)
d1_pam3_18_19 <- cluster::pam(dados1_18_19, k = 3)

set.seed(1504)
d1_pam4_18_19 <- cluster::pam(dados1_18_19, k = 4)
```

5.1.2.2 2020-2021

```r
# Ploting the k-medians knee-plots for the 2020-2021 period
fviz_nbclust(dados1_20_21, cluster::pam, method = "wss")
```

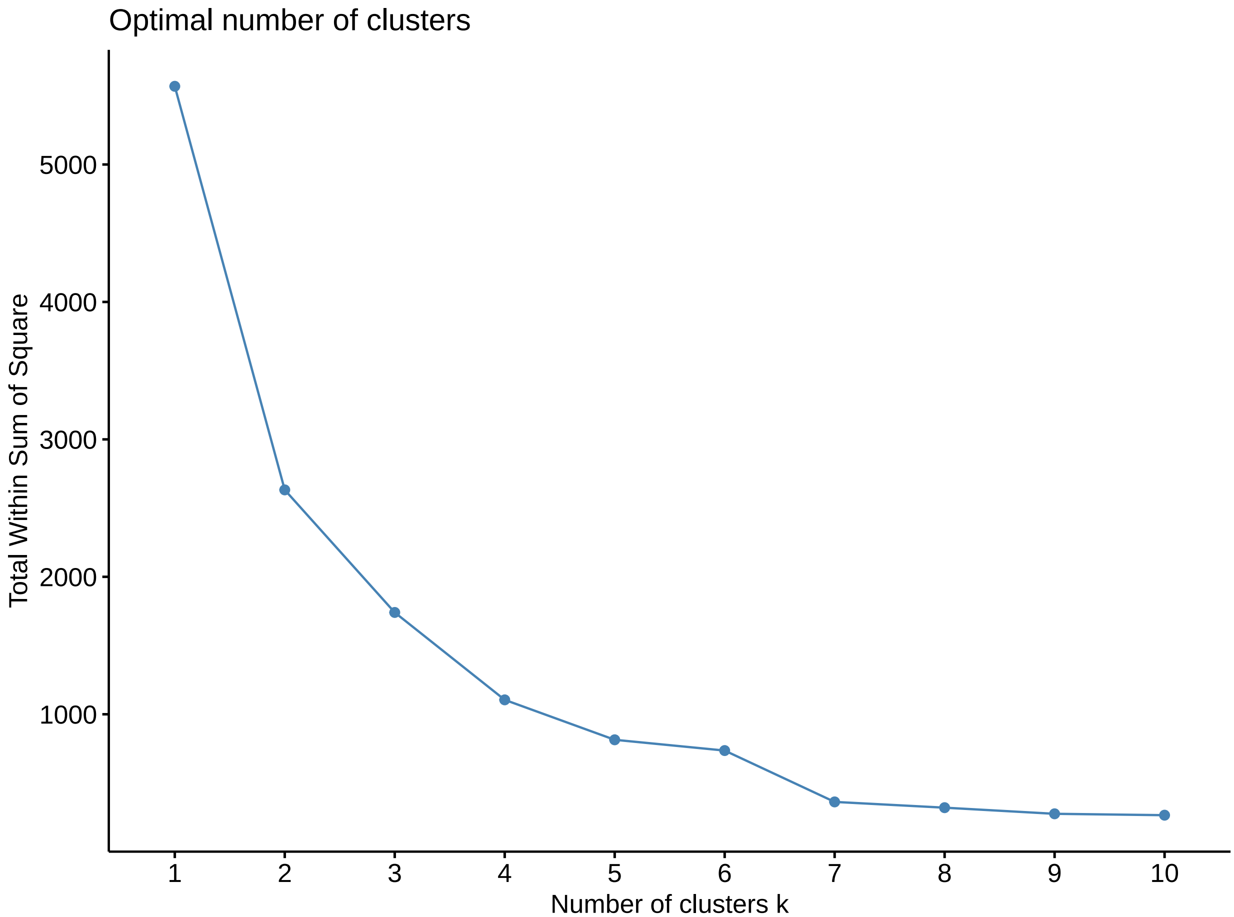

Figure 5: K-Medians knee-plot for the 10 to 14 age group and 2020-2021 period

- **Candidates:** K = 3 and K = 4 (an argument that the total within-group variance does not decrease significantly after these values can be made for both numbers of groups).

```
# Adjusting the k-medians method with the chosen numbers of clusters
set.seed(1504)
d1_pam3_20_21 <- cluster::pam(dados1_20_21, k = 3)

set.seed(1504)
d1_pam4_20_21 <- cluster::pam(dados1_20_21, k = 4)
```

5.1.3 Ward's method

5.1.3.1 2018-2019

```
# Ploting the drodrogram for the 2018-2019 period
set.seed(1504)
d1_ward_18_19 <- hclust(dados1_18_19_dist, method = "ward.D2")
plot(as.dendrogram(d1_ward_18_19))
```

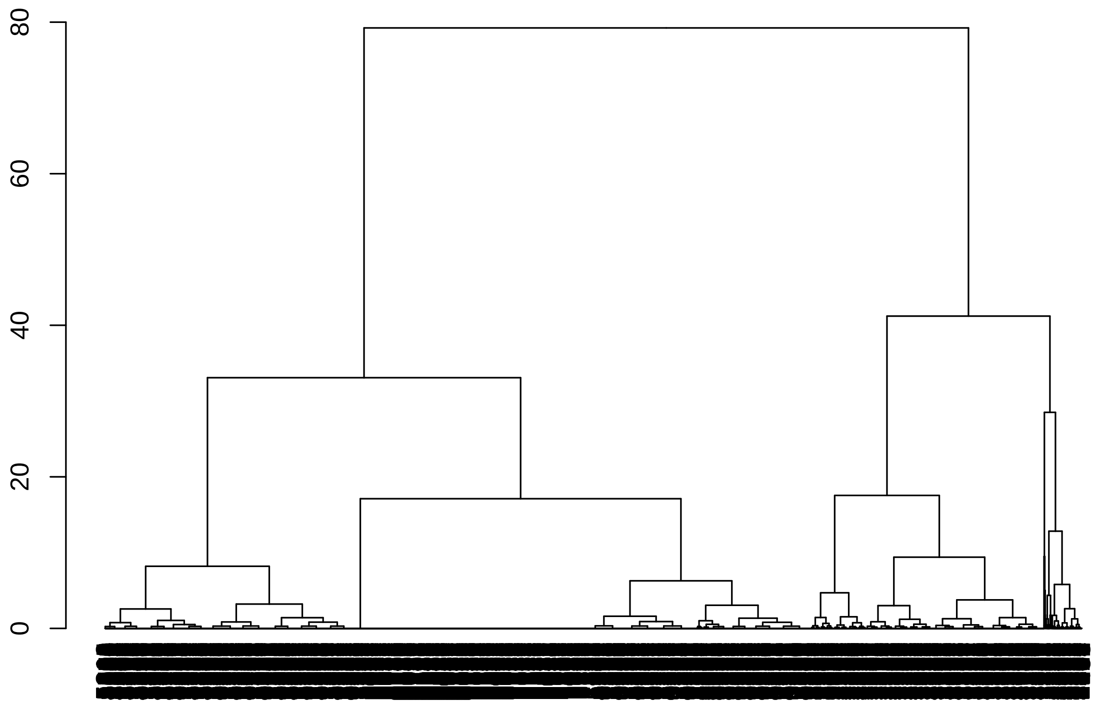

Figure 6: Dendrogram for Ward's method for the 10 to 14 age group and 2018-2019 period

- **Candidates:** K = 2 and K = 3.

```
# Adjusting Ward's method with the chosen numbers of clusters
d1_ward2_18_19_class <- cutree(d1_ward_18_19, k = 2)
d1_ward3_18_19_class <- cutree(d1_ward_18_19, k = 3)
```

5.1.3.2 2020-2021

```
# Ploting the drodrogram for the 2020-2021 period
set.seed(1504)
d1_ward_20_21 <- hclust(dados1_20_21_dist, method = "ward.D2")
plot(as.dendrogram(d1_ward_20_21))
```

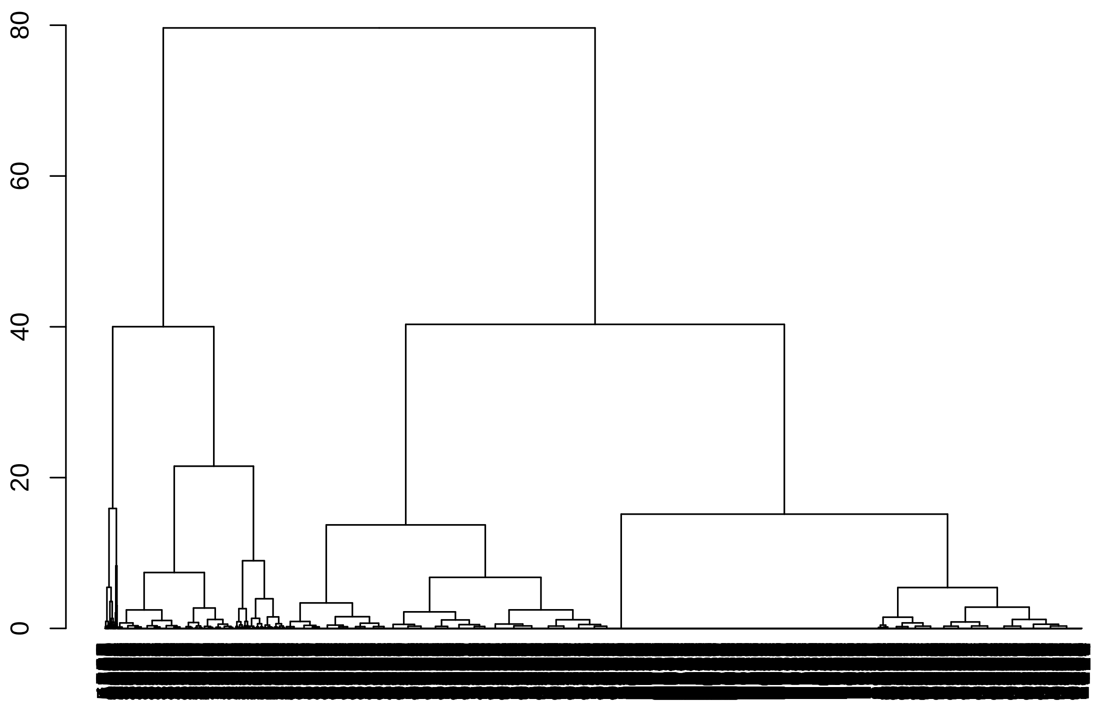

Figure 7: Dendrogram for Ward's method for the 10 to 14 age group and 2020-2021 period

- **Candidates:** K = 2 and K = 4.

```
# Adjusting Ward's method with the chosen numbers of clusters
d1_ward2_20_21_class <- cutree(d1_ward_20_21, k = 2)
d1_ward4_20_21_class <- cutree(d1_ward_20_21, k = 4)
```

5.1.4 Single linkage

5.1.4.1 2018-2019

```
# Ploting the dendrogram for the 2018-2019 period
set.seed(1504)
d1_single_18_19 <- hclust(dados1_18_19_dist, method = "single")
plot(as.dendrogram(d1_single_18_19))
```

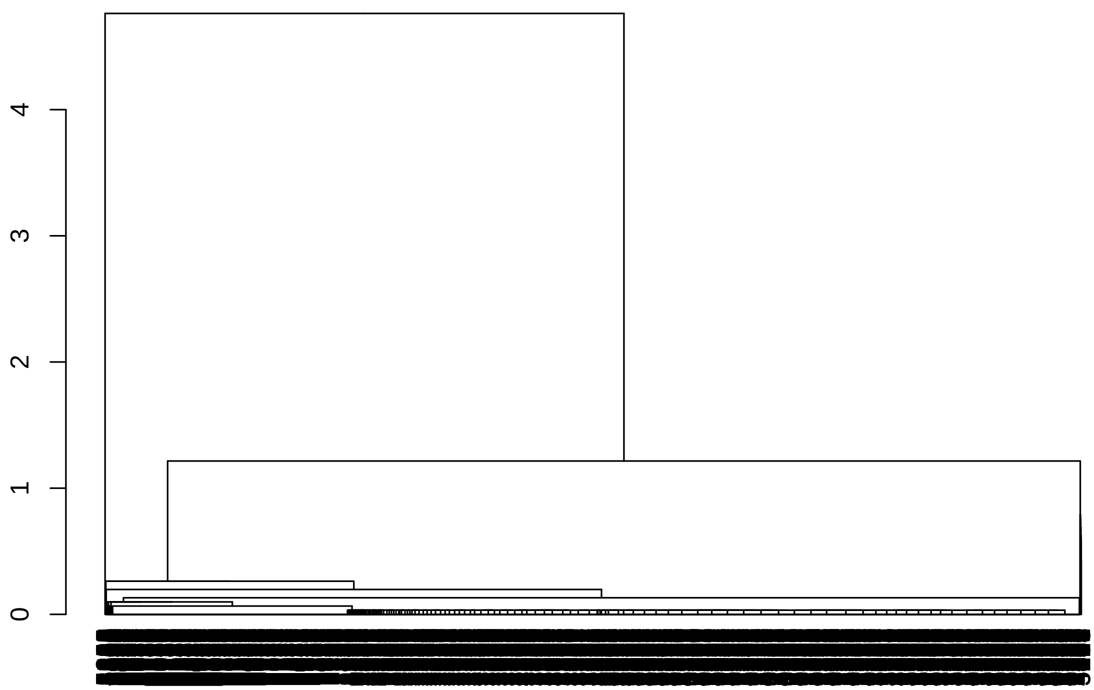

Figure 8: Dendrogram for the single linkage method for the 10 to 14 age group and 2018-2019 period

- **Candidates:** No candidates. A bad fit overall.

5.1.4.2 2020-2021

```
# Ploting the dendrogram for the 2020-2021 period
set.seed(1504)
d1_single_20_21 <- hclust(dados1_20_21_dist, method = "single")
plot(as.dendrogram(d1_single_20_21))
```

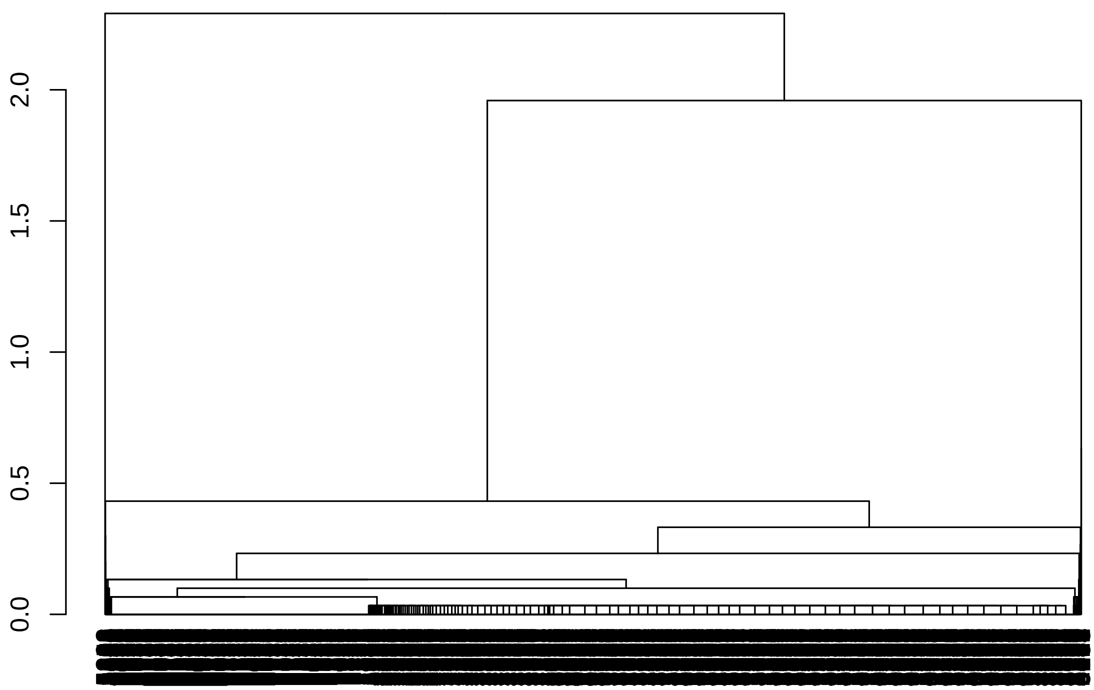

Figure 9: Dendrogram for the single linkage method for the 10 to 14 age group and 2020-2021 period

- **Candidates:** No candidates. A bad fit overall.

5.1.5 Complete linkage

5.1.5.1 2018-2019

```
# Ploting the dendrogram for the complete linkage method for the 2018-2019 period
set.seed(1504)
d1_complete_18_19 <- hclust(dados1_18_19_dist, method = "complete")
plot(as.dendrogram(d1_complete_18_19))
```

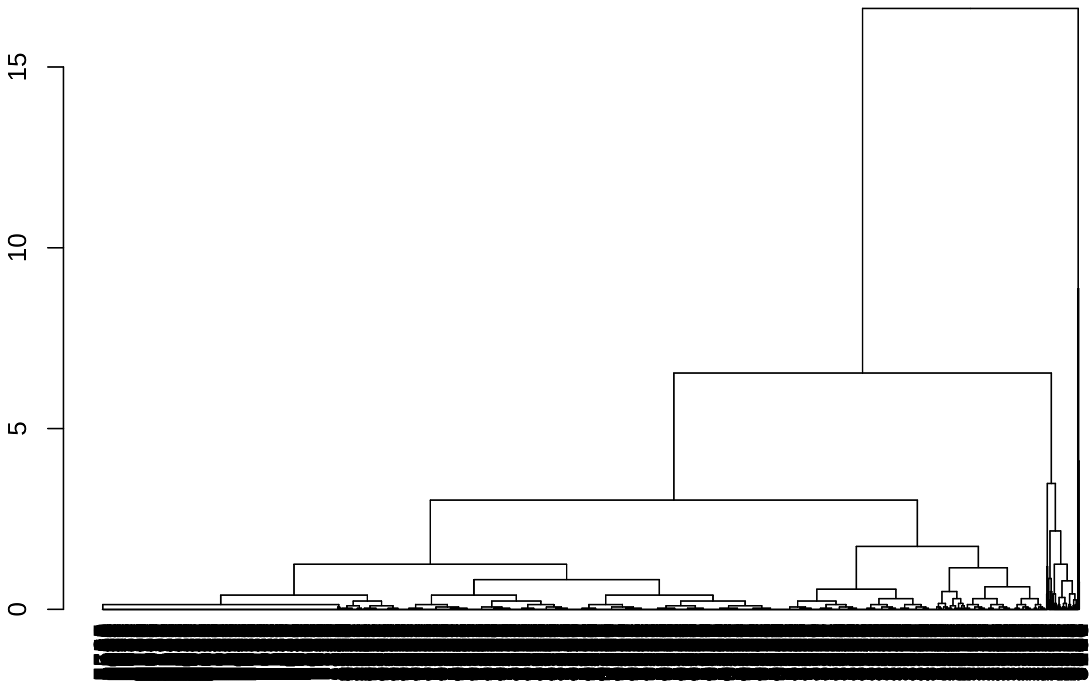

Figure 10: Dendrogram for the complete linkage method for the 10 to 14 age group and 2018-2019 period

- **Candidates:** No candidates. A bad fit overall.

5.1.5.2 2020-2021

```
# Ploting the dendrogram for the complete linkage method for the 2020-2021 period
set.seed(1504)
d1_complete_20_21 <- hclust(dados1_20_21_dist, method = "complete")
plot(as.dendrogram(d1_complete_20_21))
```

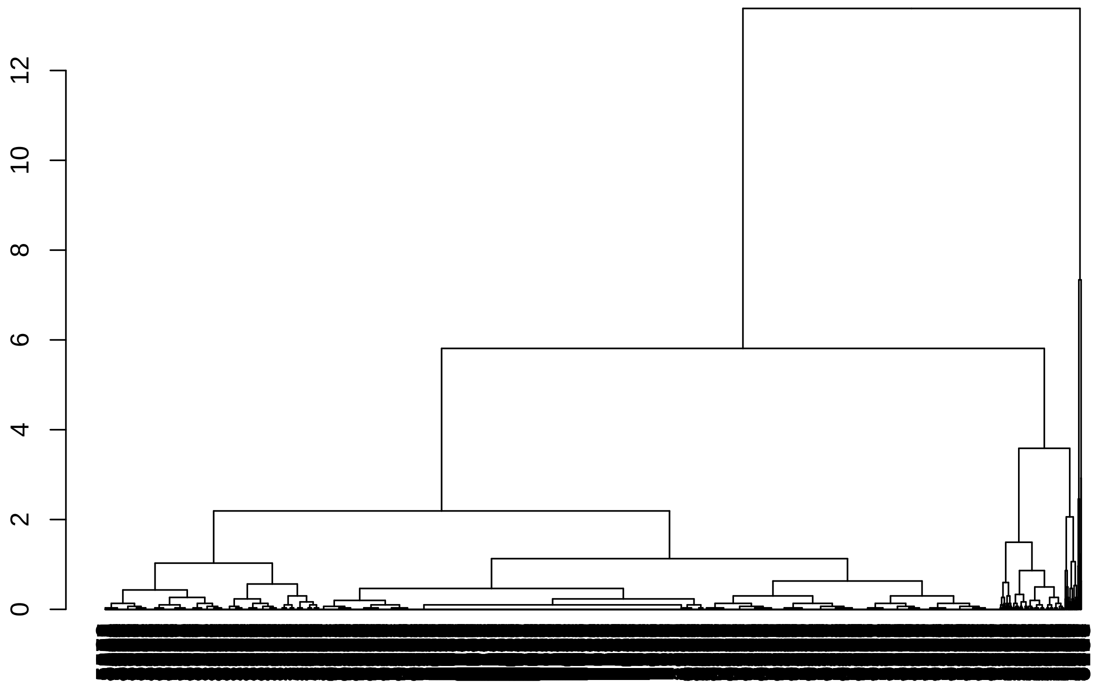

Figure 11: Dendrogram for the complete linkage method for the 10 to 14 age group and 2020-2021 period

- **Candidates:** No candidates. A bad fit overall.

5.1.6 Average linkage

5.1.6.1 2018-2019

```r
# Ploting the dendrogram for the average linkage method for the 2018-2019 period
set.seed(1504)
d1_average_18_19 <- hclust(dados1_18_19_dist, method = "average")
plot(as.dendrogram(d1_average_18_19))
```

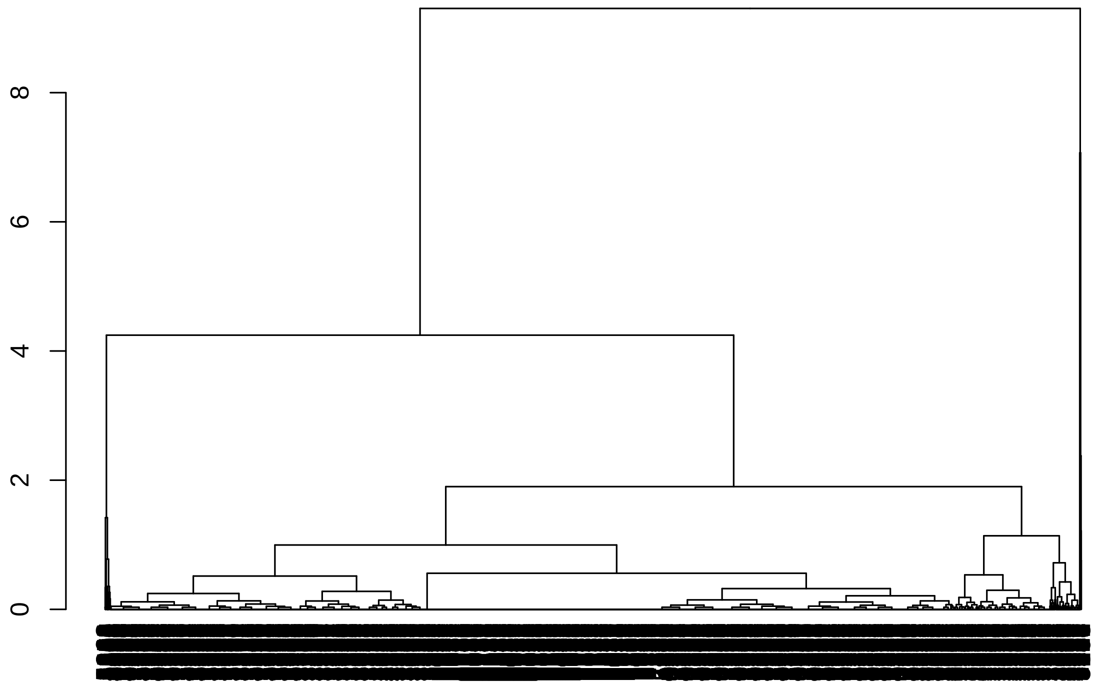

Figure 12: Dendrogram for the average linkage method for the 10 to 14 age group and 2018-2019 period

- **Candidates:** No candidates. A bad fit overall.

5.1.6.2 2020-2021

```r
# Ploting the dendrogram for the average linkage method for the 2020-2021 period
set.seed(1504)
d1_average_20_21 <- hclust(dados1_20_21_dist, method = "average")
plot(as.dendrogram(d1_average_20_21))
```

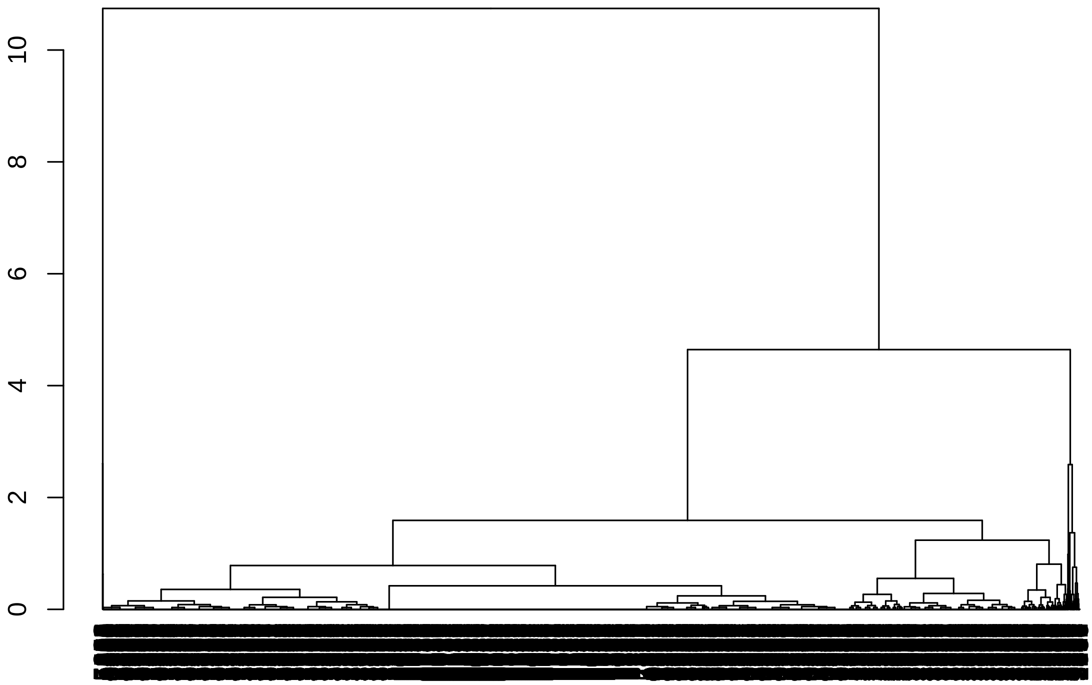

Figure 13: Dendrogram for the average linkage method for the 10 to 14 age group and 2020-2021 period

- **Candidates:** No candidates. A bad fit overall.

5.1.7 Choosing the best clustering method

5.1.7.1 2018-2019

```
# Calculating the considered indices
d1_kmeans_index_18_19 <- data.frame(
  method = unlist(lapply(3:4, function(i) paste0("kmeans", i))),
  db_index_cent = unlist(lapply(
    3:4,
    function(i)
      index.DB(dados1_18_19, get(paste0("d1_kmeans", i, "_18_19"))$cluster)$DB
  )),
  db_index_med = unlist(lapply(
    3:4,
    function(i)
      index.DB(
        dados1_18_19,
        get(paste0("d1_kmeans", i, "_18_19"))$cluster,
        d = dados1_18_19_dist,
        centrotypes = "medoids"
```

```
      )$DB
  )),
  dunn_index = unlist(lapply(
    3:4,
    function(i)
      dunn(
        distance = dados1_18_19_dist,
        get(paste0("d1_kmeans", i, "_18_19"))$cluster
      )
  )),
  silh_index = unlist(lapply(
    3:4,
    function(i)
      index.S(dados1_18_19_dist, get(paste0("d1_kmeans", i, "_18_19"))$cluster)
  )),
  ch_index_cent = unlist(lapply(
    3:4,
    function(i)
      index.G1(dados1_18_19, get(paste0("d1_kmeans", i, "_18_19"))$cluster)
  )),
  ch_index_med = unlist(lapply(
    3:4,
    function(i)
      index.G1(
        dados1_18_19,
        get(paste0("d1_kmeans", i, "_18_19"))$cluster,
        d = dados1_18_19_dist,
        centrotypes = "medoids"
      )
  ))
)

d1_pam_index_18_19 <- data.frame(
  method = unlist(lapply(3:4, function(i) paste0("pam", i))),
  db_index_cent = unlist(lapply(
    3:4,
    function(i)
      index.DB(dados1_18_19, get(paste0("d1_pam", i, "_18_19"))$cluster)$DB
  )),
  db_index_med = unlist(lapply(
    3:4,
    function(i)
      index.DB(
        dados1_18_19,
        get(paste0("d1_pam", i, "_18_19"))$cluster,
        d = dados1_18_19_dist,
        centrotypes = "medoids"
      )$DB
  )),
  dunn_index = unlist(lapply(
    3:4,
    function(i)
      dunn(
        distance = dados1_18_19_dist,
        get(paste0("d1_pam", i, "_18_19"))$cluster
      )
  )),
  silh_index = unlist(lapply(
```

```
    3:4,
    function(i)
      index.S(dados1_18_19_dist, get(paste0("d1_pam", i, "_18_19"))$cluster)
  )),
  ch_index_cent = unlist(lapply(
    3:4,
    function(i)
      index.G1(dados1_18_19, get(paste0("d1_pam", i, "_18_19"))$cluster)
  )),
  ch_index_med = unlist(lapply(
    3:4,
    function(i)
      index.G1(
        dados1_18_19,
        get(paste0("d1_pam", i, "_18_19"))$cluster,
        d = dados1_18_19_dist,
        centrotypes = "medoids"
      )
  ))
)

d1_ward_index_18_19 <- data.frame(
  method = unlist(lapply(2:3, function(i) paste0("ward", i))),
  db_index_cent = unlist(lapply(
    2:3,
    function(i)
      index.DB(dados1_18_19, get(paste0("d1_ward", i, "_18_19", "_class")))$DB
  )),
  db_index_med = unlist(lapply(
    2:3,
    function(i)
      index.DB(
        dados1_18_19,
        get(paste0("d1_ward", i, "_18_19", "_class")),
        d = dados1_18_19_dist,
        centrotypes = "medoids"
      )$DB
  )),
  dunn_index = unlist(lapply(
    2:3,
    function(i)
      dunn(
        distance = dados1_18_19_dist,
        get(paste0("d1_ward", i, "_18_19", "_class"))
      )
  )),
  silh_index = unlist(lapply(
    2:3,
    function(i)
      index.S(dados1_18_19_dist, get(paste0("d1_ward", i, "_18_19", "_class")))
  )),
  ch_index_cent = unlist(lapply(
    2:3,
    function(i)
      index.G1(dados1_18_19, get(paste0("d1_ward", i, "_18_19", "_class")))
  )),
  ch_index_med = unlist(lapply(
    2:3,
```

```
    function(i)
      index.G1(
        dados1_18_19,
        get(paste0("d1_ward", i, "_18_19", "_class")),
        d = dados1_18_19_dist,
        centrotypes = "medoids"
      )
  ))
)

# Printing the table
d1_avaliacao_18_19 <- rbind(
  d1_pam_index_18_19,
  d1_kmeans_index_18_19,
  d1_ward_index_18_19
)
d1_avaliacao_18_19 |>
  kable() |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE)
```

method	db_index_cent	db_index_med	dunn_index	silh_index	ch_index_cent	ch_index_med
pam3	0.8198007	1.010448	0.0021692	0.5658505	6140.890	5859.125
pam4	0.8037138	1.063107	0.0022371	0.5885946	6398.828	6004.266
kmeans3	0.8336077	1.002882	0.0023148	0.5800877	7973.284	7776.422
kmeans4	0.7828237	1.013612	0.0024876	0.5864388	9010.328	8048.691
ward2	0.8790065	1.096638	0.0021505	0.6092797	7193.089	6985.303
ward3	0.8480933	1.060615	0.0023981	0.5786352	7022.586	6973.459

Comments:

- db_index_cent: the lower the better (kmeans4, followed by pam4);
- db_index_med: the lower the better (kmeans3, followed by pam3 and kmeans4);
- dunn_index: the higher the better (kmeans4, followed by ward3 and kmeans3);
- silh_index: the higher the better (ward2, followed by pam4, kmeans4 and kmeans3);
- ch_index_cent: the higher the better (kmeans4, followed by kmeans3);
- ch_index_med: the higher the better (kmeans4, followed by kmeans3).

Choice: despite the better perfomance of k-means with $K = 4$, one of its clusters contained only a few municipalities (127).

```
d1_kmeans4_18_19$size
```

```
## [1]  998 2176  127 2269
```

```
d1_kmeans3_18_19$size
```

```
## [1] 2162 3012  396
```

Thus, the chosen method is the **k-means with $\mathbf{K = 3}$**, since it presented the best Davies-Bouldin (medoid) index, the second best Calinski-Harabasz indexes and good Dunn and Silhouette indexes.

```
# Adding the column with the clusters information to the original dataframe
df_indicadores_18_19$cluster_10_a_14 <- d1_kmeans3_18_19$cluster

# Ordering by mean fertility rate
ordem_taxa1_18_19 <- df_indicadores_18_19 |>
  group_by(cluster_10_a_14) |>
  summarise(tx_fecundidade_10_a_14 = mean(tx_fecundidade_10_a_14)) |>
  ungroup() |>
  arrange(tx_fecundidade_10_a_14) |>
  pull(cluster_10_a_14)

df_indicadores_18_19$cluster_10_a_14 <- factor(case_when(
  df_indicadores_18_19$cluster_10_a_14 == ordem_taxa1_18_19[1] ~
    "1: lowest fertility rates",
  df_indicadores_18_19$cluster_10_a_14 == ordem_taxa1_18_19[2] ~
    "2: intermediary fertility rates",
  df_indicadores_18_19$cluster_10_a_14 == ordem_taxa1_18_19[3] ~
    "3: highest fertility rates"
))
```

5.1.7.2 2020-2021

```
# Calculating the considered indices
d1_kmeans_index_20_21 <- data.frame(
  metodo = unlist(lapply(3:4, function(i) paste0("kmeans", i))),
  db_index_cent = unlist(lapply(
    3:4,
    function(i)
      index.DB(dados1_20_21, get(paste0("d1_kmeans", i, "_20_21"))$cluster)$DB
  )),
  db_index_med = unlist(lapply(
    3:4,
    function(i)
      index.DB(
        dados1_20_21,
        get(paste0("d1_kmeans", i, "_20_21"))$cluster,
        d = dados1_20_21_dist,
        centrotypes = "medoids"
      )$DB
  )),
  dunn_index = unlist(lapply(
    3:4,
    function(i)
      dunn(
        distance = dados1_20_21_dist,
        get(paste0("d1_kmeans", i, "_20_21"))$cluster
      )
  )),
  silh_index = unlist(lapply(
    3:4,
    function(i)
      index.S(dados1_20_21_dist, get(paste0("d1_kmeans", i, "_20_21"))$cluster)
  )),
  ch_index_cent = unlist(lapply(
    3:4,
    function(i)
      index.G1(dados1_20_21, get(paste0("d1_kmeans", i, "_20_21"))$cluster)
```

```r
  )),
  ch_index_med = unlist(lapply(
    3:4,
    function(i)
      index.G1(
        dados1_20_21,
        get(paste0("d1_kmeans", i, "_20_21"))$cluster,
        d = dados1_20_21_dist,
        centrotypes = "medoids"
      )
  ))
)

d1_pam_index_20_21 <- data.frame(
  metodo = unlist(lapply(3:4, function(i) paste0("pam", i))),
  db_index_cent = unlist(lapply(
    3:4,
    function(i)
      index.DB(dados1_20_21, get(paste0("d1_pam", i, "_20_21"))$cluster)$DB
  )),
  db_index_med = unlist(lapply(
    3:4,
    function(i)
      index.DB(
        dados1_20_21,
        get(paste0("d1_pam", i, "_20_21"))$cluster,
        d = dados1_20_21_dist,
        centrotypes = "medoids"
      )$DB
  )),
  dunn_index = unlist(lapply(
    3:4,
    function(i)
      dunn(
        distance = dados1_20_21_dist,
        get(paste0("d1_pam", i, "_20_21"))$cluster
      )
  )),
  silh_index = unlist(lapply(
    3:4,
    function(i)
      index.S(dados1_20_21_dist, get(paste0("d1_pam", i, "_20_21"))$cluster)
  )),
  ch_index_cent = unlist(lapply(
    3:4,
    function(i)
      index.G1(dados1_20_21, get(paste0("d1_pam", i, "_20_21"))$cluster)
  )),
  ch_index_med = unlist(lapply(
    3:4,
    function(i)
      index.G1(
        dados1_20_21,
        get(paste0("d1_pam", i, "_20_21"))$cluster,
        d = dados1_20_21_dist,
        centrotypes = "medoids"
      )
  ))
```

```
)

d1_ward_index_20_21 <- data.frame(
  metodo = unlist(lapply(c(2, 4), function(i) paste0("ward", i))),
  db_index_cent = unlist(lapply(
    c(2, 4),
    function(i)
      index.DB(dados1_20_21, get(paste0("d1_ward", i, "_20_21", "_class")))$DB
  )),
  db_index_med = unlist(lapply(
    c(2, 4),
    function(i)
      index.DB(
        dados1_20_21,
        get(paste0("d1_ward", i, "_20_21", "_class")),
        d = dados1_20_21_dist,
        centrotypes = "medoids"
      )$DB
  )),
  dunn_index = unlist(lapply(
    c(2, 4),
    function(i)
      dunn(
        distance = dados1_20_21_dist,
        get(paste0("d1_ward", i, "_20_21", "_class"))
      )
  )),
  silh_index = unlist(lapply(
    c(2, 4),
    function(i)
      index.S(dados1_20_21_dist, get(paste0("d1_ward", i, "_20_21", "_class")))
  )),
  ch_index_cent = unlist(lapply(
    c(2, 4),
    function(i)
      index.G1(dados1_20_21, get(paste0("d1_ward", i, "_20_21", "_class")))
  )),
  ch_index_med = unlist(lapply(
    c(2, 4),
    function(i)
      index.G1(
        dados1_20_21,
        get(paste0("d1_ward", i, "_20_21", "_class")),
        d = dados1_20_21_dist,
        centrotypes = "medoids"
      )
  ))
)

# Printing the table
d1_avaliacao_20_21 <- rbind(
  d1_pam_index_20_21,
  d1_kmeans_index_20_21,
  d1_ward_index_20_21
)
d1_avaliacao_20_21 |> kable() |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE)
```

metodo	db_index_cent	db_index_med	dunn_index	silh_index	ch_index_cent	ch_index_med
pam3	0.8172548	1.0793661	0.0027701	0.5656204	6121.451	5684.212
pam4	0.7682952	1.0212080	0.0029326	0.5952804	7495.894	7063.963
kmeans3	0.7644580	0.9449188	0.0030769	0.5973804	9097.737	8762.034
kmeans4	0.6547798	0.7422809	0.0034722	0.5978453	11615.918	10492.895
ward2	0.8329748	1.0298080	0.0028169	0.6397901	7363.678	7155.641
ward4	0.6518735	0.7245958	0.0036496	0.5881173	11320.427	9823.375

Comments:

- db_index_cent: the lower the better (ward4, followed by kmeans4 and kmeans3);
- db_index_med: the lower the better (ward4, followed by kmeans4 and kmeans3);
- dunn_index: the higher the better (ward4, followed by kmeans4 and kmeans3);
- silh_index: the higher the better (ward2, followed by kmeans4 and kmeans3);
- ch_index_cent: the higher the better (kmeans4, followed by ward4 and kmeans3);
- ch_index_med: the higher the better (kmeans4, followed by ward4 and kmeans3).

Choice: again, despite the better perfomance of k-means with K = 4 and Ward's method with K = 4, one of their clusters contained only a few municipalities (90 and 70, respectively).

```
d1_kmeans4_20_21$size
```

```
## [1] 2109 2459   90  912
```

```
table(d1_ward4_20_21_class)
```

```
## d1_ward4_20_21_class
##    1    2    3    4
## 2627 1909  964   70
```

```
d1_kmeans3_18_19$size
```

```
## [1] 2162 3012  396
```

Thus, the chosen method is the **K-Means with K = 3**, since it had the best overall performance after these two.

```
# Adding the column with the clusters information to the original dataframe
df_indicadores_20_21$cluster_10_a_14 <- d1_kmeans3_20_21$cluster

# Ordering by mean fertility rate
ordem_taxa1_20_21 <- df_indicadores_20_21 |>
  group_by(cluster_10_a_14) |>
  summarise(tx_fecundidade_10_a_14 = mean(tx_fecundidade_10_a_14)) |>
  ungroup() |>
  arrange(tx_fecundidade_10_a_14) |>
  pull(cluster_10_a_14)

df_indicadores_20_21$cluster_10_a_14 <- factor(case_when(
  df_indicadores_20_21$cluster_10_a_14 == ordem_taxa1_20_21[1] ~
    "1: lowest fertility rates",
  df_indicadores_20_21$cluster_10_a_14 == ordem_taxa1_20_21[2] ~
    "2: intermediary fertility rates",
  df_indicadores_20_21$cluster_10_a_14 == ordem_taxa1_20_21[3] ~
    "3: highest fertility rates"
))
```

5.2 For the 15 to 19 age group

5.2.1 K-Means

5.2.1.1 2018-2019

```
# Ploting the k-means knee-plot for the 2018-2019 period
set.seed(1504)
fviz_nbclust(dados2_18_19, kmeans, method = "wss")
```

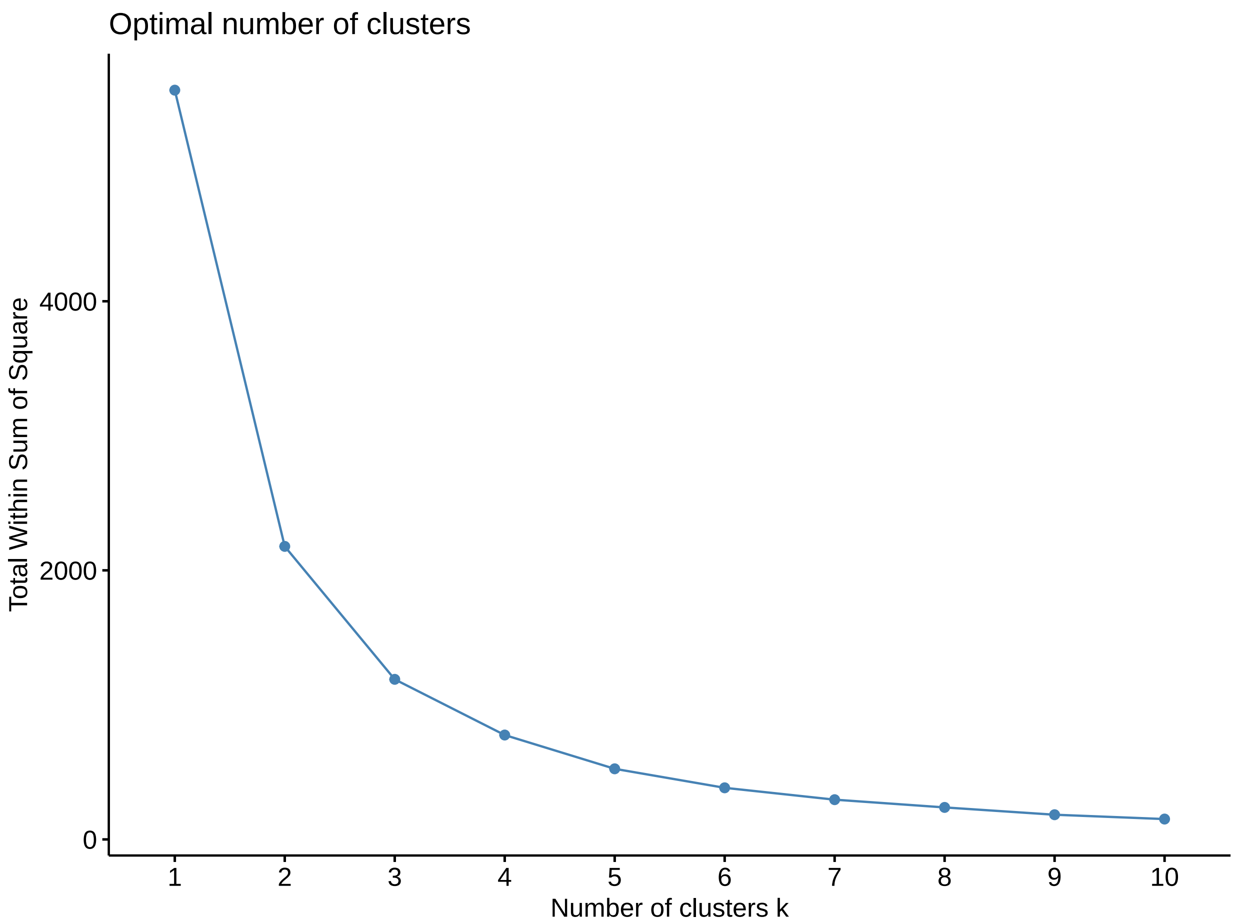

Figure 14: K-Means knee-plot for the 15 to 19 age group and 2018-2019 period

- **Candidates:** K = 3 and K = 4 (an argument that the total within-group variance does not decrease significantly after these values can be made for both numbers of groups).

```
# Adjusting the k-means method with the chosen numbers of clusters
set.seed(1504)
d2_kmeans3_18_19 <- kmeans(dados2_18_19, 3)

set.seed(1504)
d2_kmeans4_18_19 <- kmeans(dados2_18_19, 4)
```

5.2.1.2 2020-2021

```
# Ploting the k-means knee-plot for the 2020-2021 period
set.seed(1504)
fviz_nbclust(dados2_20_21, kmeans, method = "wss")
```

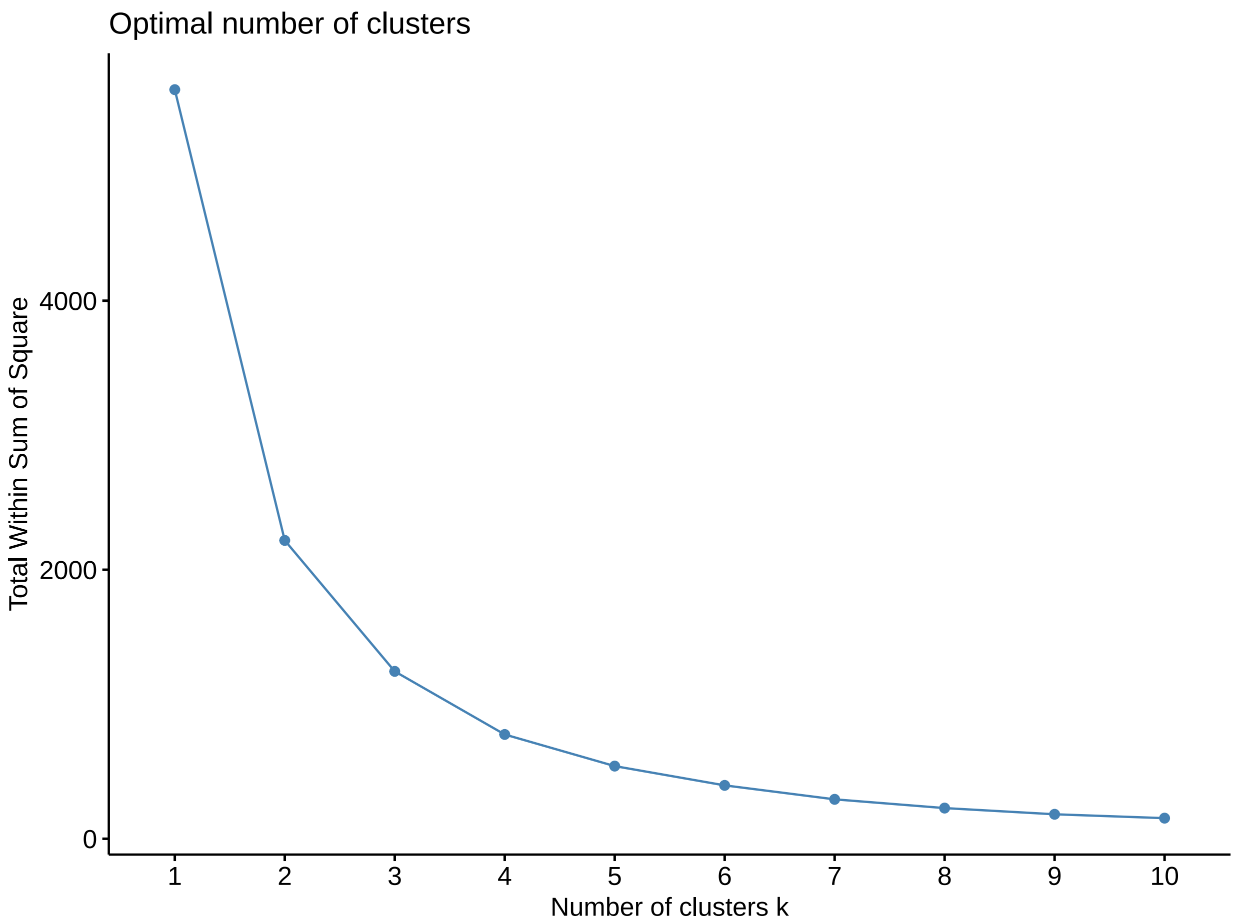

Figure 15: K-Means knee-plot for the 15 to 19 age group and 2020-2021 period

- **Candidates:** K = 3 and K = 4 (an argument that the total within-group variance does not decrease significantly after these values can be made for both numbers of groups).

```
# Adjusting the k-means method with the chosen numbers of clusters
set.seed(1504)
d2_kmeans3_20_21 <- kmeans(dados2_20_21, 3)

set.seed(1504)
d2_kmeans4_20_21 <- kmeans(dados2_20_21, 4)
```

5.2.2 K-Medians

5.2.2.1 2018-2019

```r
# Ploting the k-medoids knee-plot for the 2018-2019 period
set.seed(1504)
fviz_nbclust(dados2_18_19, cluster::pam, method = "wss")
```

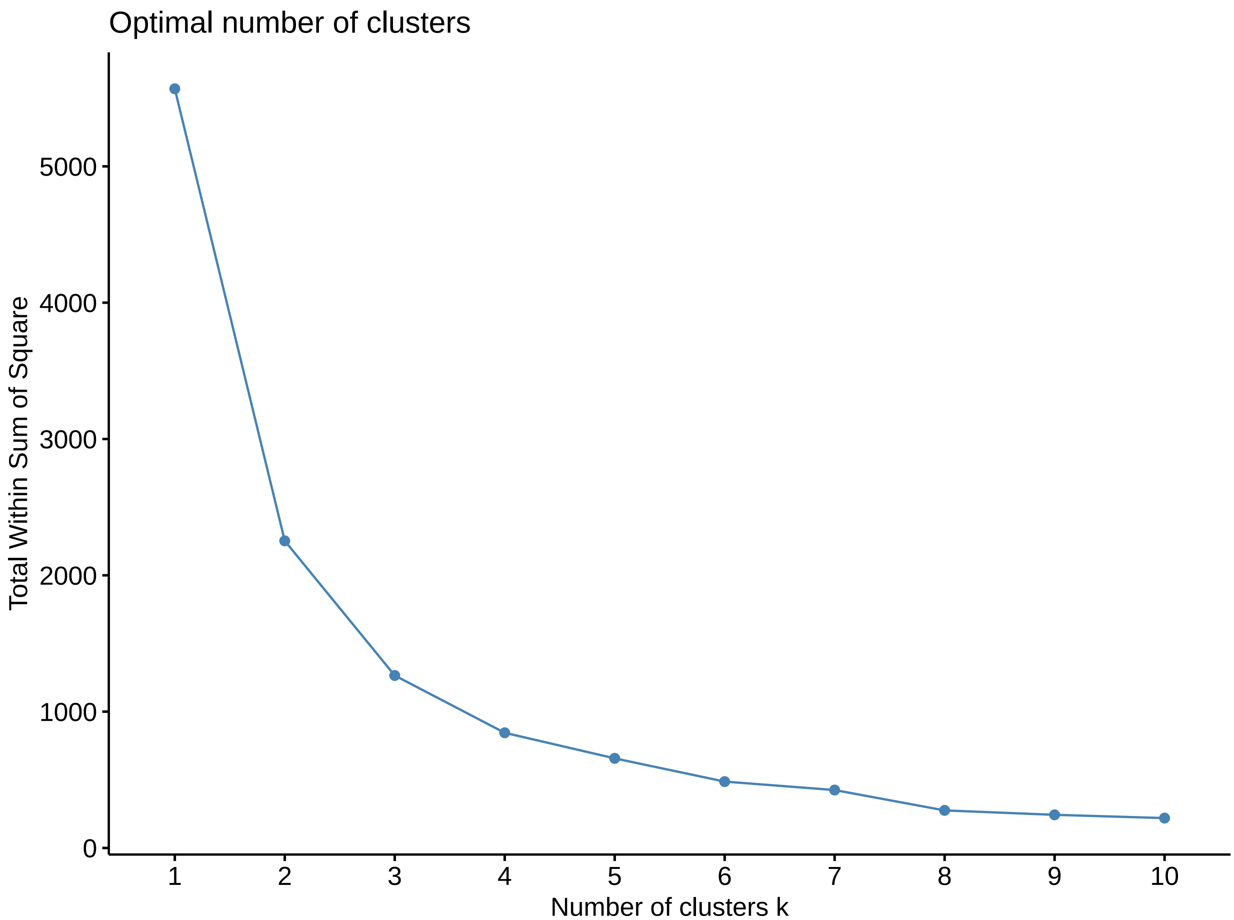

Figure 16: K-Medians knee-plot for the 15 to 19 age group and 2018-2019 period

- **Candidates:** K = 3 (an argument can be made that only after this value the total within-group variance does not decrease significantly).

```r
# Adjusting the k-medians method with the chosen numbers of clusters
set.seed(1504)
d2_pam3_18_19 <- cluster::pam(dados2_18_19, k = 3)
```

5.2.2.2 2020-2021

```r
# Ploting the k-medoids knee-plot for the 2020-2021 period
set.seed(1504)
fviz_nbclust(dados2_20_21, cluster::pam, method = "wss")
```

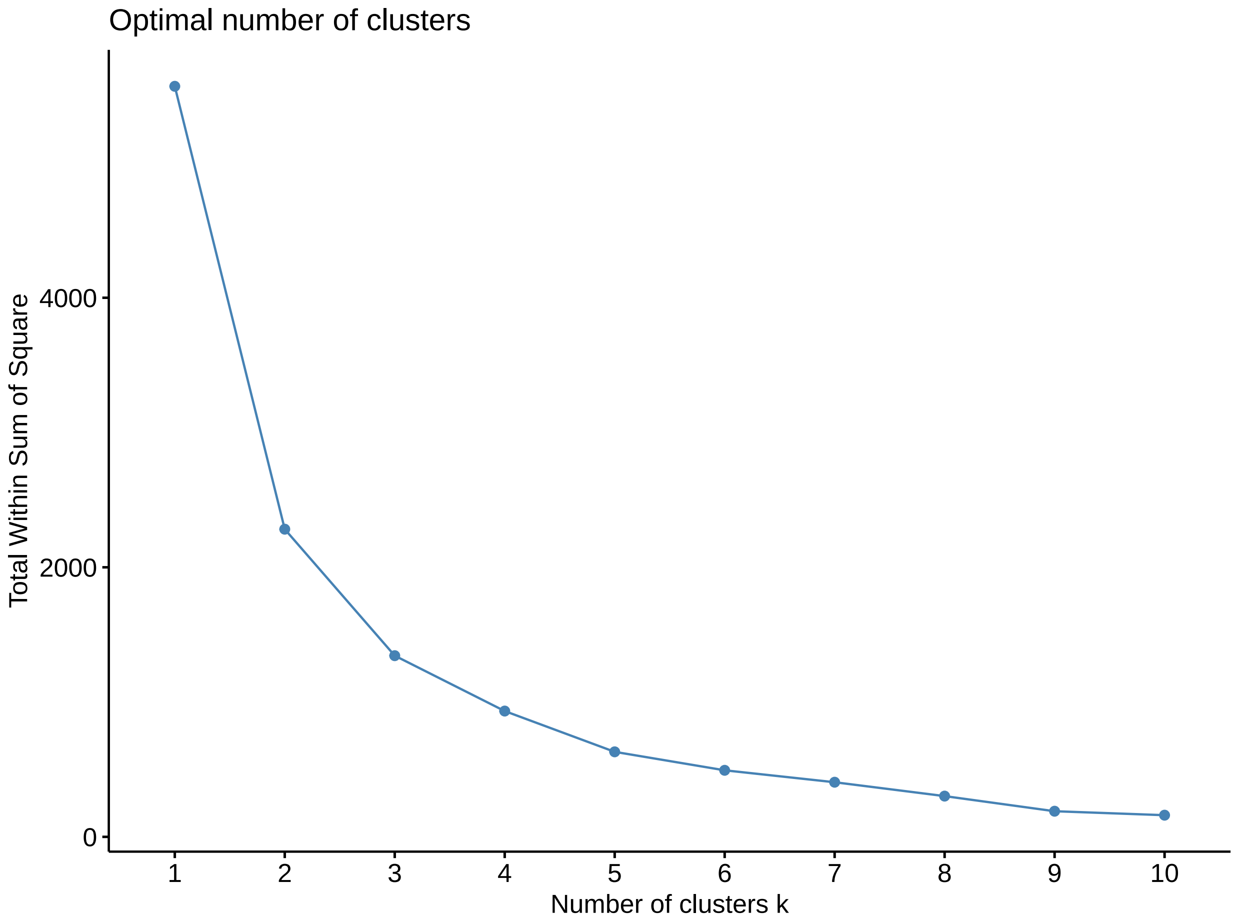

Figure 17: K-Medians knee-plot for the 15 to 19 age group and 2020-2021 period

- **Candidates:** K = 3 and K = 4 (an argument that the total within-group variance does not decrease significantly after these values can be made for both numbers of groups).

```r
# Adjusting the k-medians method with the chosen numbers of clusters
set.seed(1504)
d2_pam3_20_21 <- cluster::pam(dados2_20_21, k = 3)

set.seed(1504)
d2_pam4_20_21 <- cluster::pam(dados2_20_21, k = 4)
```

5.2.3 Ward's method

5.2.3.1 2018-2019

```r
# Ploting the drodrogram for Ward's method for the 2018-2019 period
set.seed(1504)
d2_ward_18_19 <- hclust(dados2_18_19_dist, method = "ward.D2")
plot(as.dendrogram(d2_ward_18_19), ylab = "Altura")
rect.hclust(d2_ward_18_19, k = 3, border = 2:5)
```

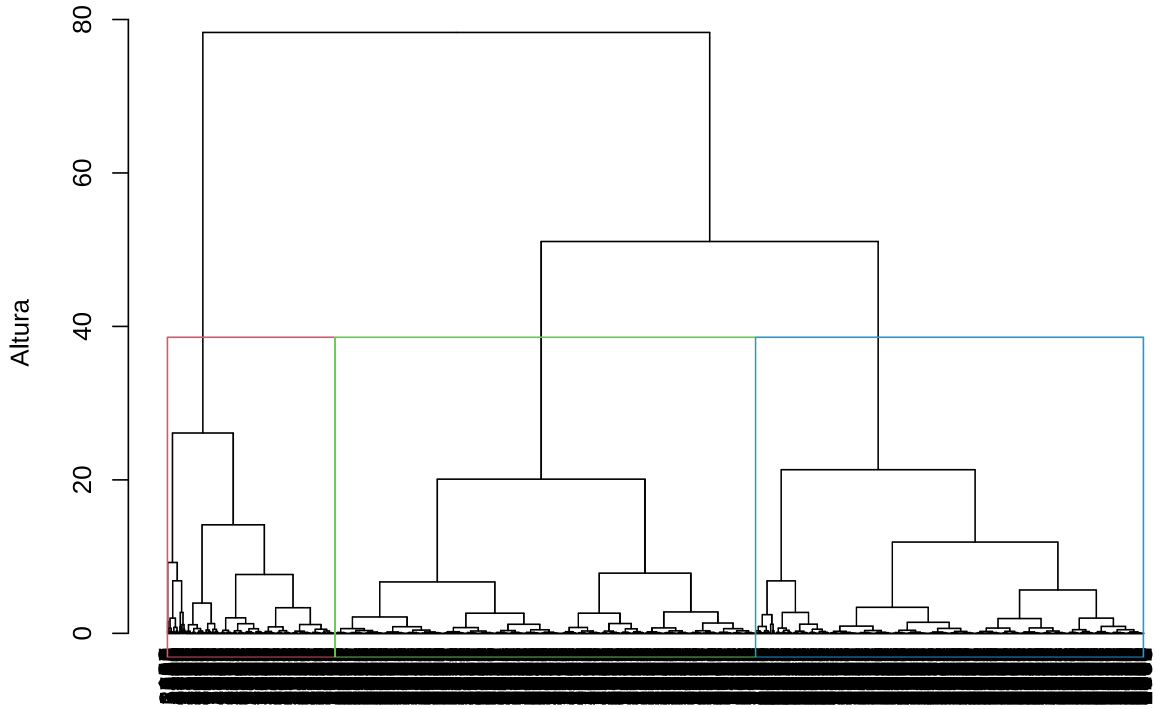

Figure 18: Dendrogram for Ward's method for the 15 to 19 age group and 2018-2019 period

- **Candidates:** K = 3.

```
# Adjusting Ward's method with the chosen numbers of clusters
d2_ward3_18_19_class <- cutree(d2_ward_18_19, k = 3)
```

5.2.3.2 2020-2021

```
# Ploting the drodrogram for Ward's method for the 2020-2021 period
set.seed(1504)
d2_ward_20_21 <- hclust(dados2_20_21_dist, method = "ward.D2")
plot(as.dendrogram(d2_ward_20_21), ylab = "Altura")
rect.hclust(d2_ward_20_21, k = 3, border = 2:5)
```

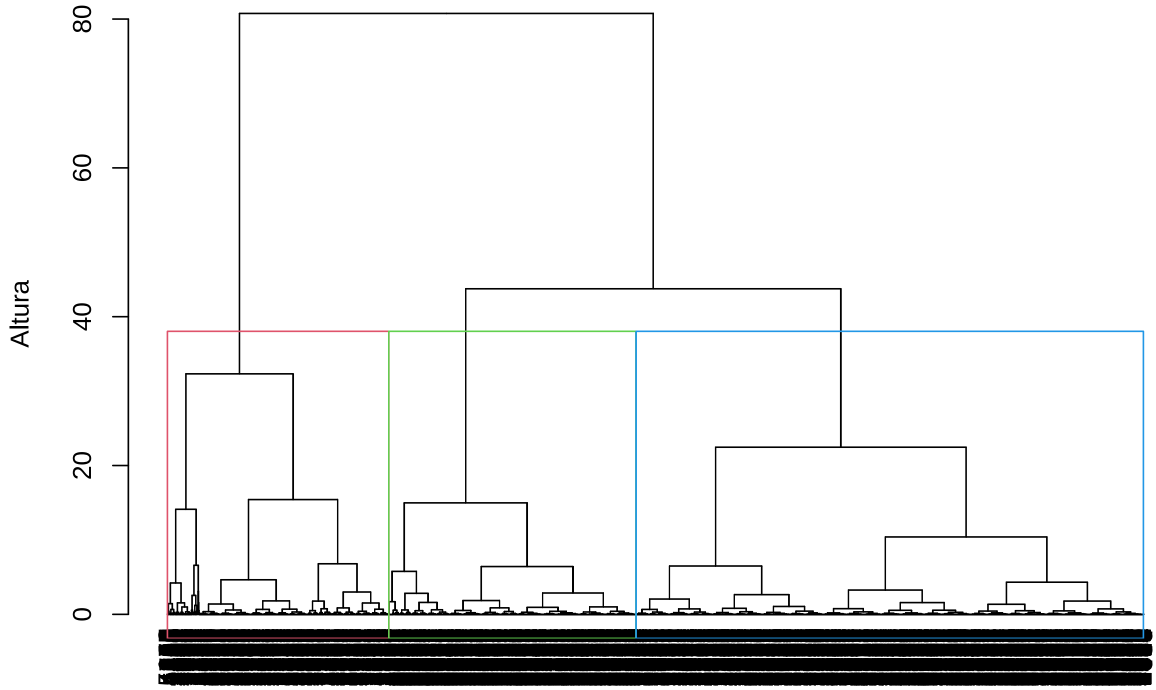

Figure 19: Dendrogram for Ward's method for the 15 to 19 age group and 2020-2021 period

- **Candidates:** K = 3.

```r
# Adjusting Ward's method with the chosen numbers of clusters
d2_ward3_20_21_class <- cutree(d2_ward_20_21, k = 3)
```

5.2.4 Single linkage

5.2.4.1 2018-2019

```r
# Ploting the drodrogram for the single linkage method for the 2018-2019 period
set.seed(1504)
d2_single_18_19 <- hclust(dados2_18_19_dist, method = "single")
plot(as.dendrogram(d2_single_18_19))
```

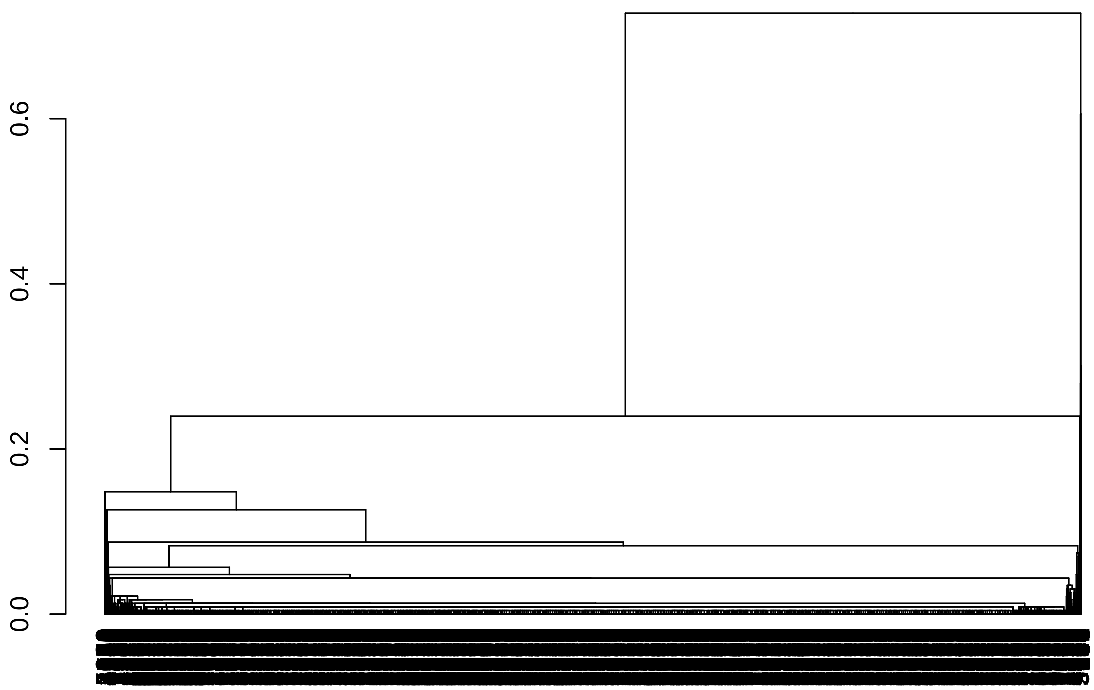

Figure 20: Dendrogram for the single linkage method for the 15 to 19 age group and 2018-2019 period

- **Candidates:** No candidates. A bad fit overall.

5.2.4.2 2020-2021

```r
# Ploting the drodrogram for the single linkage method for the 2020-2021 period
set.seed(1504)
d2_single_20_21 <- hclust(dados2_20_21_dist, method = "single")
plot(as.dendrogram(d2_single_20_21))
```

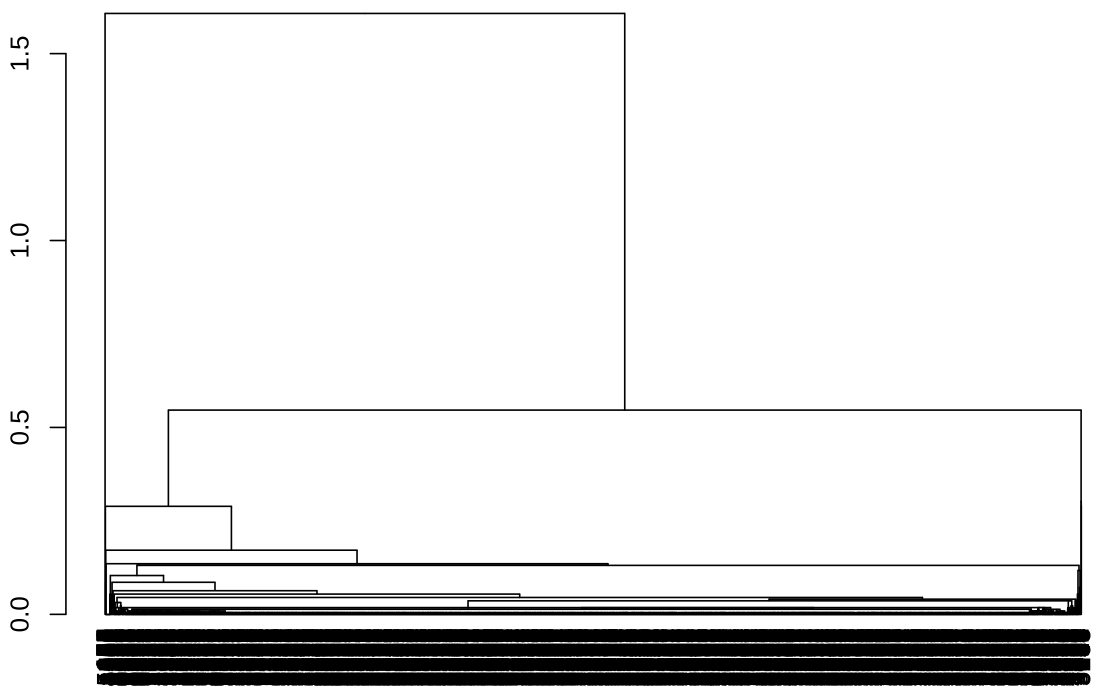

Figure 21: Dendrogram for the single linkage method for the 15 to 19 age group and 2020-2021 period

- **Candidates:** No candidates. A bad fit overall.

5.2.5 Complete linkage

5.2.5.1 2018-2019

```
# Ploting the drodrogram for the complete linkage method for the 2018-2019 period
set.seed(1504)
d2_complete_18_19 <- hclust(dados2_18_19_dist, method = "complete")
plot(as.dendrogram(d2_complete_18_19))
```

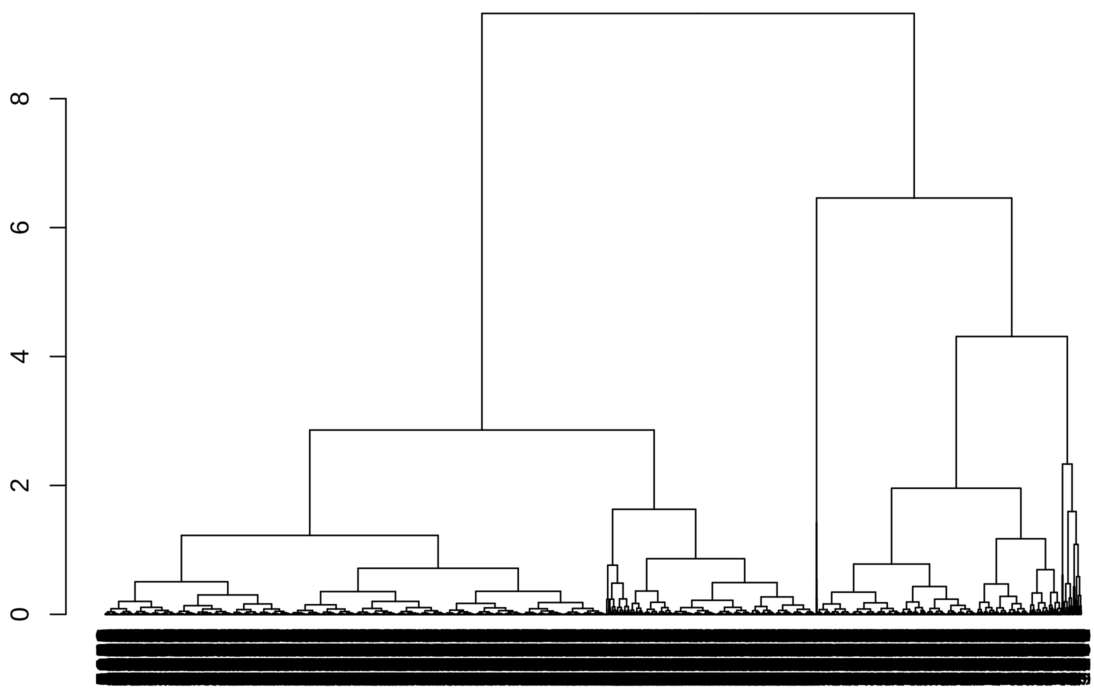

Figure 22: Dendrogram for the complete linkage method for the 15 to 19 age group and 2018-2019 period

- **Candidates:** No candidates. A bad fit overall.

5.2.5.2 2020-2021

```
# Ploting the drodrogram for the complete linkage method for the 2020-2021 period
set.seed(1504)
d2_complete_20_21 <- hclust(dados2_20_21_dist, method = "complete")
plot(as.dendrogram(d2_complete_20_21))
```

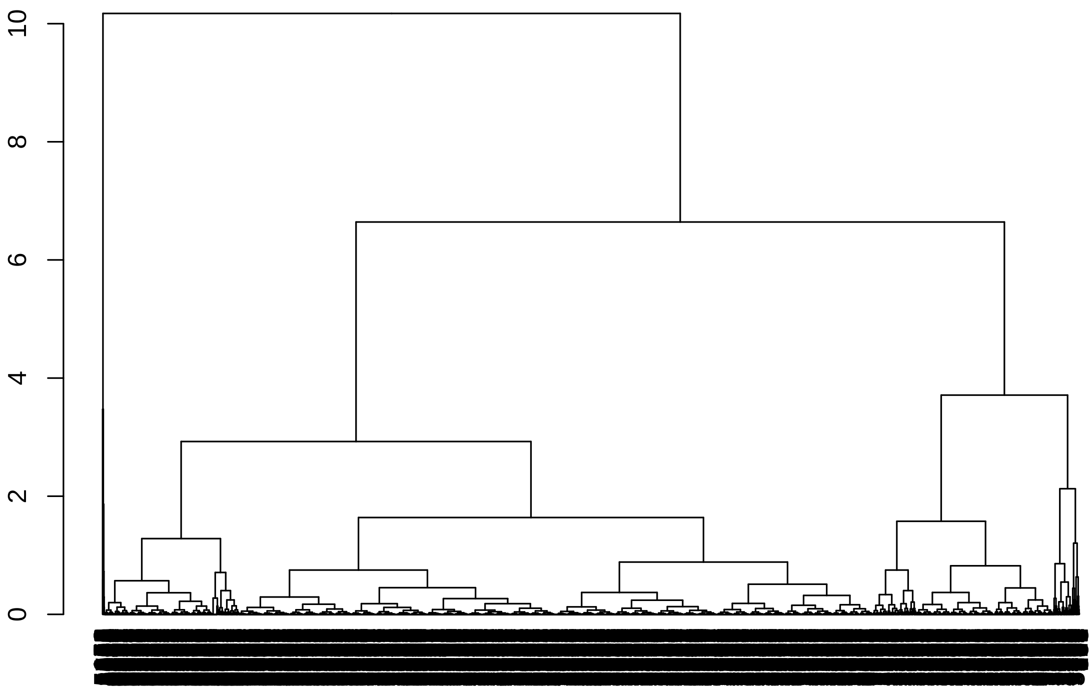

Figure 23: Dendrogram for the complete linkage method for the 15 to 19 age group and 2020-2021 period

- **Candidates:** No candidates. A bad fit overall.

5.2.6 Average linkage

5.2.6.1 2018-2019

```r
# Ploting the drodrogram for the average linkage method for the 2018-2019 period
set.seed(1504)
d2_average_18_19 <- hclust(dados2_18_19_dist, method = "average")
plot(as.dendrogram(d2_average_18_19))
```

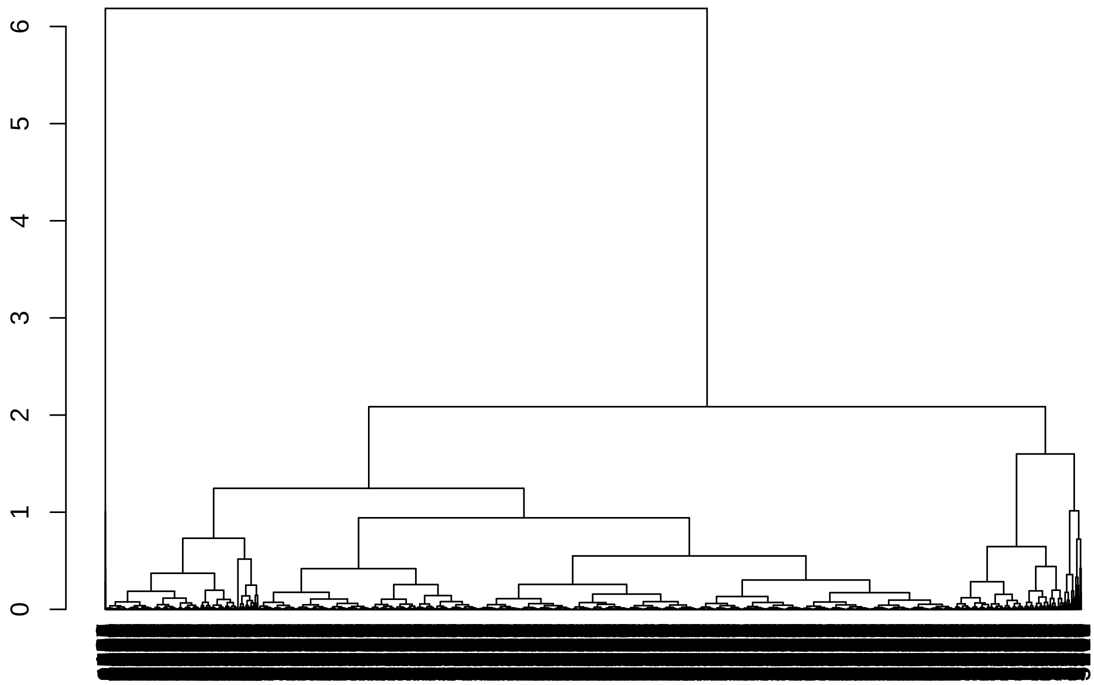

Figure 24: Dendrogram for the average linkage method for the 15 to 19 age group and 2018-2019 period

- **Candidates:** No candidates. A bad fit overall.

5.2.6.2 2020-2021

```
# Ploting the drodrogram for the average linkage method for the 2020-2021 period
set.seed(1504)
d2_average_20_21 <- hclust(dados2_20_21_dist, method = "average")
plot(as.dendrogram(d2_average_20_21))
```

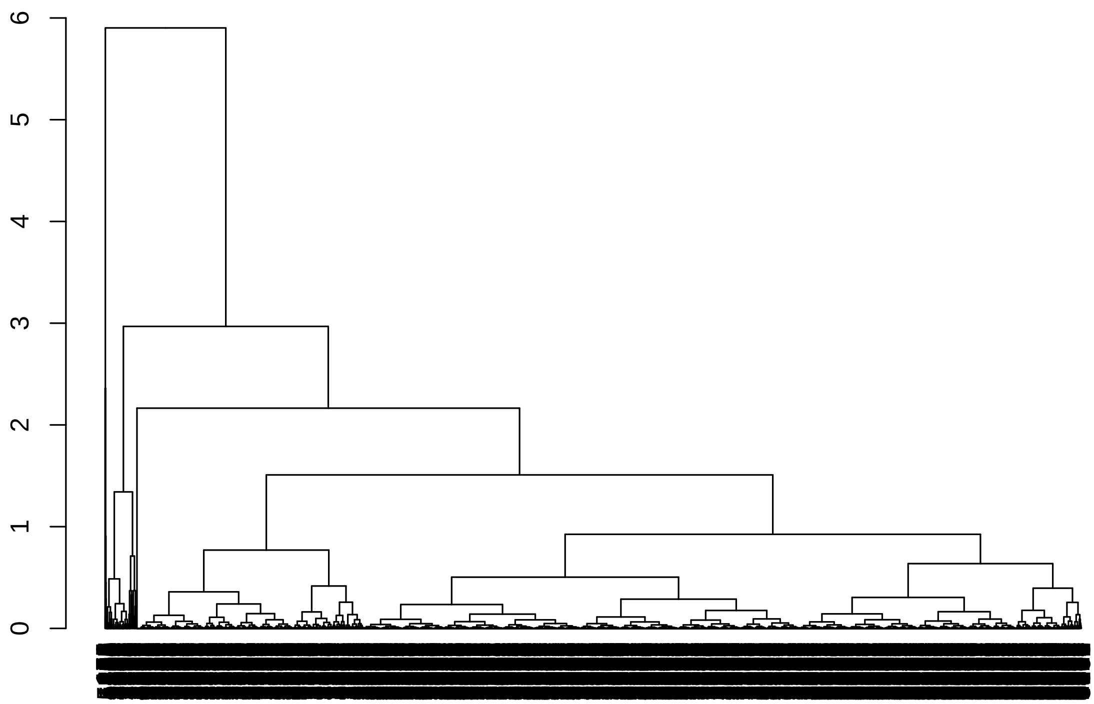

Figure 25: Dendrogram for the average linkage method for the 15 to 19 age group and 2020-2021 period

- **Candidates:** No candidates. A bad fit overall.

5.2.7 Choosing the best clustering method

5.2.7.1 2018-2019

```
# Calculating the considered indices
d2_kmeans_index_18_19 <- data.frame(
  metodo = unlist(lapply(3:4, function(i) paste0("kmeans", i))),
  db_index_cent = unlist(lapply(
    3:4,
    function(i)
      index.DB(dados2_18_19, get(paste0("d2_kmeans", i, "_18_19"))$cluster)$DB
  )),
  db_index_med = unlist(lapply(
    3:4,
    function(i)
      index.DB(
        dados2_18_19,
        get(paste0("d2_kmeans", i, "_18_19"))$cluster,
        d = dados2_18_19_dist,
        centrotypes = "medoids"
```

```
      )$DB
  )),
  dunn_index = unlist(lapply(
    3:4,
    function(i)
      dunn(
        distance = dados2_18_19_dist,
        get(paste0("d2_kmeans", i, "_18_19"))$cluster
      )
  )),
  silh_index = unlist(lapply(
    3:4,
    function(i)
      index.S(dados2_18_19_dist, get(paste0("d2_kmeans", i, "_18_19"))$cluster)
  )),
  ch_index_cent = unlist(lapply(
    3:4,
    function(i)
      index.G1(dados2_18_19, get(paste0("d2_kmeans", i, "_18_19"))$cluster)
  )),
  ch_index_med = unlist(lapply(
    3:4,
    function(i)
      index.G1(
        dados2_18_19,
        get(paste0("d2_kmeans", i, "_18_19"))$cluster,
        d = dados2_18_19_dist,
        centrotypes = "medoids"
      )
  ))
)

d2_pam_index_18_19 <- data.frame(
  metodo = unlist(lapply(3, function(i) paste0("pam", i))),
  db_index_cent = unlist(lapply(
    3,
    function(i)
      index.DB(dados2_18_19, get(paste0("d2_pam", i, "_18_19"))$cluster)$DB
  )),
  db_index_med = unlist(lapply(
    3,
    function(i)
      index.DB(
        dados2_18_19,
        get(paste0("d2_pam", i, "_18_19"))$cluster,
        d = dados2_18_19_dist,
        centrotypes = "medoids"
      )$DB
  )),
  dunn_index = unlist(lapply(
    3,
    function(i)
      dunn(
        distance = dados2_18_19_dist,
        get(paste0("d2_pam", i, "_18_19"))$cluster
      )
  )),
  silh_index = unlist(lapply(
```

```
    3,
    function(i)
      index.S(dados2_18_19_dist, get(paste0("d2_pam", i, "_18_19"))$cluster)
  )),
  ch_index_cent = unlist(lapply(
    3,
    function(i)
      index.G1(dados2_18_19, get(paste0("d2_pam", i, "_18_19"))$cluster)
  )),
  ch_index_med = unlist(lapply(
    3,
    function(i)
      index.G1(
        dados2_18_19,
        get(paste0("d2_pam", i, "_18_19"))$cluster,
        d = dados2_18_19_dist,
        centrotypes = "medoids"
      )
  ))
)

d2_ward_index_18_19 <- data.frame(
  metodo = unlist(lapply(3, function(i) paste0("ward", i))),
  db_index_cent = unlist(lapply(
    3,
    function(i)
      index.DB(dados2_18_19, get(paste0("d2_ward", i, "_18_19", "_class")))$DB
  )),
  db_index_med = unlist(lapply(
    3,
    function(i)
      index.DB(
        dados2_18_19,
        get(paste0("d2_ward", i, "_18_19", "_class")),
        d = dados2_18_19_dist,
        centrotypes = "medoids"
      )$DB
  )),
  dunn_index = unlist(lapply(
    3,
    function(i)
      dunn(
        distance = dados2_18_19_dist,
        get(paste0("d2_ward", i, "_18_19", "_class"))
      )
  )),
  silh_index = unlist(lapply(
    3,
    function(i)
      index.S(dados2_18_19_dist, get(paste0("d2_ward", i, "_18_19", "_class")))
  )),
  ch_index_cent = unlist(lapply(
    3,
    function(i)
      index.G1(dados2_18_19, get(paste0("d2_ward", i, "_18_19", "_class")))
  )),
  ch_index_med = unlist(lapply(
    3,
```

```
    function(i)
      index.G1(
        dados2_18_19,
        get(paste0("d2_ward", i, "_18_19", "_class")),
        d = dados2_18_19_dist,
        centrotypes = "medoids"
      )
  ))
)

# Printing the table
d2_avaliacao_18_19 <- rbind(
  d2_pam_index_18_19,
  d2_kmeans_index_18_19,
  d2_ward_index_18_19
)

d2_avaliacao_18_19 |>
  kable() |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE)
```

metodo	**db_index_cent**	**db_index_med**	**dunn_index**	**silh_index**	**ch_index_cent**	**ch_index_med**
pam3	0.7557952	0.8824612	0.0006920	0.5152591	9469.798	9049.865
kmeans3	0.7248163	0.8444368	0.0007418	0.5366937	10245.147	9822.237
kmeans4	0.7385346	0.8752130	0.0008006	0.5165412	11461.608	10872.393
ward3	0.7382885	0.8592175	0.0007210	0.5320165	10148.539	9710.915

Comments:

- db_index_cent: the lower the better (kmeans3, followed by ward3 and kmeans4)
- db_index_med: the lower the better (kmeans3, followed by ward3)
- dunn_index: the higher the better (kmeans3, followed by ward3)
- silh_index: the higher the better (kmeans3, followed by kmeans4)
- ch_index_cent: the higher the better (kmeans4, followed by kmeans3)
- ch_index_med: the higher the better (kmeans4, followed by kmeans3)

Choice: K-Means with K = 3, since it performed better in most of the metrics.

```
# Adding the column with the clusters information to the original dataframe
df_indicadores_18_19$cluster_15_a_19 <- d2_kmeans3_18_19$cluster

# Ordering by mean fertility rate
ordem_taxa2_18_19 <- df_indicadores_18_19 |>
  group_by(cluster_15_a_19) |>
  summarise(tx_fecundidade_15_a_19 = mean(tx_fecundidade_15_a_19)) |>
  ungroup() |>
  arrange(tx_fecundidade_15_a_19) |>
  pull(cluster_15_a_19)

df_indicadores_18_19$cluster_15_a_19 <- factor(case_when(
  df_indicadores_18_19$cluster_15_a_19 == ordem_taxa2_18_19[1] ~
    "1: lowest fertility rates",
  df_indicadores_18_19$cluster_15_a_19 == ordem_taxa2_18_19[2] ~
    "2: intermediary fertility rates",
```

```
  df_indicadores_18_19$cluster_15_a_19 == ordem_taxa2_18_19[3] ~
    "3: highest fertility rates"
))
```

5.2.7.2 2020-2021

```
# Calculating the considered indices
d2_kmeans_index_20_21 <- data.frame(
  metodo = unlist(lapply(3:4, function(i) paste0("kmeans", i))),
  db_index_cent = unlist(lapply(
    3:4,
    function(i)
      index.DB(dados2_20_21, get(paste0("d2_kmeans", i, "_20_21"))$cluster)$DB
  )),
  db_index_med = unlist(lapply(
    3:4,
    function(i)
      index.DB(
        dados2_20_21,
        get(paste0("d2_kmeans", i, "_20_21"))$cluster,
        d = dados2_20_21_dist,
        centrotypes = "medoids"
      )$DB
  )),
  dunn_index = unlist(lapply(
    3:4,
    function(i)
      dunn(
        distance = dados2_20_21_dist,
        get(paste0("d2_kmeans", i, "_20_21"))$cluster
      )
  )),
  silh_index = unlist(lapply(
    3:4,
    function(i)
      index.S(dados2_20_21_dist, get(paste0("d2_kmeans", i, "_20_21"))$cluster)
  )),
  ch_index_cent = unlist(lapply(
    3:4,
    function(i)
      index.G1(dados2_20_21, get(paste0("d2_kmeans", i, "_20_21"))$cluster)
  )),
  ch_index_med = unlist(lapply(
    3:4,
    function(i)
      index.G1(
        dados2_20_21,
        get(paste0("d2_kmeans", i, "_20_21"))$cluster,
        d = dados2_20_21_dist,
        centrotypes = "medoids"
      )
  ))
)

d2_pam_index_20_21 <- data.frame(
  metodo = unlist(lapply(3:4, function(i) paste0("pam", i))),
  db_index_cent = unlist(lapply(
```

```
    3:4,
    function(i)
      index.DB(dados2_20_21, get(paste0("d2_pam", i, "_20_21"))$cluster)$DB
  )),
  db_index_med = unlist(lapply(
    3:4,
    function(i)
      index.DB(
        dados2_20_21,
        get(paste0("d2_pam", i, "_20_21"))$cluster,
        d = dados2_20_21_dist,
        centrotypes = "medoids"
      )$DB
  )),
  dunn_index = unlist(lapply(
    3:4,
    function(i)
      dunn(
        distance = dados2_20_21_dist,
        get(paste0("d2_pam", i, "_20_21"))$cluster
      )
  )),
  silh_index = unlist(lapply(
    3:4,
    function(i)
      index.S(dados2_20_21_dist, get(paste0("d2_pam", i, "_20_21"))$cluster)
  )),
  ch_index_cent = unlist(lapply(
    3:4,
    function(i)
      index.G1(dados2_20_21, get(paste0("d2_pam", i, "_20_21"))$cluster)
  )),
  ch_index_med = unlist(lapply(
    3:4,
    function(i)
      index.G1(
        dados2_20_21,
        get(paste0("d2_pam", i, "_20_21"))$cluster,
        d = dados2_20_21_dist,
        centrotypes = "medoids"
      )
  ))
)

d2_ward_index_20_21 <- data.frame(
  metodo = unlist(lapply(3, function(i) paste0("ward", i))),
  db_index_cent = unlist(lapply(
    3,
    function(i)
      index.DB(dados2_20_21, get(paste0("d2_ward", i, "_20_21", "_class")))$DB
  )),
  db_index_med = unlist(lapply(
    3,
    function(i)
      index.DB(
        dados2_20_21,
        get(paste0("d2_ward", i, "_20_21", "_class")),
        d = dados2_20_21_dist,
```

```r
        centrotypes = "medoids"
      )$DB
  )),
  dunn_index = unlist(lapply(
    3,
    function(i)
      dunn(
        distance = dados2_20_21_dist,
        get(paste0("d2_ward", i, "_20_21", "_class"))
      )
  )),
  silh_index = unlist(lapply(
    3,
    function(i)
      index.S(dados2_20_21_dist, get(paste0("d2_ward", i, "_20_21", "_class")))
  )),
  ch_index_cent = unlist(lapply(
    3,
    function(i)
      index.G1(dados2_20_21, get(paste0("d2_ward", i, "_20_21", "_class")))
  )),
  ch_index_med = unlist(lapply(
    3,
    function(i)
      index.G1(
        dados2_20_21,
        get(paste0("d2_ward", i, "_20_21", "_class")),
        d = dados2_20_21_dist,
        centrotypes = "medoids"
      )
  ))
)

# Printing the table
d2_avaliacao_20_21 <- rbind(
  d2_pam_index_20_21,
  d2_kmeans_index_20_21,
  d2_ward_index_20_21
)

d2_avaliacao_20_21 |>
  kable() |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE)
```

metodo	**db_index_cent**	**db_index_med**	**dunn_index**	**silh_index**	**ch_index_cent**	**ch_index_med**
pam3	0.7678656	0.9280219	0.0006131	0.5107424	8747.872	8189.097
pam4	0.7720505	0.9338004	0.0006452	0.4942900	9215.322	8629.786
kmeans3	0.7490219	0.8861012	0.0006506	0.5280106	9674.998	9194.743
kmeans4	0.7205759	0.8296064	0.0007305	0.5311639	11473.176	11175.003
ward3	0.7465943	0.8912756	0.0006188	0.5129649	8687.645	8171.134

Comments:

- db_index_cent: the lower the better (kmeans4, followed by ward3 and kmeans3);
- db_index_med: the lower the better (kmeans4, followed by kmeans3 and ward3);
- dunn_index: the higher the better (kmeans4, followed by kmeans3);

- silh_index: the higher the better (kmeans4, followed by kmeans3);
- ch_index_cent: the higher the better (kmeans4, followed by kmeans3);
- ch_index_med: the higher the better (kmeans4, followed by kmeans3).

Choice: even though, this time around, the k-means with K = 4 didn't have problems with too little municipalities in a cluster, we chose the **K-Means with K = 3** to make it "comparable" with the other chosen clustering methods.

```
d2_kmeans4_20_21$size
```

```
## [1] 1300 1647 2323  300
```

```
# Adding the column with the clusters information to the original dataframe
df_indicadores_20_21$cluster_15_a_19 <- d2_kmeans3_20_21$cluster

# Ordering by mean fertility rate
ordem_taxa2_20_21 <- df_indicadores_20_21 |>
  group_by(cluster_15_a_19) |>
  summarise(tx_fecundidade_15_a_19 = mean(tx_fecundidade_15_a_19)) |>
  ungroup() |>
  arrange(tx_fecundidade_15_a_19) |>
  pull(cluster_15_a_19)

df_indicadores_20_21$cluster_15_a_19 <- factor(case_when(
  df_indicadores_20_21$cluster_15_a_19 == ordem_taxa2_20_21[1] ~
    "1: lowest fertility rates",
  df_indicadores_20_21$cluster_15_a_19 == ordem_taxa2_20_21[2] ~
    "2: intermediary fertility rates",
  df_indicadores_20_21$cluster_15_a_19 == ordem_taxa2_20_21[3] ~
    "3: highest fertility rates"
))
```

5.3 Visualizing the clusters in a map

```
# Downloading the geometry data
df_muni_sf <- read_municipality(year = 2019, showProgress = FALSE) |>
  mutate(codmunres = as.numeric(substr(code_muni, 1, 6)))

df_ufs_sf <- read_state(year = 2019, showProgress = FALSE)

# Joining the geometry and the clustering data for each period
df_dados_mapa_18_19 <- left_join(df_indicadores_18_19, df_muni_sf) |>
  mutate(periodo = "2018-2019")

df_dados_mapa_20_21 <- left_join(df_indicadores_20_21, df_muni_sf) |>
  mutate(periodo = "2020-2021")

df_dados_mapa_completo <- full_join(df_dados_mapa_18_19, df_dados_mapa_20_21) |>
  st_as_sf()
```

For the 10 to 14 years age group, group 3 (municipalities with the highest fertility rates) is predominantly formed by municipalities in the North region of Brazil, while group 1 (municipalities with the lowest fertility rates) is mainly composed of municipalities in the Southeast and South regions, with a similar pattern for the two periods of analysis.

```r
# Plotting the map for the 10 to 14 age group
ggplot() +
  geom_sf(
    data = df_dados_mapa_completo,
    aes(fill = cluster_10_a_14),
    color = NA
  ) +
  facet_wrap(vars(periodo)) +
  scale_fill_viridis_d(
    name = "Groups",
    end = 0.8,
    alpha = 0.6,
    direction = -1
  ) +
  geom_sf(data = df_ufs_sf, fill = NA, linewidth = 0.08, color = "black") +
  theme_bw() +
  theme(legend.position = "bottom")
```

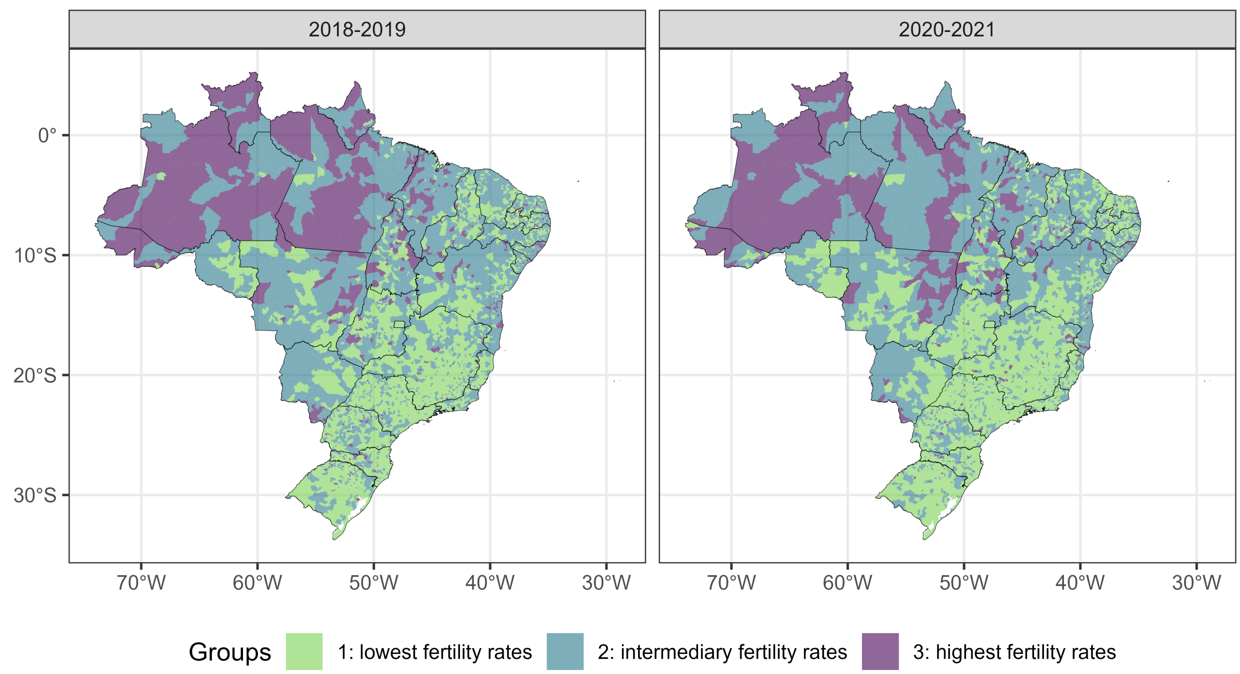

Figure 26: Distribution of groups of municipalities formed by municipalities with similar fertility rate of women aged 10 to 14 years in the pre-pandemic period (on the left) and in the pandemic period (on the right).

For the 15 to 19 years old age group, group 3 (municipalities with the highest fertility rates) is predominantly formed by municipalities located in the North of Brazil and in the Southeast of the state of Mato Grosso. Group 1 (municipalities with the lowest fertility rates) is predominantly composed of municipalities in the Southeast and South regions (Figure 3).

```r
# Plotting the map for the 15 to 19 age group
ggplot() +
  geom_sf(
    data = df_dados_mapa_completo,
    aes(fill = cluster_15_a_19),
```

```r
    color = NA
  ) +
  facet_wrap(vars(periodo)) +
  scale_fill_viridis_d(
    name = "Groups",
    end = 0.8,
    alpha = 0.6,
    direction = -1
  ) +
  geom_sf(data = df_ufs_sf, fill = NA, linewidth = 0.08, color = "black") +
  theme_bw() +
  theme(legend.position = "bottom")
```

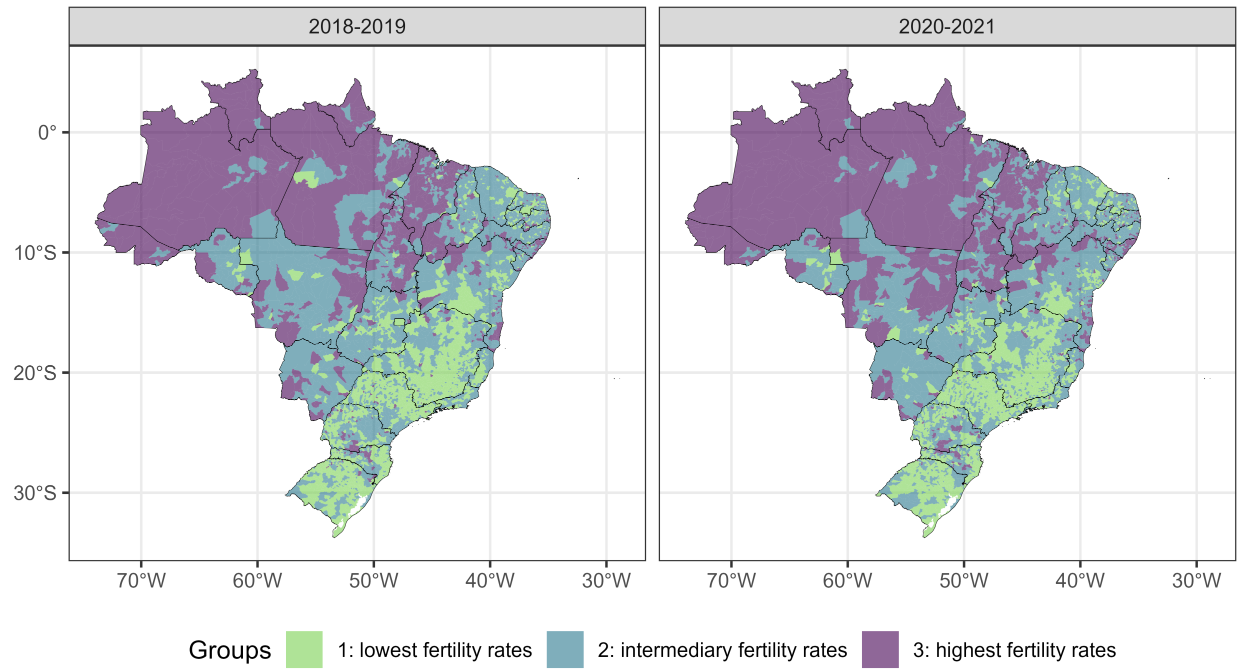

Figure 27: Distribution of groups of municipalities formed by municipalities with similar fertility rate of women aged 15 to 19 years in the pre-pandemic period (on the left) and in the pandemic period (on the right).

5.4 Summary tables

```r
# Creating a function that creates a table with the summary measures
cria_tabelas <- function(
  variaveis,
  df,
  var_grupos,
  label_tabela,
  faixa_etaria,
  periodo
) {
  grupos <- levels(df[[var_grupos]])
```

```r
for (variavel in variaveis) {
  tabela <- data.frame(
    grupo = grupos,
    n = unlist(lapply(
      grupos,
      function(grupo) df[df[[var_grupos]] == grupo, ] |> nrow()
    )),
    minimo = round(
      unlist(lapply(
        grupos,
        function(grupo)
          df[df[[var_grupos]] == grupo, ] |>
            pull(variavel) |>
            min(na.rm = TRUE)
      )),
      2
    ),
    primeiro_qt = round(
      unlist(lapply(
        grupos,
        function(grupo)
          df[df[[var_grupos]] == grupo, ] |>
            pull(variavel) |>
            quantile(0.25, na.rm = TRUE)
      )),
      2
    ),
    media = round(
      unlist(lapply(
        grupos,
        function(grupo)
          df[df[[var_grupos]] == grupo, ] |>
            pull(variavel) |>
            mean(na.rm = TRUE)
      )),
      2
    ),
    mediana = round(
      unlist(lapply(
        grupos,
        function(grupo)
          df[df[[var_grupos]] == grupo, ] |>
            pull(variavel) |>
            median(na.rm = TRUE)
      )),
      2
    ),
    dp = round(
      unlist(lapply(
        grupos,
        function(grupo)
          df[df[[var_grupos]] == grupo, ] |>
            pull(variavel) |>
            sd(na.rm = TRUE)
      )),
      2
    ),
    terceiro_qt = round(
```

```
      unlist(lapply(
        grupos,
        function(grupo)
          df[df[[var_grupos]] == grupo, ] |>
            pull(variavel) |>
            quantile(0.75, na.rm = TRUE)
      )),
      2
    ),
    maximo = round(
      unlist(lapply(
        grupos,
        function(grupo)
          df[df[[var_grupos]] == grupo, ] |>
            pull(variavel) |>
            max(na.rm = TRUE)
      )),
      2
    )
  )

  if (variavel == variaveis[1]) {
    print(
      kable(
        tabela,
        align = "cccccccc",
        col.names = c(
          "Grupo (cluster)",
          "n",
          "Mín.",
          "1º Quartil",
          "Média",
          "Mediana",
          "D.P.",
          "3º Quartil",
          "Máx."
        ),
        caption = glue::glue(
          "Summary measures of the variables of interest for the clusters of
          municipalities grouped by the fertility rate of women aged
          {faixa_etaria} (per thousand) in the period of {periodo}.
          \\\\ \\\\ \\textbf{{{variavel}}}"
        ),
        label = paste0("tabela", label_tabela)
      ) |>
        kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
        row_spec(1, bold = TRUE)
    )
  } else {
    cat(knitr::asis_output(paste0("**", variavel, "**\n\n")))
    print(
      kable(
        tabela,
        align = "cccccccc",
        col.names = c(
          "Grupo (cluster)",
          "n",
          "Mín.",
```

```
            "1º Quartil",
            "Média",
            "Mediana",
            "D.P.",
            "3º Quartil",
            "Máx."
          )
        ) |>
          kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
          row_spec(0, bold = TRUE)
      )
    }
  }
}
nomes_variaveis1 <- c(
  "idhm",
  "cobertura_ab",
  "porc_exclusivas_sus",
  "tx_fecundidade_10_a_14"
)
nomes_variaveis2 <- c(
  "idhm",
  "cobertura_ab",
  "porc_exclusivas_sus",
  "tx_fecundidade_15_a_19"
)
```

5.4.1 For the 10 to 14 age group

5.4.1.1 2018-2019

```
# Creating the summary tables for the 10 to 14 age group for the 2018-2019 period
cria_tabelas(
  df = df_indicadores_18_19,
  var_grupos = "cluster_10_a_14",
  label_tabela = "1",
  faixa_etaria = "aged 10 to 14",
  variaveis = nomes_variaveis1,
  periodo = "2018 to 2019"
)
```

Table 3: Summary measures of the variables of interest for the clusters of municipalities grouped by the fertility rate of women aged aged 10 to 14 (per thousand) in the period of 2018 to 2019.

idhm

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	3012	0.48	0.64	0.68	0.70	0.07	0.74	0.86
2: intermediary fertility rates	2162	0.42	0.58	0.63	0.63	0.06	0.69	0.77
3: highest fertility rates	396	0.44	0.56	0.61	0.60	0.07	0.66	0.76

cobertura_ab

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	3012	0	90.80	91.44	99.8	16.07	100	100
2: intermediary fertility rates	2162	0	93.90	93.38	100.0	13.30	100	100
3: highest fertility rates	396	25	93.83	94.08	100.0	12.06	100	100

porc_exclusivas_sus

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	3012	0	2.3	11.94	7.4	12.36	18.20	84.3
2: intermediary fertility rates	2162	0	1.0	6.16	2.6	8.62	7.80	98.5
3: highest fertility rates	396	0	0.5	2.94	1.2	4.24	3.42	24.8

tx_fecundidade_10_a_14

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	3012	0.0	0.0	0.94	0.9	0.95	1.80	2.6
2: intermediary fertility rates	2162	2.7	3.3	4.40	4.2	1.26	5.30	7.3
3: highest fertility rates	396	7.4	8.1	10.34	9.1	4.28	10.93	50.6

5.4.1.2 2020-2021

```r
# Creating the summary tables for the 10 to 14 age group for the 2020-2021 period
cria_tabelas(
  df = df_indicadores_20_21,
  var_grupos = "cluster_10_a_14",
  label_tabela = "2",
  faixa_etaria = "aged 10 to 14",
  variaveis = nomes_variaveis1,
  periodo = "2020 to 2021"
)
```

Table 4: Summary measures of the variables of interest for the clusters of municipalities grouped by the fertility rate of women aged aged 10 to 14 (per thousand) in the period of 2020 to 2021.

idhm

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	3274	0.45	0.63	0.68	0.70	0.07	0.73	0.86
2: intermediary fertility rates	1986	0.45	0.58	0.63	0.62	0.06	0.68	0.81
3: highest fertility rates	310	0.42	0.55	0.60	0.59	0.07	0.66	0.76

cobertura_ab

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	3274	21.6	89.9	91.77	100	14.69	100	100
2: intermediary fertility rates	1986	30.0	96.4	95.31	100	10.09	100	100
3: highest fertility rates	310	43.5	95.7	94.89	100	10.59	100	100

porc_exclusivas_sus

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	3274	0	2.5	12.42	8.1	12.28	18.88	76.2
2: intermediary fertility rates	1986	0	0.9	5.77	2.4	8.08	7.20	98.5
3: highest fertility rates	310	0	0.4	3.57	1.3	5.64	4.30	50.1

tx__fecundidade__10__a__14

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	3274	0.0	0.00	0.90	0.8	0.93	1.7	2.6
2: intermediary fertility rates	1986	2.7	3.30	4.44	4.2	1.32	5.4	7.7
3: highest fertility rates	310	7.8	8.53	11.14	9.6	4.24	12.0	40.3

5.4.2 For the 15 to 19 age group

5.4.2.1 2018-2019

```r
# Creating the summary tables for the 15 to 19 age group for the 2018-2019 period
cria_tabelas(
  df = df_indicadores_18_19,
  var_grupos = "cluster_15_a_19",
  label_tabela = "3",
  faixa_etaria = "aged 15 to 19",
  variaveis = nomes_variaveis2,
  periodo = "2018 to 2019"
)
```

Table 5: Summary measures of the variables of interest for the clusters of municipalities grouped by the fertility rate of women aged aged 15 to 19 (per thousand) in the period of 2018 to 2019.

idhm

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	2357	0.48	0.66	0.70	0.71	0.06	0.74	0.86
2: intermediary fertility rates	2466	0.45	0.59	0.64	0.64	0.06	0.70	0.78
3: highest fertility rates	747	0.42	0.56	0.60	0.59	0.06	0.64	0.77

cobertura__ab

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	2357	0.0	89.70	90.95	99.8	16.68	100	100
2: intermediary fertility rates	2466	0.0	93.80	93.47	100.0	13.15	100	100
3: highest fertility rates	747	19.8	93.85	93.33	100.0	13.35	100	100

porc__exclusivas__sus

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	2357	0	3.0	13.44	8.7	13.01	20.6	84.3
2: intermediary fertility rates	2466	0	1.2	6.79	3.2	8.60	9.3	98.5
3: highest fertility rates	747	0	0.4	2.68	1.1	4.66	2.9	61.6

tx__fecundidade__15__a__19

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	2357	0.0	30.1	35.44	37.2	9.49	42.90	48.3
2: intermediary fertility rates	2466	48.4	53.8	61.25	60.2	8.57	67.97	78.9
3: highest fertility rates	747	79.1	84.6	96.83	91.4	17.69	103.10	213.9

5.4.2.2 2020-2021

```r
# Creating the summary tables for the 15 to 19 age group for the 2020-2021 period
cria_tabelas(
  df = df_indicadores_20_21,
  var_grupos = "cluster_15_a_19",
  label_tabela = "4",
  faixa_etaria = "aged 15 to 19",
  variaveis = nomes_variaveis2,
  periodo = "2020 to 2021"
)
```

Table 6: Summary measures of the variables of interest for the clusters of municipalities grouped by the fertility rate of women aged aged 15 to 19 (per thousand) in the period of 2020 to 2021.

idhm

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	2244	0.49	0.66	0.70	0.71	0.06	0.75	0.86
2: intermediary fertility rates	2499	0.48	0.59	0.64	0.64	0.06	0.70	0.85
3: highest fertility rates	827	0.42	0.56	0.60	0.59	0.06	0.64	0.75

cobertura_ab

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	2244	21.6	87.27	90.62	99.55	15.69	100	100
2: intermediary fertility rates	2499	30.9	96.20	95.09	100.00	10.63	100	100
3: highest fertility rates	827	35.8	94.75	94.56	100.00	10.97	100	100

porc_exclusivas_sus

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	2244	0	3.9	14.75	10.75	13.15	22.83	74.5
2: intermediary fertility rates	2499	0	1.2	7.12	3.70	8.52	10.05	98.5
3: highest fertility rates	827	0	0.5	2.83	1.20	4.57	3.00	46.6

tx_fecundidade_15_a_19

Grupo (cluster)	n	Mín.	1º Quartil	Média	Mediana	D.P.	3º Quartil	Máx.
1: lowest fertility rates	2244	0.0	25.8	30.68	32.4	8.80	37.6	42.5
2: intermediary fertility rates	2499	42.6	47.7	54.50	53.4	8.01	60.6	71.5
3: highest fertility rates	827	71.6	76.1	88.61	82.9	18.29	94.4	225.3

5.5 Multiple comparisions tables

```r
# Creating a function that creates a table with the multiple comparisions
cria_tabelas_testes <- function(
  variaveis,
  df,
  var_grupos,
```

```
  label_tabela,
  faixa_etaria,
  periodo
) {
  for (variavel in variaveis) {
    formula <- as.formula(glue::glue("{variavel} ~ {var_grupos}"))
    resultados_dunn <- dunn_test(
      data = df,
      formula = formula,
      p.adjust.method = "bonferroni"
    )

    # Calcula a medida de efeito d de Cohen
    efeito_d <- cohens_d(data = df, formula = formula)

monta_linhas <- function(num_grupo) {
  conteudo <- paste(
    -1 *
      round(
        resultados_dunn$statistic[which(
          resultados_dunn$group1 == unique(resultados_dunn$group1)[num_grupo]
        )],
        3
      ),
    "(Z) \\\\",
    ifelse(
      round(
        resultados_dunn$p.adj[which(
          resultados_dunn$group1 == unique(resultados_dunn$group1)[num_grupo]
        )],
        3
      ) < 0.05,
      ifelse(
        round(
          resultados_dunn$p.adj[which(
            resultados_dunn$group1 == unique(resultados_dunn$group1)[num_grupo]
          )],
          3
        ) == 0,
        "< 0.001*",
        paste0(
          round(
            resultados_dunn$p.adj[which(
              resultados_dunn$group1 == unique(resultados_dunn$group1)[num_grupo]
            )],
            3
          ),
          "*"
        )
      ),
      round(
        resultados_dunn$p.adj[which(
          resultados_dunn$group1 == unique(resultados_dunn$group1)[num_grupo]
        )],
        3
      )
    ),
    "(valor-p) \\\\",
```

```
    round(
      efeito_d$effsize[
        efeito_d$group1 == unique(efeito_d$group1)[num_grupo]
      ],
      3
    ),
    "(D de Cohen)"
  )

  glue::glue("\\makecell{{{conteudo}}}")
}

    df_teste <- data.frame(
      grupo1 = monta_linhas(1),
      grupo2 = c(rep("", 1), monta_linhas(2)),
      row.names = unique(resultados_dunn$group2)
    )

    rownames(df_teste) <- paste0("\\textbf{", rownames(df_teste), "}")
    colnames(df_teste) <- unique(resultados_dunn$group1)

    if (variavel == variaveis[1]) {
      print(
        kbl(
          df_teste,
          align = "cc",
          caption = glue::glue(
            "Results of Dunn's tests (with Bonferroni correction) for multiple
            comparisons between pairs of groups of municipalities grouped by the
            fertility rate of women {faixa_etaria} (per thousand) in the period
            {periodo}. \\\\ \\\\ \\textbf{{{variavel}}}"
          ),
          label = paste0("tabela-comparacoes-", label_tabela),
          escape = FALSE
        ) |>
          kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
          row_spec(1, bold = TRUE)
      )
    } else {
      cat(knitr::asis_output(paste0("**", variavel, "**\n\n")))
      print(
        kbl(
          df_teste,
          align = "cc",
          escape = FALSE
        ) |>
          kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
          row_spec(0, bold = TRUE)
      )
    }
  }
}
nomes_variaveis1 <- c(
  "idhm",
  "cobertura_ab",
  "porc_exclusivas_sus",
  "tx_fecundidade_10_a_14"
)
```

```r
nomes_variaveis2 <- c(
  "idhm",
  "cobertura_ab",
  "porc_exclusivas_sus",
  "tx_fecundidade_15_a_19"
)
```

5.5.1 For the 10 to 14 age group

5.5.1.1 2018-2019

```r
# Creating the multiple comparisions tables for the 10 to 14 age group and 2018-2019 period
cria_tabelas_testes(
  df = df_indicadores_18_19,
  var_grupos = "cluster_10_a_14",
  label_tabela = "1",
  faixa_etaria = "aged 10 to 14",
  variaveis = nomes_variaveis1,
  periodo = "2018 to 2019"
)
```

Table 7: Results of Dunn's tests (with Bonferroni correction) for multiple comparisons between pairs of groups of municipalities grouped by the fertility rate of women aged 10 to 14 (per thousand) in the period 2018 to 2019.

idhm

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	24.864 (Z) < 0.001* (valor-p) 0.757 (D de Cohen)	
3: highest fertility rates	19.661 (Z) < 0.001* (valor-p) 1.171 (D de Cohen)	6.402 (Z) < 0.001* (valor-p) 0.431 (D de Cohen)

cobertura_ab

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	-3.749 (Z) 0.001* (valor-p) -0.132 (D de Cohen)	
3: highest fertility rates	-3.02 (Z) 0.008* (valor-p) -0.186 (D de Cohen)	-1.02 (Z) 0.923 (valor-p) -0.055 (D de Cohen)

porc_exclusivas_sus

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	20.206 (Z) < 0.001* (valor-p) 0.543 (D de Cohen)	
3: highest fertility rates	19.258 (Z) < 0.001* (valor-p) 0.974 (D de Cohen)	8.412 (Z) < 0.001* (valor-p) 0.474 (D de Cohen)

tx_fecundidade_10_a_14

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	-57.478 (Z) < 0.001* (valor-p) -3.099 (D de Cohen)	
3: highest fertility rates	-45.295 (Z) < 0.001* (valor-p) -3.033 (D de Cohen)	-14.654 (Z) < 0.001* (valor-p) -1.884 (D de Cohen)

5.5.1.2 2020-2021

```
# Creating the multiple comparisions tables for the 10 to 14 age group and 2020-2021 period
cria_tabelas_testes(
  df = df_indicadores_20_21,
  var_grupos = "cluster_10_a_14",
  label_tabela = "2",
  faixa_etaria = "aged 10 to 14",
  variaveis = nomes_variaveis1,
  periodo = "2020 to 2021"
)
```

Table 8: Results of Dunn's tests (with Bonferroni correction) for multiple comparisons between pairs of groups of municipalities grouped by the fertility rate of women aged 10 to 14 (per thousand) in the period 2020 to 2021.

idhm

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	27.357 (Z) < 0.001* (valor-p) 0.85 (D de Cohen)	
3: highest fertility rates	17.661 (Z) < 0.001* (valor-p) 1.159 (D de Cohen)	4.439 (Z) < 0.001* (valor-p) 0.361 (D de Cohen)

cobertura_ab

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	-7.638 (Z) < 0.001* (valor-p) -0.281 (D de Cohen)	
3: highest fertility rates	-3.145 (Z) 0.005* (valor-p) -0.243 (D de Cohen)	0.497 (Z) 1 (valor-p) 0.041 (D de Cohen)

porc__exclusivas__sus

	1: lowest fertility rates	**2: intermediary fertility rates**
2: intermediary fertility rates	23.519 (Z) < 0.001* (valor-p) 0.639 (D de Cohen)	
3: highest fertility rates	16.742 (Z) < 0.001* (valor-p) 0.925 (D de Cohen)	5.338 (Z) < 0.001* (valor-p) 0.316 (D de Cohen)

tx__fecundidade__10__a__14

	1: lowest fertility rates	**2: intermediary fertility rates**
2: intermediary fertility rates	-58.04 (Z) < 0.001* (valor-p) -3.103 (D de Cohen)	
3: highest fertility rates	-39.905 (Z) < 0.001* (valor-p) -3.333 (D de Cohen)	-11.799 (Z) < 0.001* (valor-p) -2.132 (D de Cohen)

5.5.1.3 For the 15 to 19 age group

```
# Creating the multiple comparisions tables for the 15 to 19 age group and 2018-2019 period
cria_tabelas_testes(
  df = df_indicadores_18_19,
  var_grupos = "cluster_15_a_19",
  label_tabela = "3",
  faixa_etaria = "aged 15 to 19",
  variaveis = nomes_variaveis2,
  periodo = "2018 to 2019"
)
```

Table 9: Results of Dunn's tests (with Bonferroni correction) for multiple comparisons between pairs of groups of municipalities grouped by the fertility rate of women aged 15 to 19 (per thousand) in the period 2018 to 2019.

idhm

	1: lowest fertility rates	**2: intermediary fertility rates**
2: intermediary fertility rates	25.777 (Z) < 0.001* (valor-p) 0.821 (D de Cohen)	
3: highest fertility rates	31.829 (Z) < 0.001* (valor-p) 1.595 (D de Cohen)	14.211 (Z) < 0.001* (valor-p) 0.742 (D de Cohen)

cobertura__ab

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	-3.922 (Z) < 0.001* (valor-p) -0.168 (D de Cohen)	
3: highest fertility rates	-3.63 (Z) 0.001* (valor-p) -0.158 (D de Cohen)	-0.944 (Z) 1 (valor-p) 0.011 (D de Cohen)

porc_exclusivas_sus

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	20.071 (Z) < 0.001* (valor-p) 0.603 (D de Cohen)	
3: highest fertility rates	28.976 (Z) < 0.001* (valor-p) 1.101 (D de Cohen)	15.287 (Z) < 0.001* (valor-p) 0.595 (D de Cohen)

tx_fecundidade_15_a_19

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	-52.06 (Z) < 0.001* (valor-p) -2.856 (D de Cohen)	
3: highest fertility rates	-59.509 (Z) < 0.001* (valor-p) -4.326 (D de Cohen)	-23.921 (Z) < 0.001* (valor-p) -2.56 (D de Cohen)

```r
# Creating the multiple comparisions tables for the 15 to 19 age group and 2020-2021 period
cria_tabelas_testes(
  df = df_indicadores_20_21,
  var_grupos = "cluster_15_a_19",
  label_tabela = "4",
  faixa_etaria = "aged 15 to 19",
  variaveis = nomes_variaveis2,
  periodo = "2020 to 2021"
)
```

Table 10: Results of Dunn's tests (with Bonferroni correction) for multiple comparisons between pairs of groups of municipalities grouped by the fertility rate of women aged 15 to 19 (per thousand) in the period 2020 to 2021.

idhm

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	27.995 (Z) < 0.001* (valor-p) 0.922 (D de Cohen)	
3: highest fertility rates	35.326 (Z) < 0.001* (valor-p) 1.764 (D de Cohen)	15.53 (Z) < 0.001* (valor-p) 0.791 (D de Cohen)

cobertura_ab

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	-9.895 (Z) < 0.001* (valor-p) -0.333 (D de Cohen)	
3: highest fertility rates	-5.593 (Z) < 0.001* (valor-p) -0.291 (D de Cohen)	1.502 (Z) 0.399 (valor-p) 0.049 (D de Cohen)

porc_exclusivas_sus

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	22.25 (Z) < 0.001* (valor-p) 0.689 (D de Cohen)	
3: highest fertility rates	31.802 (Z) < 0.001* (valor-p) 1.21 (D de Cohen)	16.117 (Z) < 0.001* (valor-p) 0.626 (D de Cohen)

tx_fecundidade_15_a_19

	1: lowest fertility rates	2: intermediary fertility rates
2: intermediary fertility rates	-50.709 (Z) < 0.001* (valor-p) -2.831 (D de Cohen)	
3: highest fertility rates	-61.675 (Z) < 0.001* (valor-p) -4.037 (D de Cohen)	-25.779 (Z) < 0.001* (valor-p) -2.416 (D de Cohen)

6 Phase 3: GAMLSS modeling

This stage of the article aimed to find and interpret a regression model that would best explain the fertility rate in women under 20 years of age in Brazilian municipalities, initially based on covariates such as the reference year, the HDI-M, the population coverage of Primary Care, the percentage of women aged 10 to 49 who are exclusive users of the SUS, and a variable indicating the Covid-19 pandemic. Although the interest was, most likely, in modeling the entire period from 2012 to 2021, we had to make the decision to restrict the analysis period to the years 2018 to 2021, since models that considered the entire period did not present appropriate adjustments. With this consideration in mind, a brief description of the variables that will be considered in this phase can be seen in **Table 11**.

Table 11: Description of the variables considered for the GAMLSS modeling.

Variable	Description
ano	Year
tx_fecundidade_menor_20	Fertility rate in women under 20 years of age
idhm	HDI-M
cobertura_ab	Population coverage of Primary Care
porc_exclusivas_sus	Percentage of women aged 10 to 49 who are exclusive users of the SUS
pandemia	Variable indicating the Covid-19 pandemic (receives 'yes' if the reference year is 2020 or 2021 and 'no' otherwise)

6.1 Preparing the data

```r
# Reading and manipulating the data
df_indicadores <- read.csv("databases/data_fertility_rates_article.csv") |>
  filter(ano >= 2018 & ano <= 2021) |>
  mutate(
    codmunres = as.character(codmunres),
    tx_fecundidade_menor_20 = round(
      nvm_10_a_19 / pop_feminina_10_a_19 * 1000,
      1
    ),
    cobertura_ab = round(media_cobertura_ab / populacao_total * 100, 1),
    porc_exclusivas_sus = round(
      (pop_feminina_10_a_49 - pop_fem_10_49_com_plano_saude) /
        pop_feminina_10_a_49 *
        100,
      1
    ),
    pandemia = factor(
      ifelse(ano < 2020, "before", "during"),
      levels = c("before", "during")
    ),
    ano = as.factor(ano)
  ) |>
  drop_na() # There are 5 municipalities without the HDI-M information
```

6.2 About regression

Initial attempts to find regression models always involve, of course, adjustments to linear regression models, within which we make assumptions that imply the normality of the response variable in question. The histogram of the fertility rate in women under 20 years old, presented in **Figure 28**, shows, however, a relatively common scenario in which the response variable of the problem **does not** appear to be normally distributed, which can be noted by the strong asymmetry to the right present in the graph. In this scenario, linear regression models have their assumptions violated and, therefore, their conclusions are no longer valid.

```r
hist(
  df_indicadores$tx_fecundidade_menor_20[which(df_indicadores$ano == "2019")],
  freq = FALSE,
  col = "lightblue",
  main = "",
  xlab = "Fertility rate in women under 20 years old",
  ylab = "Density"
)
```

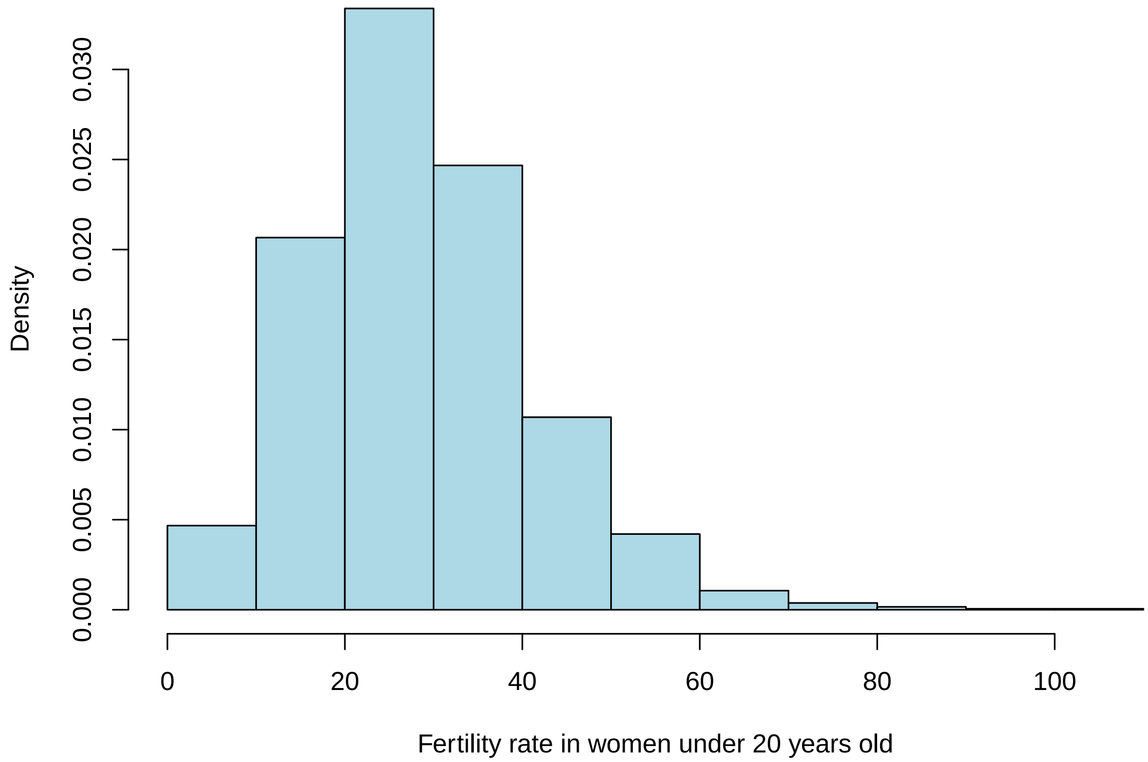

Figure 28: Histogram of the fertility rate in women under 20 years of age in 2019.

Thus, to overcome this problem, we expanded our search for regression models to GAMLSS models (R. A. Rigby and D. M. Stasinopoulos 2005), which allow us to choose from a range of different distributions for the response variable and which have as a major differential the fact that they allow us to separately model the parameters of location (related to the mean of the distribution), scale (related to the variance of the distribution) and shape (related to the asymmetry and kurtosis of the distribution). We adjusted and verified the adequacy of a series of different GAMLSS models, and decided to discuss in this work the models that: 1) assume that the response variable follows a skew t type 4 distribution, which allows us to model all four parameters discussed; and 2) assume that the response variable follows a zero-adjusted gamma distribution (ZAGA), which allows us to model the parameters of location, scale and the shape parameter related to the asymmetry of the distribution. It is important to note that, to avoid multicollinearity problems, we decided not to include in the analysis the covariate of the percentage of women aged 10 to 49 who are exclusive users of the SUS, since, as can be seen in the correlogram shown in **Figure 29**, the variables `porc_exclusivas_sus` and `idhm` have a considerable linear correlation.

```
corrplot(
  cor(
    df_indicadores |>
      filter(ano == "2019") |>
      dplyr::select_at(vars(
        starts_with("tx") | "idhm" | "cobertura_ab",
        "porc_exclusivas_sus"
      )),
```

```
    use = "complete.obs",
    method = "pearson"
  ),
  method = "color",
  type = "lower",
  addCoef.col = "black",
  tl.col = "black",
  tl.srt = 45,
  tl.cex = 0.85,
  diag = TRUE,
  cl.pos = "b",
  mar = c(0, 0, 1.5, 0)
)
```

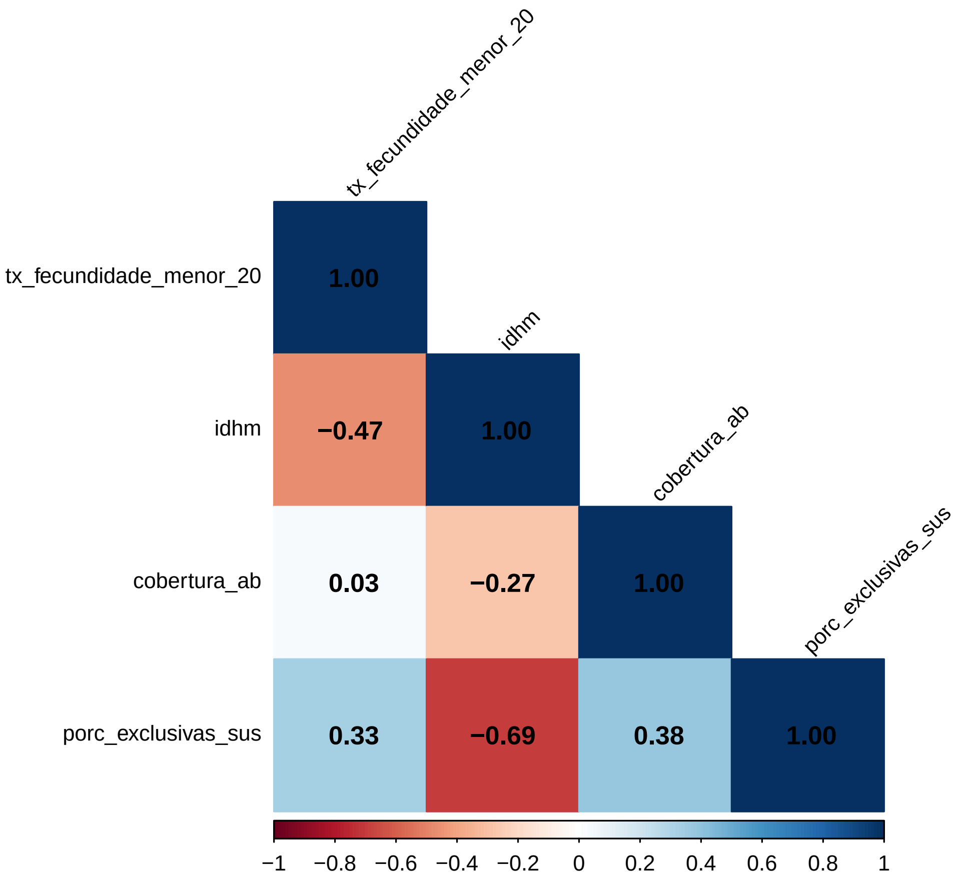

Figure 29: Pearson correlations between the variables in the dataset in 2019.

6.3 Fitting the models

Since GAMLSS models give us the freedom to model separately the location, scale and shape parameters of the distribution, a very large set of different models for a chosen distribution can be fitted. Therefore, after assuming a distribution for the response variable, we need to find the best model among all those that could be fitted considering this chosen distribution. To do this, we used two strategies described by Stasinopoulos et al. (2017, 397), which, in short, use stepwise selection methods and 1) allow different sets of covariates to be chosen to model each parameter of the distribution; or 2) force the same covariates to be selected to model all parameters of the distribution. Therefore, for each of the two selected distributions (skew t type 4 and ZAGA), we fitted:

- the full model, which has a random intercept for the reference year and which uses as covariates `coverage_ab`, `idhm`, `pandemic` and their interactions;
- the reduced model chosen by strategy 1;
- the reduced model chosen by strategy 2;
- and a third reduced model, which uses, for the μ model, only the significant terms of the full model.

To decide whether the reduced models were better than the full model, we used the generalized likelihood ratio test (GLRT) - which, in a simplified way, tests the null hypothesis that the reduced model is better than the full model (against the hypothesis that the full model is better). The **rejection** of the GLR null hypothesis, then, gives us evidence to believe that the reduced model is worse than the full model (which would therefore lead us to choose the full model). If the GLRT null hypothesis was not rejected for all the reduced models, we chose the reduced model that presented the most appropriate residual analysis or the lowest AIC value. Finally, after selecting the best model for each distribution, we chose, between the two selected models, the one that presented the most appropriate residual analysis or the lowest AIC value. All analyses were conducted using functions from the `{gamlss}` package (R. A. Rigby and D. M. Stasinopoulos 2005), available in the R software (R Core Team 2023).

It is important to note that we chose to incorporate a random effect for the reference year to address the issue that observations from the same year are possibly correlated with each other, which violates the basic assumption that the observations are all independent.

6.3.1 For the skew t type 4 distribution

```r
# Adjusting the full model
fit1_st4 <- gamlss(
  tx_fecundidade_menor_20 ~
    random(ano) +
      (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  data = df_indicadores,
  family = ST4(),
  sigma.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  nu.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  tau.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  control = gamlss.control(n.cyc = 500)
)
sumario_fit1_st4 <- summary(fit1_st4, save = TRUE)

# Utilizing the two strategies described in the book "Flexible Regression and
# Smoothing: Using GAMLSS in R", page 397, to get the best models for each parameter
## Adjusting a model with only the intercept for mu and considering sigma and nu
## as constants
fit2_st4 <- gamlss(
  tx_fecundidade_menor_20 ~ 1,
  data = df_indicadores,
  family = ST4,
  control = gamlss.control(n.cyc = 500)
)
```

```r
## Strategy 1:
### Starting from the simplest model and utilizing the forward stepwise method
### to find the model with the lowest AIC for each parameter
### Obs.: in this strategy, different variables can be selected for each
### parameter's model
fit3_st4 <- stepGAICAll.A(
  fit2_st4,
  scope = list(
    lower = ~1,
    upper = ~ random(ano) +
      (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm)
  )
)
sumario_fit3_st4 <- summary(fit3_st4, save = TRUE)

### Strategy 2:
### This strategy forces all os the models to select the same variables subsets
fit4_st4 <- stepGAICAll.B(
  fit2_st4,
  scope = list(
    lower = ~1,
    upper = ~ random(ano) +
      (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm)
  )
)
sumario_fit4_st4 <- summary(fit4_st4, save = TRUE)

## Adjusting a third model with only the significant terms for mu
fit5_st4 <- gamlss(
  tx_fecundidade_menor_20 ~
    random(ano) +
      pandemia +
      cobertura_ab +
      idhm +
      pandemia * cobertura_ab +
      cobertura_ab * idhm,
  data = df_indicadores,
  family = ST4(),
  sigma.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  nu.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  tau.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  control = gamlss.control(n.cyc = 500)
)
sumario_fit5_st4 <- summary(fit5_st4, save = TRUE)
```

Tables 12 to **15** show the estimates, standard errors, and significance tests for the models adjusted for μ in each of the four settings, as well as the additional random intercept terms for each year in each model. The applications of the GLRT and a comparison between the AIC of the three models can be seen below the tables.

- **For the complete model:**

Table 12: Measures related to the estimated coefficients for the location parameter in the full model (for the skew t type 4 distribution).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	138.7366	4.1172	33.6965	0.0000
pandemiaduring	-4.2729	1.8892	-2.2618	0.0237
cobertura_ab	-0.6024	0.0433	-13.9234	0.0000
idhm	-150.9868	5.4621	-27.6427	0.0000
pandemiaduring:cobertura_ab	0.0238	0.0074	3.2010	0.0014
pandemiaduring:idhm	-1.0223	2.1262	-0.4808	0.6306
cobertura_ab:idhm	0.7453	0.0584	12.7643	0.0000

```
##      2018       2019       2020       2021
##  0.8221415 -0.8226965  0.1621959 -0.1629049
```

- **For the reduced model chosen by the first variable selection strategy:**

Table 13: Measures related to the estimated coefficients for the location parameter in the first reduced model (for the skew t type 4 distribution).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	136.2914	4.2732	31.8947	0
idhm	-150.9912	5.6766	-26.5987	0
cobertura_ab	-0.5865	0.0453	-12.9436	0
idhm:cobertura_ab	0.7388	0.0608	12.1517	0

```
##      2018       2019       2020       2021
##  2.3412697  0.6101234 -1.3193291 -1.5666371
```

- **For the reduced model chosen by the second variable selection strategy:**

Table 14: Measures related to the estimated coefficients for the location parameter in the second reduced model (for the skew t type 4 distribution).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	138.4183	4.2498	32.5704	0.0000
idhm	-150.4722	5.6645	-26.5641	0.0000
cobertura_ab	-0.5948	0.0452	-13.1711	0.0000
pandemiaduring	-4.9100	0.5504	-8.9205	0.0000
idhm:cobertura_ab	0.7336	0.0608	12.0684	0.0000
cobertura_ab:pandemiaduring	0.0231	0.0063	3.6428	0.0003

```
##      2018       2019       2020       2021
##  0.8195032 -0.8201027  0.1304171 -0.1300818
```

- **For the reduced model chosen by the third variable selection strategy:**

Table 15: Measures related to the estimated coefficients for the location parameter in the third reduced model (for the skew t type 4 distribution).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	139.0234	4.1120	33.8089	0
pandemiaduring	-5.1416	0.5466	-9.4068	0
cobertura_ab	-0.6015	0.0437	-13.7600	0
idhm	-151.2952	5.4772	-27.6227	0
pandemiaduring:cobertura_ab	0.0257	0.0063	4.0672	0
cobertura_ab:idhm	0.7428	0.0588	12.6256	0

```
##       2018       2019       2020       2021
##  0.8195032 -0.8201027  0.1304171 -0.1300818
```

- **GLRT applications and comparison between AICs:**

```
# Using the LR test to verify if the simpler models are better than the full model
## The model selected by the first strategy IS better than the full model
LR.test(fit3_st4, fit1_st4)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 167640.6 with  26.21832 deg. of freedom
##  Altenative model: deviance= 167621.7 with  38.09582 deg. of freedom
##
##  LRT = 18.85653 with 11.87751 deg. of freedom and p-value= 0.08819718
```

```
## The model selected by the second strategy IS better than the full model
LR.test(fit4_st4, fit1_st4)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 167623.9 with  32.36235 deg. of freedom
##  Altenative model: deviance= 167621.7 with  38.09582 deg. of freedom
##
##  LRT = 2.161286 with 5.733477 deg. of freedom and p-value= 0.8871167
```

```
## The model selected by the third strategy IS better than the full model
LR.test(fit5_st4, fit1_st4)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 167622 with  37.10088 deg. of freedom
##  Altenative model: deviance= 167621.7 with  38.09582 deg. of freedom
##
##  LRT = 0.2250064 with 0.9949396 deg. of freedom and p-value= 0.6331505
```

```r
# Comparing the AICs
GAIC(fit1_st4, fit3_st4, fit4_st4, fit5_st4)
```

```
##                df      AIC
## fit4_st4 32.36235 167688.6
## fit3_st4 26.21832 167693.0
## fit5_st4 37.10088 167696.2
## fit1_st4 38.09582 167697.9
```

As we can see, the application of GLRT to compare the full model with the reduced models gives us evidence to believe that all the reduced models are better than the full model. As for the decision of which reduced model to choose, we can base ourselves on both the AIC values and a new application of GLRT, which can be seen below (this application can be done since the models chosen by the first and second strategies are nested in the one chosen by the third). In either case, we choose the reduced model chosen by the second strategy, `fit4_st4`.

```r
# Since the fit3_st4 model is nested within fit5_st4, we can compare them using
# GLRT (fit_st3's model for mu is contained in the fit5_st4 one, and fit5_st4 has
# the complete model for all other parameters)
LR.test(fit3_st4, fit5_st4)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 167640.6 with  26.21832 deg. of freedom
##  Altenative model: deviance= 167622 with  37.10088 deg. of freedom
##
##  LRT = 18.63152 with 10.88257 deg. of freedom and p-value= 0.06500807
```

```r
# Since the fit4_st4 model is nested within fit5_st4, we can compare them using
# GLRT (both have the same model for mu, but fit5_st4 has the complete model for
# all other parameters)
LR.test(fit4_st4, fit5_st4)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 167623.9 with  32.36235 deg. of freedom
##  Altenative model: deviance= 167622 with  37.10088 deg. of freedom
##
##  LRT = 1.936279 with 4.738538 deg. of freedom and p-value= 0.8336545
```

Typical graphs related to the analysis of residuals of this model can be seen in **Figure 30**. It is important to note here that the residuals considered by the GAMLSS models are the so-called *normalized (randomized) quantile residuals*, which, **when the distribution assumed for the response variable is adequate**, always follow a standard normal distribution. As we can see by observing the graphs presented in this figure, the assumptions of the model seem to be respected, since:

1. in the first two graphs, of residuals against adjusted values and of residuals against the indexes of the observations, the residuals appear to be distributed randomly around the horizontal line at $y = 0$;
2. in the third graph, of the estimated kernel density of the residuals, the density resembles the density of the standard normal distribution;

3. in the fourth graph, of the normal Q-Q Plot of the residuals, the observed quantiles of the residuals are in agreement with the theoretical quantiles of the standard normal distribution (to a large extent).

Furthermore, observing the summary measures present in the R output below the graphs, we can see that the mean of the residuals is approximately zero, their variance is approximately one, their skewness coefficient is close to zero and their kurtosis coefficient is close to 3. Thus, both the graphs and the summary statistics suggest that the residuals follow, approximately, the standard normal distribution, which would be necessary for the model to be considered adequate.

```r
# Residual analysis of the selected model
plot(fit4_st4)
```

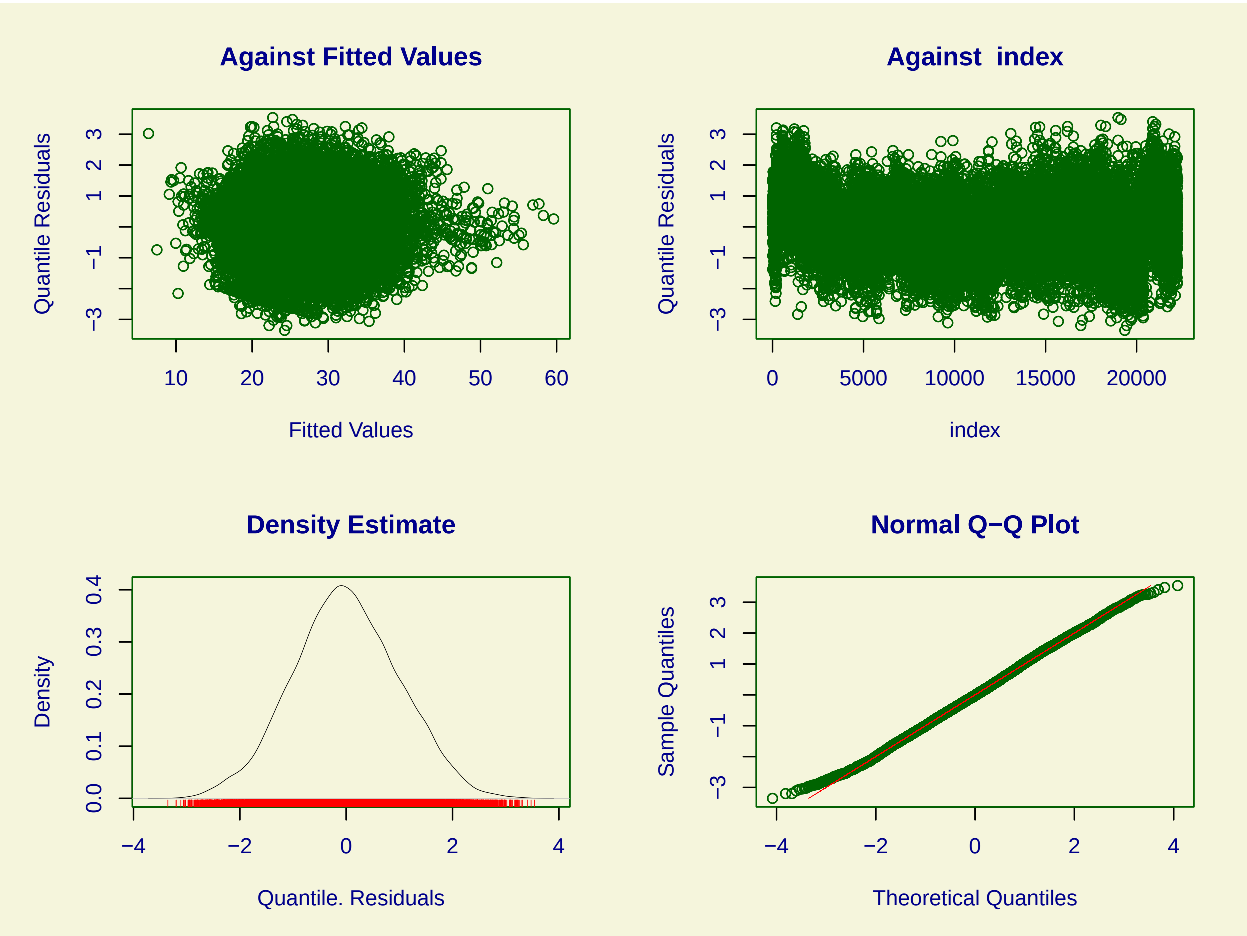

Figure 30: Usual graphs related to the residual analysis of the selected model for the skew t type 4 distribution.

```
## ******************************************************************
##        Summary of the Quantile Residuals
##                            mean    =  0.001643343
##                        variance    =  0.99794
##                coef. of skewness   =  0.04010143
##                coef. of kurtosis   =  2.865816
## Filliben correlation coefficient   =  0.9996984
## ******************************************************************
```

An additional check of the model's adequacy can be done through *worm plots*. In simplified terms, we would expect an adequate model that:

- the yellow points on the graph, which represent the model's residuals, are as close as possible to the dotted horizontal line at $y = 0$, which represents their approximate expected values;
- 95% of the yellow points are between the elliptical curves on the graph, since they represent an approximate 95% confidence interval (*point-wise*) for the model's residuals;
- and that the curve fitted to the points on the graph (in red) is as straight as possible and is as close as possible to the horizontal line at $y = 0$.

As we can see, the graph shown in **Figure 31** does indeed present points that are outside the graph's elliptical curves, which reveal that the fit was not as perfect as one would like it to be. However, as the deviations do not appear to be so numerous, and as other distributions resulted in adjustments that were extremely inferior to what was obtained through the asymmetric t distribution of type 4, we chose to consider the model found as adequate (as far as possible).

```
wp(fit4_st4, ylim.all = 0.6)
```

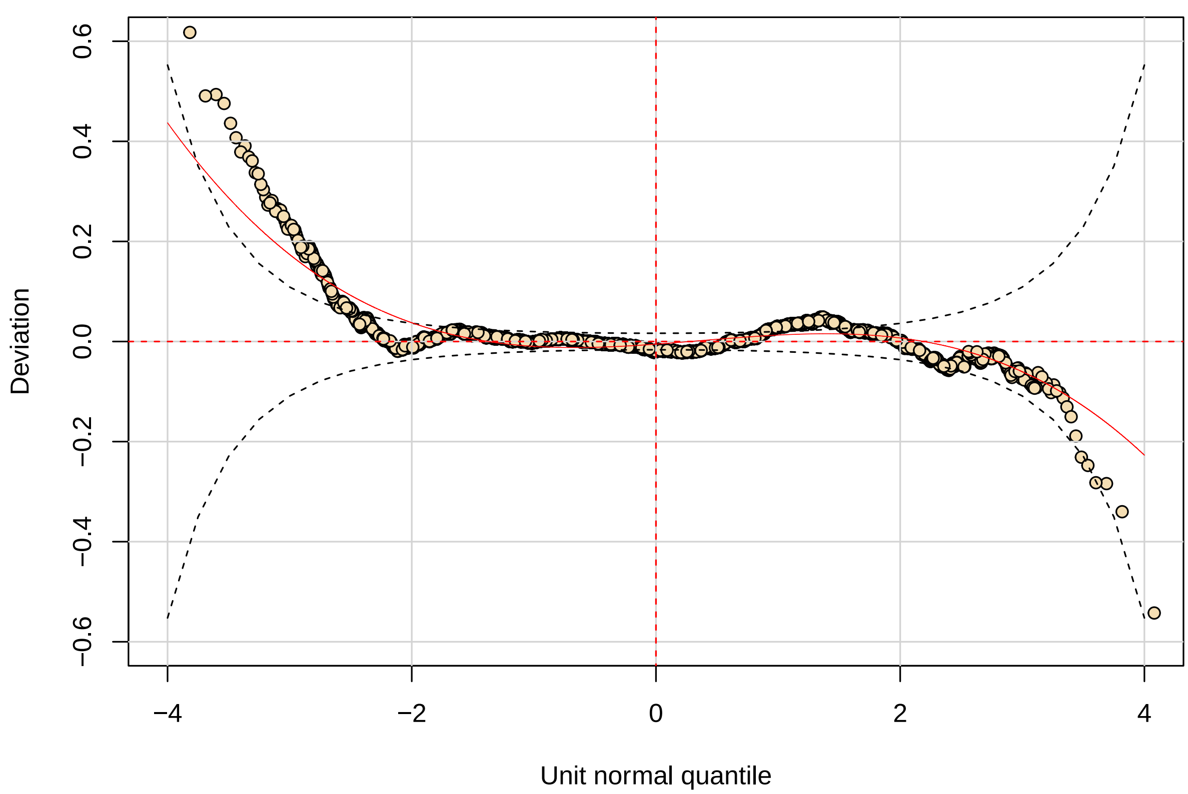

Figure 31: Worm plot of the selected model for the skew t type 4 distribution.

6.3.2 For the ZAGA distribution

```
# Adjusting the full model
fit1_zaga <- gamlss(
  tx_fecundidade_menor_20 ~
    random(ano) +
      (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  data = df_indicadores,
  family = ZAGA(),
  sigma.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  nu.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  control = gamlss.control(n.cyc = 500)
)
sumario_fit1_zaga <- summary(fit1_zaga, save = TRUE)

# Utilizing the two strategies described in the book "Flexible Regression and
# Smoothing: Using GAMLSS in R", page 397, to get the best models for each parameter
## Adjusting a model with only the intercept for mu and considering sigma and nu
## as constants
fit2_zaga <- gamlss(
  tx_fecundidade_menor_20 ~ 1,
  data = df_indicadores,
  family = ZAGA,
  control = gamlss.control(n.cyc = 500)
)

## Strategy 1:
### Starting from the simplest model and utilizing the forward stepwise method to
### find the model with the lowest AIC for each parameter
### Obs.: in this strategy, different variables can be selected for each
### parameter's model
fit3_zaga <- stepGAICAll.A(
  fit2_zaga,
  scope = list(
    lower = ~1,
    upper = ~ random(ano) +
      (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm)
  )
)
sumario_fit3_zaga <- summary(fit3_zaga, save = TRUE)

## Strategy 2:
### This strategy forces all os the models to select the same variables subsets
fit4_zaga <- stepGAICAll.B(
  fit2_zaga,
  scope = list(
    lower = ~1,
    upper = ~ random(ano) +
      (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm)
  )
)
sumario_fit4_zaga <- summary(fit4_zaga, save = TRUE)

## Adjusting a third model with only the significant terms for mu
fit5_zaga <- gamlss(
  tx_fecundidade_menor_20 ~
```

```
    random(ano) +
      pandemia +
      cobertura_ab +
      idhm +
      pandemia * idhm +
      cobertura_ab * idhm,
  data = df_indicadores,
  family = ZAGA(),
  sigma.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  nu.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  tau.formula = ~ random(ano) +
    (pandemia + cobertura_ab + idhm) * (pandemia + cobertura_ab + idhm),
  control = gamlss.control(n.cyc = 500)
)
sumario_fit5_zaga <- summary(fit5_zaga, save = TRUE)
```

Tables 16 to **19** show the estimates, standard errors, and significance tests for the models adjusted for μ in each of the four settings, as well as the additional random intercept terms for each year in each model. The applications of the GLRT and a comparison between the AIC of the three models can be seen below the tables.

- **For the complete model:**

Table 16: Measures related to the estimated coefficients for the location parameter in the full model (for the ZAGA distribution).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	7.3810	0.1608	45.9154	0.0000
pandemiaduring	0.1525	0.0691	2.2074	0.0273
cobertura_ab	-0.0231	0.0017	-13.7433	0.0000
idhm	-5.6596	0.2214	-25.5640	0.0000
pandemiaduring:cobertura_ab	0.0003	0.0003	1.0609	0.2888
pandemiaduring:idhm	-0.4239	0.0785	-5.3999	0.0000
cobertura_ab:idhm	0.0306	0.0023	13.0669	0.0000

```
##          2018         2019         2020         2021
##   0.028981948 -0.029035521 -0.002536271  0.002524325
```

- **For the reduced model chosen by the first variable selection strategy:**

Table 17: Measures related to the estimated coefficients for the location parameter in the first reduced model (for the ZAGA distribution).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	7.4360	0.1594	46.6467	0
idhm	-5.8374	0.2204	-26.4832	0
cobertura_ab	-0.0227	0.0017	-13.4841	0
idhm:cobertura_ab	0.0303	0.0024	12.8729	0

```
##         2018         2019         2020         2021
##  0.07740670  0.01684515 -0.05079312 -0.04396797
```

- **For the reduced model chosen by the second variable selection strategy:**

Table 18: Measures related to the estimated coefficients for the location parameter in the second reduced model (for the ZAGA distribution).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	7.3653	0.1596	46.1623	0
idhm	-5.6552	0.2210	-25.5898	0
cobertura_ab	-0.0230	0.0017	-13.7355	0
pandemiaduring	0.2049	0.0474	4.3199	0
idhm:cobertura_ab	0.0307	0.0023	13.1290	0
idhm:pandemiaduring	-0.4566	0.0716	-6.3724	0

```
##          2018          2019          2020          2021
##  0.028938741 -0.028995204 -0.002883041  0.002868946
```

- **For the reduced model chosen by the third variable selection strategy:**

Table 19: Measures related to the estimated coefficients for the location parameter in the third reduced model (for the ZAGA distribution).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	7.3631	0.1596	46.1481	0
pandemiaduring	0.2055	0.0475	4.3315	0
cobertura_ab	-0.0230	0.0017	-13.7147	0
idhm	-5.6520	0.2210	-25.5786	0
pandemiaduring:idhm	-0.4576	0.0717	-6.3826	0
cobertura_ab:idhm	0.0307	0.0023	13.1089	0

```
##          2018          2019          2020          2021
##  0.028938785 -0.028995235 -0.002907182  0.002895095
```

- **GLRT applications and comparison between AICs:**

```
# Using the LR test to verify if the simpler models are better than the full model
## The model selected by the first strategy (fit3_zaga) ISN'T better than the full model
LR.test(fit3_zaga, fit1_zaga)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 167837.1 with  19.68155 deg. of freedom
##  Altenative model: deviance= 167793.9 with  25.74238 deg. of freedom
##
##  LRT = 43.22143 with 6.060835 deg. of freedom and p-value= 0.0000001127193
```

```r
## The model selected by the second strategy (fit4_zaga) IS better than the full model
LR.test(fit4_zaga, fit1_zaga)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 167796.6 with  22.78337 deg. of freedom
##  Altenative model: deviance= 167793.9 with  25.74238 deg. of freedom
##
##  LRT = 2.7297 with 2.959016 deg. of freedom and p-value= 0.4278576
```

```r
## The model selected by the third strategy (fit5_zaga) IS better than the full model
LR.test(fit5_zaga, fit1_zaga)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 167795 with  24.7419 deg. of freedom
##  Altenative model: deviance= 167793.9 with  25.74238 deg. of freedom
##
##  LRT = 1.11138 with 1.000488 deg. of freedom and p-value= 0.2919317
```

```r
# Comparing the AICs
GAIC(fit1_zaga, fit3_zaga, fit4_zaga, fit5_zaga)
```

```
##                 df      AIC
## fit4_zaga 22.78337 167842.2
## fit5_zaga 24.74190 167844.5
## fit1_zaga 25.74238 167845.3
## fit3_zaga 19.68155 167876.4
```

As we can see, the application of GLRT to compare the full model with the reduced models gives us evidence to believe that the reduced models chosen by the second and third strategy are better than the full model. Again, as for the decision of which reduced model to choose, we can base ourselves on both the AIC values and a new application of GLRT, which can be seen below (this application can be done since the model chosen by the second strategy is nested in the one chosen by the third). In either case, we choose the reduced model chosen by the second strategy, `fit4_zaga`.

```r
# Since the fit4_zaga model is nested within fit5_zaga, we can compare them
# using GLRT (both have the same model for mu, but fit5_zaga has the complete
# model for all other parameters)
LR.test(fit4_zaga, fit5_zaga)
```

```
##  Likelihood Ratio Test for nested GAMLSS models.
##  (No check whether the models are nested is performed).
##
##        Null model: deviance= 167796.6 with  22.78337 deg. of freedom
##  Altenative model: deviance= 167795 with  24.7419 deg. of freedom
##
##  LRT = 1.618321 with 1.958529 deg. of freedom and p-value= 0.4354927
```

As in the case of the type 4 asymmetric t distribution, typical graphs related to the residual analysis of this model can be seen **Figure 32**. As we can see by observing the graphs presented in this figure and the summary measures present in the R output below them, the model's assumptions do not seem to be being respected. This statement becomes evident when we observe the normal Q-Q Plot of the residuals - which has many values far from the red line as we approach the right tail of the distribution, indicating that the distribution of the residuals appears to be skewed to the right - and when we observe its kurtosis coefficient, which is considerably greater than 3, indicating that the distribution of the residuals appears to have heavier tails than the standard normal distribution.

```r
# Residual analysis of the selected model
plot(fit4_zaga)
```

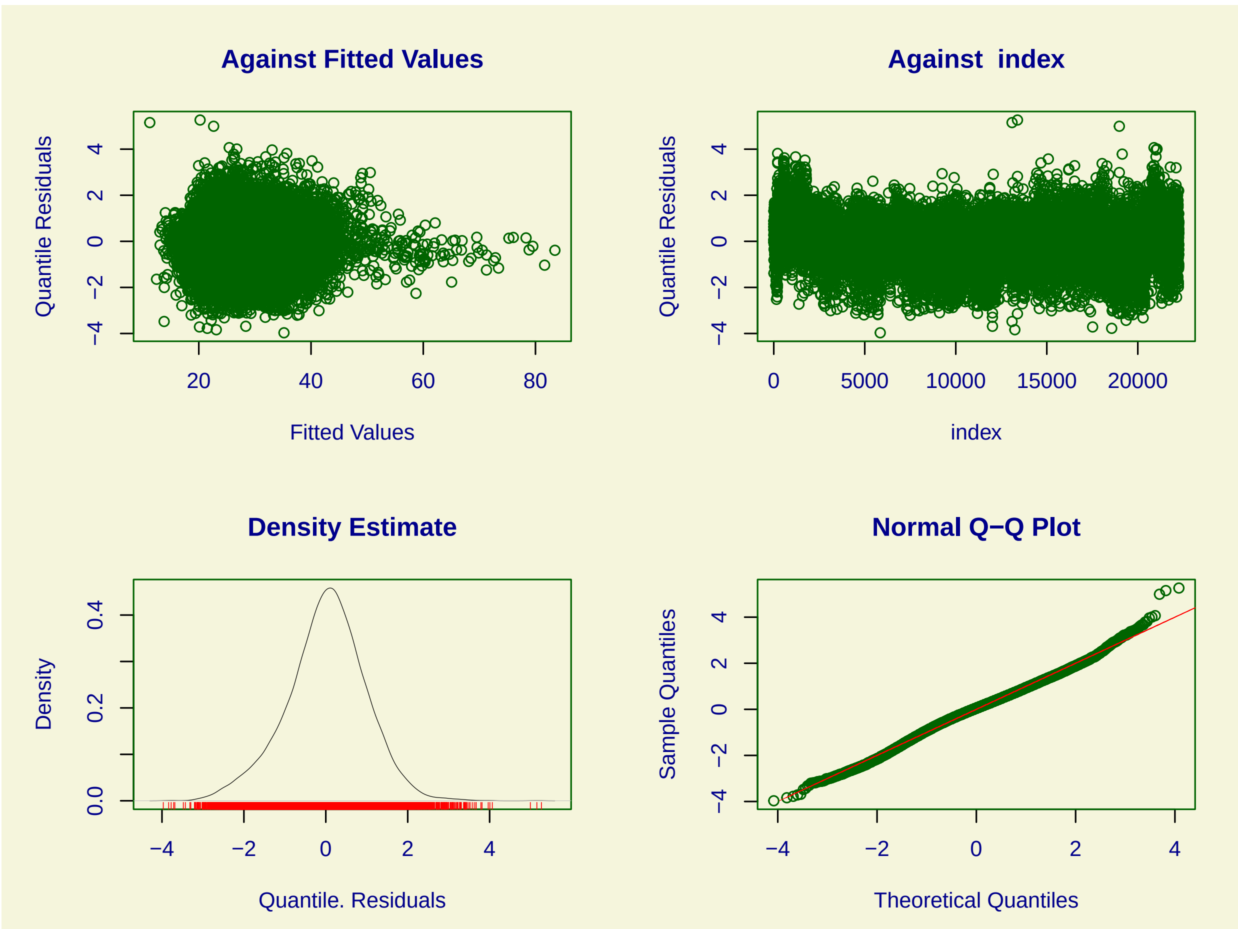

Figure 32: Usual graphs related to the residual analysis of the selected model for the ZAGA distribution.

```
## ******************************************************************
##   Summary of the Randomised Quantile Residuals
##                            mean   =  0.008808315
##                        variance   =  0.9548615
##                coef. of skewness  =  -0.1674866
##                coef. of kurtosis  =  3.473772
## Filliben correlation coefficient  =  0.9973979
## ******************************************************************
```

Furthermore, observing the worm plot shown in **Figure 33**, we can see that most of the points are outside the space formed between the two elliptic curves, giving us even more evidence that the model's residuals do not follow a standard normal distribution and that, therefore, the model is not suitable. Therefore, we conducted the remaining analyses using the reduced model chosen for the skew t type 4 distribution, since its residual analyses showed that that model appears to be, with its reservations, the most suitable.

```r
wp(fit4_zaga, ylim.all = 0.6)
```

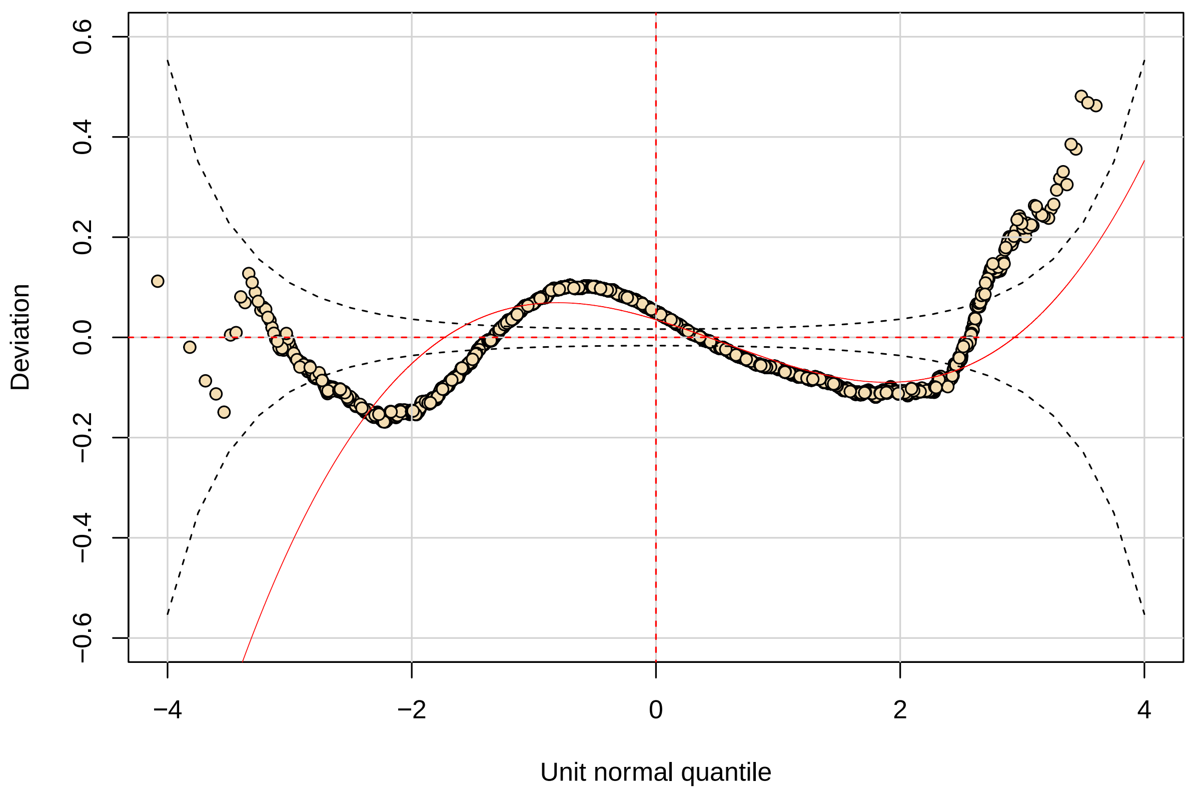

Figure 33: Worm plot of the selected model for the ZAGA distribution.

6.4 Interpreting the selected model

```r
fit_final <- fit4_st4
sumario_fit_final <- sumario_fit4_st4
```

Starting the analysis of the selected model, which assumes that the fertility rate in women under 20 years of age follows a skew t type 4 distribution, we present, in **Table 20**, the estimates, standard errors and hypothesis tests for the coefficients of the model adjusted for the location parameter - related to the mean of the distribution of the response variable - followed by the additional terms of random intercepts for each year of the model.

```r
kbl(
  round(sumario_fit_final$coef.table[c(1:6), ], 3),
  align = "cccc",
  caption = "Measures related to the estimated coefficients of the model for the
    location parameter."
) |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE)
```

Table 20: Measures related to the estimated coefficients of the model for the location parameter.

	Estimate	**Std. Error**	**t value**	**Pr(>|t|)**
(Intercept)	138.418	4.250	32.570	0
idhm	-150.472	5.664	-26.564	0
cobertura_ab	-0.595	0.045	-13.171	0
pandemiaduring	-4.910	0.550	-8.920	0
idhm:cobertura_ab	0.734	0.061	12.068	0
cobertura_ab:pandemiaduring	0.023	0.006	3.643	0

```r
getSmo(fit_final)$coef
```

```
##       2018       2019       2020       2021
##  0.8195032 -0.8201027  0.1304171 -0.1300818
```

As we can see from the sign of the estimates (and from the results of the coefficient significance tests), we can say that:

- individually, the increase in the HDI-M generates a **decrease** in the value of the fertility rate in women under 20 years of age in the municipality;
- individually, the increase in Primary Care coverage generates a **decrease** in the value of the fertility rate in women under 20 years of age in the municipality;
- individually, being a year of the Covid-19 pandemic generates a **decrease** in the value of the fertility rate in women under 20 years of age in the municipality;
- for years of the COVID-19 pandemic, the increase in PCP coverage generates an increase in the value of the fertility rate in women under 20 years of age in the municipality;
- the joint increase in the HDI-M and the PCP coverage generates an increase in the value of the municipality's fertility rate.

However, unlike linear regression models, in which making more informative interpretations about the effect of each covariate on the response variable is simple and straightforward, similar interpretations from GAMLSS models are not available. Therefore, in order to better understand how the fertility rate in women under 20 years of age behaves according to the variation of the covariates in the data set, we used the following strategy:

- we set 3 HDI-M values (quantile 0.25, median value and quantile 0.75, resulting in HDI-M values of, respectively, 0.599, 0.665 and 0.718);
- for each HDI-M value set, we varied the value of the reference year (2018 to 2021) and, therefore, the value of the variable indicating the Covid-19 pandemic;

- for each value of the HDI-M, the year and the variable indicating the Covid-19 pandemic, we used the selected model to obtain predictions for the value of the fertility rate that a municipality with Primary Care coverage ranging from 0%, 10%, 20%, . . . , 100% would have;

- We constructed a line graph that receives the value of Primary Care coverage on the x axis and the predicted value of the fertility rate on the y axis, whose lines represent the predictions for each of the three HDI-M values set in the corresponding period (the "non-pandemic" period being an average of the years 2018 and 2019 and the "with pandemic" period being an average of the years 2020 and 2021).

The graphs resulting from this process can be seen in **Figure 34**. The average variation in the predicted values of the fertility rate caused by the increase of 10 units in the population coverage of AB for each value of the HDI-M and the pandemic indicator variable can be seen in **Table 21**.

```r
# Trying to better understand the impact of each variable on the fertility rate
## Creating a dataframe with the desired observations to predict
idhms <- c(
  as.numeric(quantile(df_indicadores$idhm, 0.25)),
  median(df_indicadores$idhm),
  as.numeric(quantile(df_indicadores$idhm, 0.75))
)
anos <- 2018:2021
coberturas_ab <- seq(0, 100, by = 10)

df_newdata <- expand.grid(idhms, anos, coberturas_ab) |>
  arrange(Var1) |>
  mutate(
    pandemia = factor(
      ifelse(Var2 < 2020, "before", "during"),
      levels = c("before", "during")
    )
  )

colnames(df_newdata) <- c("idhm", "ano", "cobertura_ab", "pandemia")

## Making the predictions
predicoes <- predict(fit_final, newdata = df_newdata, type = "response")
```

```
## New way of prediction in random()  (starting from GAMLSS version 5.0-6)
```

```r
df_newdata_completo <- df_newdata |>
  mutate(
    idhm = as.factor(idhm),
    predicoes = predicoes
  ) |>
  group_by(idhm, cobertura_ab, pandemia) |>
  summarise(predicoes = mean(predicoes))
```

```r
## Plotting the predictions
ggplot(
  data = df_newdata_completo,
  mapping = aes(
    x = cobertura_ab,
    y = predicoes,
    colour = idhm,
    linetype = pandemia
  )
```

```
) +
  geom_line(linewidth = 1) +
  geom_point() +
  scale_x_continuous(breaks = unique(df_newdata_completo$cobertura_ab)) +
  labs(
    x = "Primary care coverage",
    y = "Predicted fertility rate per thousand women aged 10 to 19",
    colour = "HDI-M",
    linetype = "COVID-19 pandemic"
  ) +
  theme_bw(base_size = 14) +
  theme(legend.position = "bottom")
```

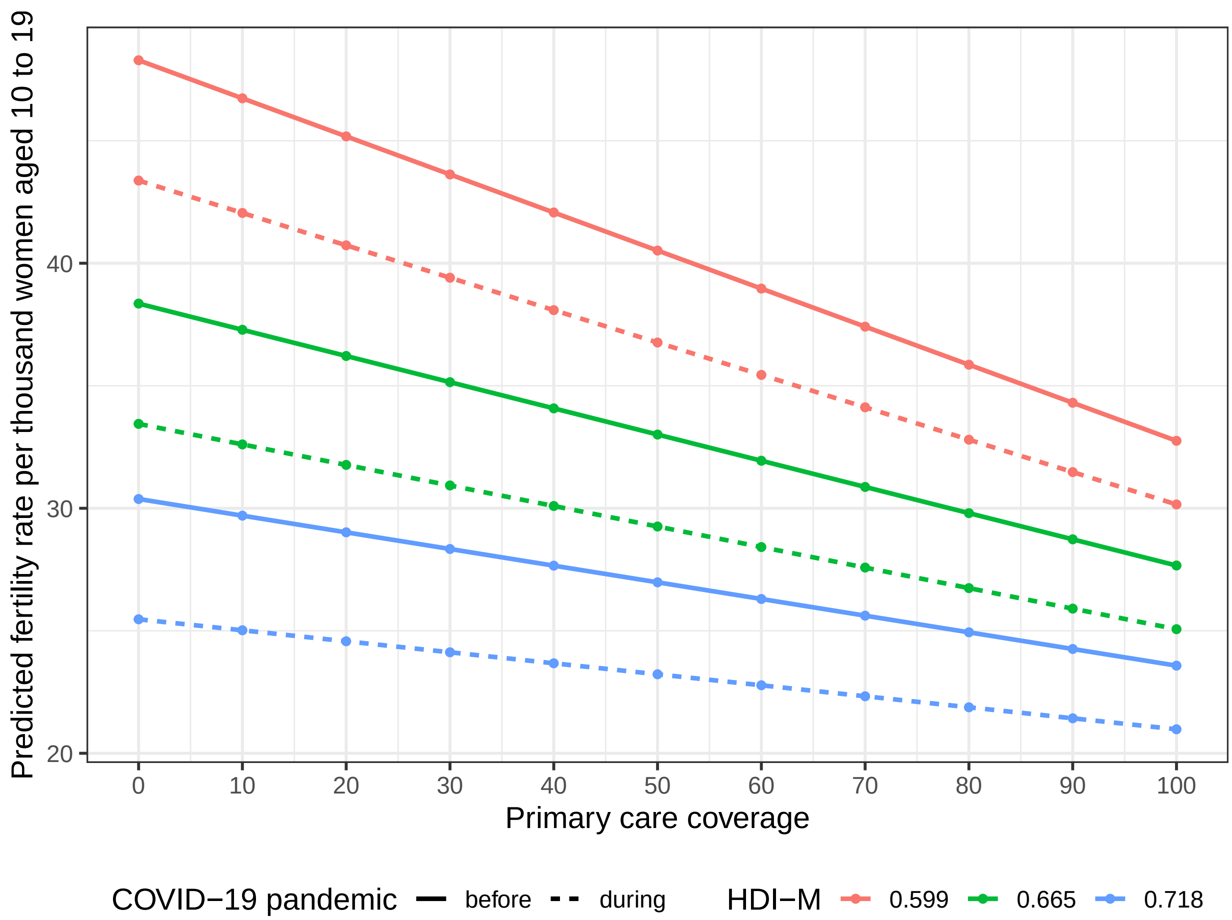

Figure 34: Predicted values of the fertility rate of women under 20 years of age according to HDI-M values, Primary Care Population coverage and information on the year of the pandemic. Brazil, 2018-2021.

```
## Creating a dataframe with the mean variations of the predicted fertility rates
## for each combination of HDI-M and the pandemic indicator when the primary care
## coverage is increased by 10
df_variacao <- df_newdata_completo |>
  group_by(idhm, pandemia) |>
  mutate(variacao = (predicoes - lag(predicoes))) |>
  summarise(variacao_media = round(mean(variacao, na.rm = T), 3)) |>
  arrange(idhm, pandemia)
```

```r
kbl(
  df_variacao,
  align = "ccc",
  col.names = c(
    "HDI-M",
    "Covid-19 pandemic",
    "Mean variation on the predictions"
  ),
  caption = "Average variation in predicted fertility rate values caused by an
    increase of 10 units in Primary Care coverage for each fixed value of the
    HDI-M and the pandemic indicator variable."
) |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE)
```

Table 21: Average variation in predicted fertility rate values caused by an increase of 10 units in Primary Care coverage for each fixed value of the HDI-M and the pandemic indicator variable.

HDI-M	Covid-19 pandemic	Mean variation on the predictions
0.599	before	-1.553
0.599	during	-1.322
0.665	before	-1.069
0.665	during	-0.838
0.718	before	-0.680
0.718	during	-0.449

Analyzing the estimates used to construct the graph presented in more detail, we can say, based on the table above and descriptively, that:

- For a municipality with an HDI-M in the 0.25 quantile of the HDI-Ms of Brazilian municipalities (from 0.599), an increase of 10 units in population coverage with Family Health teams generates:
  1. for years not affected by the Covid-19 pandemic, an average decrease of 1.553 units in the fertility rate of women under 20 years of age in the municipality;
  2. for years affected by the pandemic, an average decrease of 1.322 units in this same rate.
- For a municipality with the median HDI-M among municipalities in Brazil (of 0.665), an increase of 10 units in population coverage with Family Health teams generates:
  1. for years not affected by the Covid-19 pandemic, an average decrease of 1.069 units in the fertility rate of women under 20 years of age in the municipality;
  2. for years affected by the pandemic, an average decrease of 0.838 units in this same rate.
- For a municipality with HDI-M in the 0.75 quantile of the HDI-Ms of Brazilian municipalities (from 0.718), an increase of 10 units in population coverage with Family Health teams generates:
  1. for years not affected by the Covid-19 pandemic, an average decrease of 0.68 units in the fertility rate of women under 20 years of age in the municipality;
  2. for years affected by the pandemic, an average decrease of 0.449 units in this same rate.

Furthermore, for the entire period of 2018-2021 - as it can be seen in **Table 22-**, an increase of 10 units in the PCP coverage in a municipality with HDI-M in quantile 0.25 generates a decrease of 1.438 units in the fertility rate of women under 20 years of age. For a municipality with a median HDI-M, this increase of 10 units in PCP coverage generates a decrease of 0.953 units in the fertility rate of women under 20 years of age. Finally, in a municipality with HDI-M in quantile 0.75, an increase of 10 units in the PCP coverage generates a decrease of 0.565 units in the fertility rate of women under 20 years of age in the municipality.

```
## Creating a dataframe with the mean variations of the predicted fertility rates
## for each value of HDI-M when the primary care coverage is increased by 10
df_variacao_sem_pandemia <- df_newdata_completo |>
  group_by(idhm, pandemia) |>
  mutate(variacao = (predicoes - lag(predicoes))) |>
  ungroup() |>
  group_by(idhm) |>
  summarise(variacao_media = round(mean(variacao, na.rm = T), 3)) |>
  arrange(idhm)

kbl(
  df_variacao_sem_pandemia,
  align = "cc",
  col.names = c("HDI-M", "Mean variation on the predictions"),
  caption = "Average variation in predicted fertility rate values caused by an
    increase of 10 units in Primary Care coverage for each fixed HDI-M value."
) |>
  kable_styling(latex_options = c("scale_down", "HOLD_position")) |>
  row_spec(0, bold = TRUE)
```

Table 22: Average variation in predicted fertility rate values caused by an increase of 10 units in Primary Care coverage for each fixed HDI-M value.

HDI-M	Mean variation on the predictions
0.599	-1.438
0.665	-0.953
0.718	-0.565

References

R Core Team. 2023. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/.

R. A. Rigby, and D. M. Stasinopoulos. 2005. "Generalized Additive Models for Location, Scale and Shape,(with Discussion)." *Applied Statistics* 54.3: 507–54.

Stasinopoulos, Mikis D, Robert A Rigby, Gillian Z Heller, Vlasios Voudouris, and Fernanda De Bastiani. 2017. *Flexible Regression and Smoothing: Using GAMLSS in r*. CRC Press.